

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

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Originating OWNER	Epic Atlantic Limited
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ENGINEERING, PROCUREMENT & CONSTRUCTION
of Gas Gathering Pipelines & Associated Infrastructure-Phase 1

Section IV

EPC Scope of Work

TCE Comments:

- as marked on this document**
- EAL to attach complete FEED Package with Tender**
- various Exhibits referred in this document were not found attached, EAL to furnish**
- EAL to attach Pipeline Route Survey and Geotechnical Report**
- BFPCL's Approved Vendor List is attached**

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1.2. Purpose

This Purpose of this document is to describe and specify the scope of work required for the Engineering, Procurement, Construction, Commissioning and handover of the Pipeline and stations for BFPCL Gas Gathering Project-Phase 1.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Brief Description of Project Sites.

The Pipeline Right of way, BBOU Manifold Platform, Elepa manifold Platform and Akabel Piping station are all situated in remote environment composed of dense jungle of mangroves with matured and tall trees and thick undergrowth, shallow rivers, swamps and lagoons, which are fed by rainwater. Akabel Piping station is located inside Akabel methanol complex situated near the Atlantic Ocean.

The project area falls within the localities the kingdoms of Bubouwe Bou (BBOU), Elepa and Odioma, among others in the Niger Delta Area of Bayelsa State of Nigeria. Many of the local people are fishermen, farmers and hunters who rely on natural resources for their livelihood.

2.1. Bubouwe Bou (BBOU) Manifold Platform.

The footprint of the platform is about 39.5m x 28.5m. with a UTM Zone 32 coordinate of North: 494189.597m and East: 179056.494m.

2.2. Elepa (ELEP) Manifold Platform.

The footprint of the platform is about 48.7m x 35.3m. with a UTM Zone 32 coordinate of North: 490836.965m and East: 184805.527m.

2.3. Akabel Pig Receiver station.

Akabel pig receiver station has a footprint of 48m x 30m. The UTM Zone 32 coordinate is at North: 478440.715m, East: 211177.693.

2.4. Pipeline Right of way:

The Owner carried out a topographical survey of the Right of Way (ROW) chosen during the concept selection stage of the project totalling about 38.2km. A section of this ROW between BBOU and Elepa was found to be impacting on SPDC well head location at Elepa hence a new route of about 10.4km further south of the surveyed route is selected as an alternative to move the pipeline away from SPDC well head Location.

Also, some part of the surveyed route traverses the methanol complex at Akabel diagonally and impacted on the plot allocated for the methanol tanks. For this reason, this section of the ROW was rerouted away to the east side of the plant. The new route at the section is about 11km.

2.5. Sectionalising Valves

In accordance with the recommendations given in the international pipeline design code ASME B31.8 section 846.1 pipeline block valves are normally installed on gas pipelines at 32km (20mile) distances to allow the pipeline to be sectionalised. The distances of two pipeline system BBOU to Elepa, Elepa to Akabel do not meet the 32km distance criteria for a sectionalising valve as per ASME B31.8. However, a Quantitative Risk Assessment (QRA) will be carried out in detailed engineering to assess the pros and cons of installing sectionalising valves across some major rivers in line with Nigerian regulations. Where the pipeline systems qualify for a sectionalising valve, the Owner may approach the appropriate regulatory body (e.g. NURPC / NMDPRA) for a waiver.

Contractor may consider the cost of installation of a sectionalising valve at some major river crossings as a separate item in his quotation. the Location of this sectionalising valve shall be defined at the detailed design phase as part of the EPC contract scope. Subject sectional valve shall be installed on piled foundations and completely caged using a security Category-4 fencing material. Subject fencing shall completely enclose the riser sections of the block valve to prevent third party intrusions. The intrusion prevention and warning system shall be inclusive of anti-climbing devices, enclosed stair ways and ladders with access locks, sub-structure caging to avoid any access to risers from underneath platform.

3. General Provisions for The Scope Of Work (SOW)

3.1. Scope of Work (non exhaustive)

- The Scope of Work describes, without limitation, the nature of the Works to be executed by the Contractor pursuant to the Contract.
- The Scope of Work provides a summary and general description of the Works and highlights factors of particular importance to the Owner, which the Contractor must consider in executing the Works.
- The requirements identified and set out in this document and in the technical requirements shall be regarded as the minimum works or minimum standards of performance required to be executed or satisfied by the Contractor under the Contract.
- The Scope of Work shall not be taken to limit any of the Contractor's obligations under the Contract; neither shall it be taken to compromise any of the Contractor's obligations under the Contract. The Contract must be read as a whole.
- The Contractor is required to perform all management, engineering, surveying, procurement including purchasing, supply and transportation, construction, installation, inspection and testing, completion including mechanical completion, pre-commissioning, commissioning, start-up, **initial operation**, and performance-test and to provide training and operation assistance to the Owner, supply Services and facilities to the Owner and the Owner's personnel and perform all other Works required, to achieve the issuance of the Taking-Over Certificate and Maintenance Certificate etc. for the Project.

3.2. Scope of Work as Minimum Requirements

- The Scope of Work shall be considered as a guide for the Contractor, and the requirements referred to herein shall be taken as minimum requirements. The Scope of Work is not intended to limit or restrict the obligations of the Contractor in respect to the execution of the Works and no one part of the Scope of Work shall limit any other part of the Scope of Work or any of the technical requirements nor the responsibilities of the Contractor in respect thereof.
- Any part or aspect of the Works not expressly detailed in the technical requirements or not specified in the Contract, but necessary for the proper execution and completion of the Works and according to the provisions of the Contract are necessary to ensure that the Works are fit for the purpose for which they are intended, shall be performed by the Contractor and are deemed to be included in, and to form part of the Scope of Work.

3.3. Project Specifications

- The Project Specifications contains the technical description of the Project, including

drawings, pipeline alignment sheets and specifications as set out in the Front End Engineering and Design (FEED) documents. The Project Specification includes, but not limited to:

- the technical documentation prepared during the FEED phase, including applicable general engineering standards, guidelines and rules;
 - a) EIA study;
 - b) Topographical survey; and
 - c) Geotechnical survey.
- The Project Specification, including reports of the FEED documentation has to be endorsed by the Contractor and it remains the responsibility of the Contractor to perform any additional studies and engineering investigations as the Contractor deems necessary.
- The description and location of the structures and crossings presented in the FEED documents are provided as a guide to the Contractor and it remains the responsibility of the Contractor to verify any relevant data on the Site.

3.4. Contractor Responsibilities

- This Section is without prejudice or limitation to the obligations of the Contractor and is set out here for general overview purposes only.
- The Contractor's responsibilities include, but are not limited to:
 - a) The provision of services and facilities to the Owner as defined in the Contract;
 - b) The provision of management, co-ordination, engineering and supervision services for all aspects of the Works including:
 - (i) the development and implementation of a comprehensive control, traceability and tracking system of the Materials and equipment throughout all phases of the Works;
 - (ii) coordination of all internal and external matters with other work Contractors, Authorities of the Federal Republic of Nigeria and other Project stakeholders;
 - (iii) the planning, scheduling, execution and supervision of the Works to ensure the completion of the Works as per Contract requirements; and
 - (iv) the coordination of all internal interface matters with other Contractors and Sub-Contractors for the Project and all external interface matters with external parties to the Project.
- The provision and implementation of detailed plans, programmes and procedures for all Works, including management, engineering, procurement, construction, completion

and operation of the Project;

- The performance of and the participation in technical reviews as defined in the Contract and as required to ensure the integrity and functionality of the installations and the implementation of all findings of technical review performed by the Contractor or other Third Parties for the Project;
- The liaison with Third Parties and preparation of all required technical documents and studies to obtain necessary permits, licenses, Approvals and authorisations for the Project;
- The preparation of all documents to allow the Owner to obtain any required permits, licenses, Approvals and authorisations from Government Authorities and Third Parties;
- The completion of all engineering to allow the procurement, construction, and completion and operation of the Pipeline as specified in the FEED documents;
- The update of the Project Specifications as provided by the Owner including but not limited to, technical calculations, development of engineering drawings, incorporation of the results of design reviews and assessments performed by the Contractor, the Owner or any other Third Party;
- The development of all designs, drawings, documents, specifications and procedures for the Project to an AFC status;
- The performance of any necessary supplementary Site and geotechnical surveys and any other engineering studies as required for the confirmation of the route and station location selection and for the completion of Works in accordance with the requirements of the Contract;
- The development of all technical documents for all Sub-Contractors;
- The preparation and issue of all necessary Material Take Offs (MTO), Bill of Quantities (BOQ), Material Requisitions (MR), Material Data Sheets (MDS) and other technical documents and evaluation for the procurement of all necessary equipment including transport, handling and installation;
- The procurement of all items, including Materials, equipment, consumables, spare parts and any other items required for the construction and completion of the Project, excluding only materials, equipment, consumables, spare parts and any other items specified as being supplied by the Owner;
- The construction of the Pipelines including all linear sections, pipeline Manifold stations at BBOU and Elepa, the Pigging station at Akabel, all the ancillary installations and temporary installations as specified in the FEED documents and as required for the completion of the Pipeline systems;

- The operation, and maintenance of all the construction Sites and all Construction Material, including all security, health and safety and environment protection requirements and any temporary equipment, fixtures and facilities as required for the execution of the Works;
- The provision of inspection and testing services, procedures, facilities, equipment, Materials, consumables and documents necessary for inspecting and testing of the Project as defined in the technical requirements and in accordance with the relevant testing requirements and as required to ensure the integrity and functionality of the installations;
- The preservation and maintenance of equipment and Materials, proper storage, protection against weather influences, corrosion etc., as well as preservation of facilities under construction and partly or fully completed parts of the Works;
- The compilation, production and provision of all as-built documentation of the Project;
- The clean-up of all the Sites and the Right of Way (ROW) including the removal of all temporary attachments, supports and the like as well as the removal of waste and debris and restoration of all areas in accordance with the requirements of the Contract and as requested by the Authorities of the Federal Republic of Nigeria, affected lands and the Owner;
- The completion of the Project, including pre-commissioning, commissioning and start-up and trial operation and the performance of all tests on completion;
- The provision of operation assistance for the specified duration during the initial operation period of the Project.

3.5. Marine Operations/Warranty Surveyor (MWS)

- There are no access roads to the sites. Therefore, the only mode of transportation for moving personnel and materials to the sites is by marine transportation through the navigable waterways. The coast and river ways may require dredging to allow the movement of some cargo vessels.

3.5.1. General Requirements

- Contractor shall be responsible for performing or providing all loadout and transportation related activities including but not limited to; engineering, equipment modification, storage, marshalling, arranging pre-inspections by Company's compliance group, equipment mobilization and demobilization, equipment movement, documentation of Company's concession entry and exit routes, anchor handling plans, and other documentation such as permits, licenses, equipment clean off, statutory/regulatory approvals etc..

- CONTRACTOR shall be responsible for the safety issues, community issues and security arrangements associated with tow and House boat location
- Contractor shall provide all transportation and support equipment necessary to loadout and transport all pipeline and related components to the Work Area or Work Site(s). Said equipment shall comply with all aspects of the technical requirements, inclusive of Industry Standard marine compliance requirements.
- Contractor may use self-propelled transportation equipment. The Technical Requirements shall be interpreted to mean that references to barges apply equally to self-propelled equipment. All remedial action required for cargo barges or self-propelled equipment to comply with the Technical Requirements and marine warranty surveyor inspection shall be for Contractor's sole account.

3.5.2. Marine Engineering

- Contractor shall perform the loadout, support, and transportation analyses in accordance with best industrial standards. Such analyses shall be completed prior to transportation activities. Equipment or bare stability shall satisfy the requirements of the warranty surveyor.
- Contractor shall be responsible for protection of all equipment during the transportation period. Any damage, including water or corrosion related, shall be corrected by Contractor. Contractor shall ensure the preparation for shipping procedures are satisfied before the tow.

3.5.3. Marine Transport Journey Management

- Contractor shall implement tow monitoring and maintain contact with transport equipment at all times, and post to a website accessible by Company, the locations of all equipment en-route, at least on a regular basis. Contractor shall arrange weather forecast services, plus alternate routes and ports to avoid weather conditions exceeding transport design loads.
- Prior to the departure of the barge(s) carrying the structures from the fabrication yards, CONTRACTOR shall furnish a three (3) day weather forecast to COMPANY's site representative. During the period that any cargo barge is under tow and while on location awaiting installation, CONTRACTOR shall provide weather forecasts to COMPANY's site representative.

3.5.4. Warranty Surveyor (MWS)

- Contractor shall utilize the services of owner-approved MWS for the Pre-mobilisation inspection of Contractors sea vessels, cargo barges and torque boats. No marine vessel without appropriate certifications shall be deployed in the project

- Contractor shall utilize the services of a Owner-approved MWS for all Contractor load out to final installation designs, procedures, fabrications, installation aids, etc. needed for load outs, grillage, sea fastening, tows and marine installation activities.
- Contractor shall cooperate with the MWS, comply with MWS requirements and obtain certificates of approval from the MWS before the commencement of any load out, towing, marine transportation activities, float over installation activities and marine lifts.
- Contractor shall ensure the above mentioned requirements are sought and granted in the early stage of detailed design so they can affect any aspects of detailed design, any subsequent material orders, any fabrication schedules, etc.
- Contractor shall obtain Owner written approval before sail away.

3.5.4.1. MWS Duties

- The MWS duties will include the following as a minimum:
- Review and approve cargo and seafastening arrangement
- Review and approve cargo vessel stability calculations
- Review, approve and certify seafastening
- Inspect and approve towing vessel
- Inspect and approve cargo vessel before and after load out Witness load out operations
- Be present during pre-tow check of cargo and seafastening
- Approve decision to commence tow in conjunction with Contractor and Owner
- Issue a Certificate of Seaworthiness for the tow Review, approve and certify installation procedures

3.5.4.2. Contractor Responsibilities to MWS

- Contractor shall correct all items identified as deficient by the MWS at Contractor expense.

3.6. Services and Facilities to the Owner

- The Contractor shall accommodate representatives of the Owner and any Third Party nominated by the Owner at the Contractor's home office, the Site, and at the Sub-Contractor and Vendors premises or any other location where the Contractor is performing any portion of the Works.
- Further, detailed requirements for the provision of Services and facilities to the Owner are defined in **Exhibit C - Annex D, Attachment 1 "Services and Facilities to the Owner EPC Pipeline"**. All such Services as described shall be provided and maintained until full completion of the respective Works.
- Unless expressly stated, the defined Services and facilities are included in the Contract Price and shall be provided by the Contractor to the Owner at no extra cost.

3.7. Project Specification Endorsement

- The Project shall be designed and constructed to meet all technical requirements including those as provided in the FEED documents and those established as further optimisation during the Contractor's engineering.
- The Owner has provided the contents of the FEED documents as to minimum technical and engineering requirements for the Project. The Contractor has endorsed the contents of the FEED documents and shall execute the Works in accordance therewith, and (without limitation to the Contractor's obligations to execute the Works in accordance with the Contract) the Contractor shall take full responsibility for the FEED documents, as if the Contractor had prepared such contents by himself.
- The Owner may offer any further clarification, guidance or explanation of the contents of the FEED documents and the Contractor uses or relies on such data at its own risk.

3.8. Contract Execution Plan

- The Contractor will be responsible for producing, maintaining and updating the Contract Execution Plan throughout the execution of the Works.
- The Contract Execution Plan is the overall controlling document that will identify in general, the tiers set out for the Project. "Contract Execution Plan", including all plans, policies and procedures are specifically the activities necessary for the Contractor to implement in order to ensure that the Works are conducted in accordance with the Contract documents.
- The Contractor shall develop its Contract Execution Plan from the Project Execution Strategy submitted as part of the Tender Documents.
- The Contractor's Contract Execution Plan shall be in compliance with the Contract Milestone Schedule.
- The Contractor's Contract Execution Plan shall be subject to Approval by the Owner.
- The Contractor shall maximise the use of information management tools for efficient operations and ease of transmittal and storage of information.
- The Contract Execution Plan is a document that shall evolve as the Works proceed. The Owner will Approve plans, policies and procedures developed and implemented pursuant to the Contract Execution Plan as indicated in **Exhibit C - Annex F "Documentation Requirements"** or as otherwise stated or inferred throughout the Scope of Work.
- The Contractor is expected to apply all expertise and technical sophistication to its proposed plans, policies and procedures to be incorporated into the Contract Execution Plan as is consistent with Good Industry Practice.
- Within thirty (30) days after the Commencement Date, the Contractor shall submit a detailed Contract Execution Plan for Approval, which includes:
 - The Contractor's proposed subdivisions to be included in the Contract Execution Plan;
 - Proposed content of all plans and procedures (to the extent they are able to be produced) to be implemented in respect of the Works and collectively to comprise the Contract Execution Plan; and

- A corresponding programme of deadline dates (for all procedures yet to be finalised) by which the Contractor will submit such procedures for Approval.
- The Contractor shall update all plans and procedures (and obtain any required Approval in relation thereto) for the execution of the Works as may be necessary to ensure the Works are conducted in accordance with the Contract.
- The Contractor shall submit the Contract Execution Plan as a complete and current document.

3.9. Permits and Third Party Interfaces

- In respect of the Works, the Contractor shall be fully and solely responsible for securing in a timely and systematic manner, permits, licenses, Approvals and authorisations and any other requirements associated with the interfaces with Third Parties or Government Authorities required for the execution of the Works.
- This shall include but not limited to:
 - a) the Contractor shall in its Contract Execution Plan detail the manner and organization he shall implement for the management of communications with Governmental Authorities and Third Parties; and
 - b) the Contractor shall prepare all applications and meet all obligations and obtain all permits, licenses, Approvals and authorizations in a timely manner such that there is no delay in the Works. Where the Owner is required to obtain a permit in the Owner's name, the Contractor shall prepare the application and give reasonable advanced notice to the Owner of its obligations such that there is no delay in the Works and provide all necessary assistance to the Owner for obtaining the relevant permit.
- The Contractor shall not be responsible for EIA Approvals and Permits to Operate respectively Operating Licenses.

3.10. Contractor's Personnel and Working Hours

- All the Contractor's key personnel are subject to Approval by the Owner. The Owner may deny such Approval or request the removal of any of the Contractor's personnel from the Site which is within the sole discretion of the Owner.
- Working hours shall be kept in accordance with legal requirements of the Federal Republic of Nigeria. Shift Work and the Work during night time hours thirty (30) minutes after sunset and thirty (30) minutes prior to sunrise shall not be permitted unless specifically Approved by the Owner.

3.11. Battery Limits and Work Interfaces

- Battery limits and interfaces between the Works identified in the Scope of Work and the work being carried out by other parties (mainly the EPC Contractor for Supervisory Control and Data Acquisition (SCADA) as well as the EPC Contractor for Dredging of Access Canals to the Stations) are identified in Exhibit C, Annex G, Attachment 2 "Interface Plan and Matrix", to be developed by the contractor and approved by the Owner.

3.12. Health Safety and Security Requirements

- The implementation and adherence to good Health, Safety and Security practices are of utmost importance to the Owner. The Contractor's implementation of, and compliance with, all required procedures, regulations and policies in respect thereof will be regularly and diligently reviewed and enforced by the Owner.
- The Contractor shall implement a Health and Safety Management system in accordance with the requirements of the latest edition of OHSAS 18001 "Occupational Health and Safety Management Systems - Requirements".
- The Contractor shall observe the requirements set out in the Contract, the Scope of Work, Exhibit C, Annex B, "Project HSE Plan", and best international practice.
- The Contractor shall observe all requirements stipulated by the rules and regulations of the Federal Republic of Nigeria. Each element of the Works shall be executed in compliance with the Contractor's Health and Safety Management system.

3.13. Environmental Requirements

- The implementation and adherence to good environmental practices are of utmost importance to the Owner. The Contractor's implementation of, and compliance with, all required procedures, regulations and policies in respect thereof will be regularly and diligently reviewed and enforced by the Owner.
- The Contractor shall implement an Environmental Management system in accordance with the requirements of the latest edition of ISO 14001 "Environmental Management Systems - Requirements".
- The Contractor shall implement all requirements and recommendations derived from the Approved Environmental Impact Assessment (EIA) for the Project and the Environmental Laws of the Federal Republic of Nigeria.
- The Contractor shall observe all requirements stipulated by the rules and regulations of the Federal Republic of Nigeria.
- The Contractor shall submit the Environmental Management Plan (EMP) at least thirty (30) days prior to the commencement of any Works for Approval by the Owner.
- Each element of the Works shall be executed in compliance with the Contractor's Environmental Management System (EMS).

3.14. Quality Requirements

- The Contractor shall implement a Quality Management System in accordance with the requirements of ISO 9001:2008 “Quality Management Systems – Requirements”.
- The Contractor shall in addition observe the requirements set out in the Contract, the Scope of Work, Exhibit C, Annex C, Attachment 1 “Quality Management System Requirements” and as per best international practice.
- The Contractor shall observe all requirements stipulated by the rules and regulations of the Federal Republic of Nigeria.
- Each element of the Works shall be executed in compliance with the Contractor’s Quality Management System for the Project.

3.15. Requirements for Technical Reviews and Auditing

- The Owner will arrange for Technical Reviews as defined in the Scope of Works and Exhibit C, Annex D, Attachment 2 “Technical Reviews”, to be carried out at specific stages of the Works in order to evaluate the safety, reliability, operability and maintainability of the Project.
- The Contractor shall have due regard to its obligations in respect of technical reviews as set out in the Scope of Work and Exhibit C, Annex D, Attachment 2 “Technical Reviews” and provide all assistance and cooperation to the Owner (including relevant documentation availability) as may be required.
- The Contractor shall, at all times, keep available all records and aspects of the Works at each place where the Works are being executed for detailed auditing by the Owner or any Third Party appointed by the Owner for this purpose.
- In particular the Contractor shall maintain and make available at each place where the Works are being executed, an original copy of the latest issue of all applicable codes, norms, standards, drawings, specifications, procedures and other requirements for the performance of the Works, for reference and the Owner’s use.
- These Technical Reviews performed by the Owner shall be additional to the Contractor’s own, internal engineering reviews and technical auditing required to be performed in accordance with ISO 9001: 2008 “Quality Management Systems – Requirements” and with Scope of Work, Exhibit C, Annex C, Attachment 1 “Quality Management System Requirements”.

3.16. Data and Documentation Requirements

- The Contractor has extensive obligations to produce documents (including all drawings, plans, documents and procedures and other documents identified in the technical requirements) in the course of executing the Works.
- The Owner will require the Contractor to cooperate and comply with information management, communications processes and electronic document management systems.
- The Contractor shall comply, without limitation, with all documentation requirements

specified in the Contract, the Scope of Work and Exhibit C, Annex F “Documentation Requirements”.

3.17. Engineering Requirements

- The Contractor acknowledges that the contents of the Project Specifications and other documents provided by the Owner do not define the Project in full detail, and that the Contractor shall therefore develop and update the Project Specifications and shall carry out all additional engineering necessary for the complete and detailed definition of the Project, procurement and supply, transportation, construction and fabrication, hook-up, completion and operation of the Project in accordance with the Scope of Work.
- The requirements for the performance of the Contractor’s engineering shall include but not limited to:
 - a) The Contractor shall perform all required investigations, studies and assessments and produce all required calculations, drawings, layouts, diagrams, lists, material take-offs, specifications, datasheets and any other engineering documents required for the Specification, procurement, construction and installation, inspection and testing, completion and operation, of the Project or any part of the Project;
 - b) The Contractor shall perform all engineering activities in accordance with Approved procedures, the Approved Contract Execution Plan and in compliance with the relevant Project Specifications and procedures;
 - c) The Contractor’s engineering is subject to review and Approval by the Owner, according to the Document Review Procedure described in the Project Execution Plan. The Contractor shall co-ordinate the review of its engineering documents with the Owner. The Contractor shall incorporate all relevant comments of the Owner prior to the re-submission of the relevant documents;
 - d) The Owner’s review and Approval does not relieve the Contractor from its responsibilities whilst performing the engineering for the Project;
 - e) The Contractor shall perform all engineering activities in accordance with the Project requirements, the requirements of the applicable codes and standards, specifications, rules and regulations of the Federal Republic of Nigeria, good international engineering practice, and any other requirements identified or introduced by the Owner;
 - f) The Contractor shall recognize and implement all requirements stipulated by the Authorities of the Federal Republic of Nigeria and any other official entity of Third Party involved with the licensing or Approval of the engineering or any other part or aspect of the Works;
 - g) The Contractor shall recognize and implement all requirements stipulated by applicable national and international standards, codes and practices as well as good international engineering practice during the performance of the engineering and any other part or aspect of the Works;
 - h) The Contractor shall participate in, and action the results of Project technical and Constructability reviews, project reviews such as HAZID, HAZOP, SIL, SIMOPS, PHSER, Fire Safety Review and others, quality audits and other audits initiated by the Owner, or the Authorities of the Federal Republic of Nigeria and co-ordinate the resolution of

technical queries and prepare and apply required responses and technical inputs to its engineering;

- i) The Contractor shall observe all requirements stipulated in the Exhibit C - Annex G, Attachment 2 "Interface Plan and Matrix - EPC Pipeline and Stations" for definition of its engineering scope of work; and
- j) During the performance of the engineering for the pipeline stations (Manifold and pigging stations), the Contractor shall address the following, but not limited to these engineering disciplines:
 - (i) Process;
 - (ii) Piping;
 - (iii) Mechanical;
 - (iv) Civil / structural;
 - (v) Painting/insulation/fireproofing;
 - (vi) Electrical;
 - (vii) Cathodic protection;
 - (viii) Heating, ventilation and air-conditioning (HVAC);
 - (ix) Instrumentation and control systems including SCADA and telecommunications;
 - (x) Fire safety;
 - (xi) Health safety and environment (HSE); and
 - (xii) Security and surveillance.
 - (xiii)
- k) The Contractor shall perform or supplement all required studies and investigations for the determination of relevant design parameters for the engineering of the temporary (sacrificial anode), permanent (impressed current) cathodic protection systems and the high voltage influence protection system of the Pipeline;
- l) The Contractor shall recognize all requirements (i.e. maximum loads and sizing) of local utility providers for the engineering of permanent and temporary utility connections;
- m) The Contractor shall provide all engineering for the inspection and testing of the Works, including the performance of critical engineering assessments for the interpretation of inspection and testing results;
- n) The Contractor shall review and update all engineering documents to ensure conformance with health & safety, environment and quality requirements of the Project;
- o) The Contractor shall liaise with any other works or EPC Contractors and other Third Parties or Government Authorities for all matters interfacing with the Contractor and/or the Works;
- p) The Contractor shall recognize all interface matters resulting from, or otherwise

connected with, Project interfaces and prepare and issue all engineering documents including for tie-ins, isolation points and related MTOs;

- q) Upon completion of its engineering the Contractor shall compile all AFC documents in an engineering dossier, which shall constitute the comprehensive and detailed description of the Project;
- r) The Contractor shall update and re-issue all engineering documents with “As-Built” Status” on the basis of the as-built records, red mark-up drawings and all other relevant documents that have been generated during the construction and completion of the Works, including all necessary rectifications and Approved modifications that may have been implemented;
- s) The Contractor shall ensure full traceability of all Approved engineering modifications through the employment of an engineering database system, as part of the final documentation requirements.

- Further requirements for the Contractor’s engineering are specified in Exhibit C - Annex G, Attachment 1 “Engineering Management Requirements”.

3.18. Procurement Requirements

- The Contractor shall perform all his procurement activities in compliance with all applicable Project rules and the applicable rules and regulations for procurement in the Federal Republic of Nigeria.
- Further requirements for the Contractor’s engineering are specified in Exhibit C, Annex H, Attachment 2 “Procurement and Material Control Requirements”.

3.19. Construction Requirements

- The Contractor shall perform the construction of the Project in accordance with all applicable Project requirements, the relevant Project Specifications and procedures, codes and standards and the requirements of the Laws, Rules and Regulations of the Federal Republic of Nigeria.
- The Contractor shall observe all requirements stipulated in the Exhibit C, Annex G, Attachment 2 “Interface Plan and Matrix - EPC Pipeline and Stations” for definition of its construction scope of work.

3.20. Completion Requirements

- The Contractor shall perform the completion of the Project in accordance with the requirements set out in Exhibit C, Annex I, Attachment 1 “Completion Management Requirements - EPC Pipeline and Stations”.
- During the performance of the Trial Operation, the Contractor shall observe all requirements of Exhibit C, Annex J, Attachment 1 “Operation Management - EPC Pipeline and Stations”, and Attachment 2 “Maintenance Management - EPC Pipeline”, as well as all legal requirements of the Federal Republic of Nigeria and best industry practice.

3.21. Requirements for Operations and Maintenance (O&M) Training & O&M Assistance

- The Contractor shall perform the O&M and the provision of O&M assistance of the Project in strict compliance with all requirements stipulated in the operation license for the Project.
- The Contractor shall provide O&M training and assistance in accordance with the requirements set out in Exhibit C, Annex J, Attachment 1 "Operation Management - EPC Pipeline", Attachment 2 "Maintenance Management - EPC Pipeline" and Attachment 3 "Training Requirements".

3.22. The Nigerian Content Requirement.

- Nigerian Content is defined in the Nigerian Oil & Gas Industry Content Development Act as "the quantum of composite value added to or created in the Nigerian economy by a systematic development of capacity and capabilities through a deliberate utilisation of Nigerian human, material resources and services in the Nigerian oil and gas industry".
- Owner is committed to the Government's aspiration of utilizing as much as possible the skills and resources available in the Nigerian industry and in expression of this commitment has undertaken to carry out the following in its operations: -
 - a) Employment and personal development opportunities for Nigerians.
 - b) Maximise the use of Nigerian manufactured goods and services.
 - c) Drive structured training and skill development for effective transfer of management and technical skills.
 - d) Sustainable development of Nigerian businesses as suppliers, sub-contractors and service providers.
 - e) Promote the contribution of Nigerian contractors to the development of the national industrial base beyond the Oil and Gas Industry.
- The Contractor shall maintain a Nigerian Content Plan throughout the Contract duration. This Plan shall document all aspects of the Contractors strategy for achieving the Nigerian Content commitments in accordance with their submitted technical proposal and consistent with the minimum targets set in Schedule of the Act. Refer to Volume 02 – Annex B – "The Nigerian Content Requirement" for details of requirements.

4. Scope of The Pipeline Sections and Stations

4.1. General

The Scope of Work comprises mainly the EPC Work for the construction of the two Pipeline Sections of 24" and 36" , two manifold Platforms, a pigging station, Pipeline Ancillary Installations such as Cathodic protection systems, FOC conduit and manholes and the installation of 38.2km x 90mm² high voltage armoured cable etc.

The detailed scope of the Pipeline and Stations are identified in the FEED documents, the work packages of the Scope of Work and all other information provided in the Contract. The following information is provided without limiting or restricting the Scope of Work of the Contractor under the Contract.

4.2. Linear Sections

4.2.1. 24" Multiphase Pipeline section

- The 24" Multiphase Pipeline section is described thus;

Parameter	Description
Design Code	ASME B31.8
NPS	24"
OD / Wall Thickness	15.88mm
From (Battery Limit)	BBOU Manifold Station (Last Flange Before Monoblock Isolation)
To (Battery Limit)	ELEPA Manifold Station (First Flange After Monoblock Isolation)
Pipeline Length	6.7km
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL -SMS

4.2.2. 36" Multiphase Pipeline Section

- The 36" Multiphase Pipeline section is described thus;

Parameter	Description
Design Code	ASME B31.8
NPS	36"
OD / Wall Thickness	23.83mm
From (Battery Limit)	Elepa Manifold Platform Station (Last Flange Before Monoblock Isolation)
To (Battery Limit)	Akabel Piping Station (First Flange After Monoblock Isolation)
Pipeline Length	31.5km
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL

4.2.3. Crossings

- The List of Crossings of the Pipeline is provided in relevant attachments of **Exhibit C, Annex A, Attachment 1 "Crossing List "**.
- The provided list is indicative and all crossings as well as the identification of other crossings shall **be performed by the Contractor as part of the construction investigations as described in WSG 5.2.5 "Construction Investigations"**.

4.2.4. Special Areas

- "Not applicable".

4.3. Pipeline Ancillary Installations

4.3.1. Cathodic Protection

- The scope of the Pipeline includes the design procurement, construction, installation and completion of the temporary (sacrificial anode) and permanent (impressed current) CP system for the pipeline.

4.3.2. 4 core 90mm² x 11kVA high voltage Electrical cable an AC Mitigation / HV-Influence Protection System.

The scope of the Pipeline includes;

- Design, procurement and installation of a **4 core 90mm² x 11kVA high voltage Electrical Armoured cable** capable to withstand the harsh conditions of Niger delta marine environment for over a period of 30 years in service. The cable shall run from the high voltage electrical switch gear/power transformer at BBOU manifold platform to the high voltage electrical switch gear/power transformer at Elepa manifold platform. Then from Elepa Manifold station to the power station in Akabel Methanol Plant traveling a

total distance of about 38.2km.

- Performance of a High Voltage Influence Study and the design procurement, construction, installation and completion of the AC- Mitigation / HV-Influence Protection System as the cable is planned to be laid in parallel to pipeline along the ROW. Contractor shall advice the Owner on the best possible solution to achieve this objective.

4.3.3. Control and Instrumentation and Telecom Systems

- The scope of the Pipeline includes the supply and installation of field instrumentation including cable laying within the Pipeline stations and all other Pipeline related instrumentation, intrusion detection system, F&G detection system, SCS based on (PLC/RTU), ESD system and control containers. These are Containerized E-House/Field Auxiliary Room (FAR). There will be RTU room in each manifold that shall house SCADA, Telecom equipment, Power supply etc.
- Supply and installation of telecommunication systems as VoIP, backbone WAN network, CCTV transmission to security operation centre, etc., is the responsibility of Station EPC Contractor.

4.3.4. Fibre Optic Cable/ Conduit

- The scope of the Pipeline includes the supply, installation and testing of one (1) Fibre Optic Cable Conduit, Cable Chambers and all other appurtenances along the entire length of the Pipeline, inside the pipeline stations, manifold stations and Akabel Station/Cabinet room. At specified crossing locations one spare fibre optic cable conduit shall be installed.
- The scope of the Pipeline may include the supply, installation and testing of two (2) Fibre Optic Cables of 48 cores each, one for data in a composite cable sparated from pipeline at statutory safe distance and the other for leak detection strapped on the pipeline up to the main control cabinet of each pipeline station. At the moment, the Owner is considering it as an optional scope. See section 7 for full details of the scope.

4.3.5. Marker and Measuring Posts

- The scope of the Pipeline includes the supply, installation of all marker and measuring posts indicated by the FEED and derived from the Contractor's design.

4.4. Temporary Installations

4.4.1. Temporary Site Facilities

- The Scope of Work includes the temporary acquisition of land (through rental or lease agreements), the design, construction, installation, operation and removal of all temporary Site facilities and installations, including complete reinstatement of all affected areas of all temporary Site facilities, including but not limited to the Contractor's main camps, satellite camps, manufacturing and pre-fabrication yards,

material storage areas and pipe dumps.

- The location selection of all temporary Site facilities is subject to Approval by the Owner.
- The Contractor shall be responsible for the preparation of the main PSY prior to the first arrival of line pipes (date to be specified) as well as the operation and maintenance and complete reinstatement of the PSY.
- Areas for additional, intermediate pipe storage yards (pipe dumps) and all other temporary Site facilities have to be acquired by the Contractor with the location selection being subject to Approval by the Owner.

4.4.2. Temporary Site Utilities

- The Scope of Work includes the installation of all the temporary Site utilities and potential connection of the temporary Site utilities with local utility lines, at all the Sites where temporary Site utilities may be required during the performance of the Works.

4.4.3. Temporary Access Roads and Canals

- The Scope of Work includes the provision of all temporary access roads and canal to all Site locations, as required for the performance of the Works.

4.4.4. Temporary Security Installations

- The Scope of Work includes the provision of all temporary security installations and security measures required during the performance of the Works.

4.5. Stations

4.5.1. Bubouwe Bou (BBOU) Manifold Platform.

The platform comprises but not limited to the following;

- Field Manifold Headers – SPDC Wells & BFPCL Bulk line
- Incoming OPUG-BBOU Bulkline /Pig Receiver (Not Part of Phase 1)
- Outgoing BBOU-ELEP Bulkline / Pig Launcher
- RTU – Control Room / Field Auxilliary Room (FAR) including UPS
- Telecommunication through fibre optics cable with VSAT as back-up
- Cable Power from Akabel, with Solar as back-up
- Electrical Transformer & Electrical Room
- Chemical (Corrosion Inhibitor) Injection System
- Gas over oil actuator for valves
- Gantry & Accessories
- Jib Crane
- High Security Fencing (e.g. Citadel, Harass)
- Navigation Aid
- Earthing System

- Closed Drain Vessel
- Open Drain Tank
- Vent Stack
- Weather Shelter / Guard Hut

4.5.2. Elepa (ELEP) Manifold Platform.

The platform comprises but not limited to the following;

- Field Manifold Headers – SPDC Wells & BFPCL Bulk line
- Incoming NUNR-ELEP Bulkline / Pig Receiver (Not part of phase 1 scope)
- Outgoing ELEP-AKAB Bulkline / Pig Launcher
- RTU – Control Room / Field Auxilliary Room (FAR) including UPS
- Cable Power from Akabel
- Telecommunication through fibre optics cable with VSAT as back-up
- Electrical Transformer & Electrical Room
- Chemical (Corrosion Inhibitor) Injection System
- Gas over oil actuator for valves
- Gantry & Accessories
- Jib Crane
- High Security Fencing (e.g. Citadel, Harass)
- Navigational Aid
- Earthing System
- Closed Drain Vessel
- Open Drain Tank
- Vent Stack
- Weather Shelter / Guard Hut

4.5.3. Akabel Pig Receiver station.

The platform comprises but not limited to the following;

- Pig Receiver
- Slug Catcher – GPP Scope
- Earthing System
- Jib Crane
- Open Drain Tank
- Depressurisation through the GPP Flare system

5. Scope of Procurement

5.1. General

- The Contractor is responsible for the performance of all procurement works as defined in Section 9, WP 4 "Procurement" and requirements as specified in Exhibit C, Annex H and all attachments thereto.

5.2. Contractor's Procurement and Supply Items

5.2.1. General

- The Contractor shall procure all items, including equipment, materials and consumables required for the performance of its Scope of Work.

5.2.2. Equipment and Materials

- The Contractor's procurement of equipment and materials shall include, but not limited to:
 - a) All line pipe for the pipeline as identified in the Contractor's detailed engineering for the Pipeline;
 - b) All factory bends for the pipeline as identified in the Contractor's detailed engineering for the Pipeline;
 - c) All Structural Steel and Piles and piling sheets required for the fabrication and Installation of the Platforms
 - d) All skids or functional units identified in the detailed engineering for the Pipeline;
 - e) All tagged equipment and components identified in the detailed engineering for the Pipeline;
 - f) All non tagged materials identified in the detailed engineering for the Pipeline;
 - g) All other materials required for the performance of the Works until issuance of the Taking-Over Certificate;
 - h) All other materials required for the Works until Maintenance Certificate of the Pipeline;
 - i) All special tools and handling gear recommended by the Vendors for lifting, transportation, installation and maintenance of the relevant equipment;
 - j) All materials required for the fabrication of non-procurable items; and
- For equipment and materials required for the Instrumentation, SCADA and Telecom systems for block valve, pigging, metering and terminal gas stations. The Contractor shall observe the stipulations of Exhibit C, Annex G, Attachment 2 "Interface Plan and Matrix - EPC Pipeline and Stations".

5.2.3. Bulk Materials and Consumables

- The Contractor's procurement of bulk materials and equipment shall include, but not

limited to:

- a) all consumables and operating fluids which are required for the performance of the Works until final acceptance of the Pipeline;
- b) all operating consumables and first fills identified (except the filling of the pipeline with natural gas) including filling of all relevant systems, including but not limited to:
 - (i) fuels;
 - (ii) water;
 - (iii) coolants;
 - (iv) lubricants, oils and greases;
 - (v) foams;
 - (vi) inert gases (i.e. nitrogen).
- c) any other materials that is consumed, exhausted or extinguished during the performance of the Works;
- d) all energy that is consumed during the performance of the Works.

5.2.4. Spare Parts

- The Contractor shall provide lists for commissioning and completions spares, two years operation spares and capital spares as part of its engineering activities, according to Exhibit C, Annex H, Attachment 4 "Spare Parts and Interchangeability Philosophy".

5.2.5. Commissioning / Completion spares

- All spare parts required for the pre-commissioning, commissioning and start-up of the Pipeline shall be supplied by the Contractor and shall be included in the Contract Price.

5.2.6. Spare Parts for two years operation

- The spare parts for the two years of operation shall be supplied by the Contractor and shall be included in the Contract price.

5.2.7. Capital Spares

- The procurement of capital spares shall be carried out by the Contractor as additional works to the Contract to be paid for from the provisional sum.
- The Contractor shall provide a list of all recommended capital spares as part of its engineering, subject to Approval by the Owner.
- Following the Approval by the Owner, the Contractor shall proceed with the procurement of all Approved capital spares as part and to the same conditions as the original purchase order.
- The Contractor shall arrange the transport and delivery of the procured parts to a

specified location within the Federal Republic of Nigeria as advised by the Owner.

5.3. Owner's Procured Items

- "Not applicable".

6. Scope Of Works

The description of the Works in this Scope of Work (SOW) is broken down into discrete Work Packages. The works comprising each Work Package are set out in further detail in Work Groups (WG) and Work Sub-Groups (WSG) in the following sections of the SOW.

The requirements set out in each Work Package shall be applied for the performance of the Works.

The descriptions of the Work Packages contain in certain cases, or incorporate by reference, a listing, without limitation, of specific factors and obligations, which the Contractor must take into account in executing the Works comprising that Work Package.

The following sections set out the SOW in further detail in relation to each Work Package and incorporate into each Work Package, where appropriate, requirements contained in the other sections and Addenda to the SOW.

The division of the Works as described above is not intended to limit nor restrict the extent of the SOW in any way, nor direct or instruct the Contractor as to the detailed procedures in which the Works shall be executed.

6.1. Work Package 1 Project Management and General Work Items

6.1.1. Work Group 1.1 Project Management

- The Contractor is responsible for the management of all aspects of the Works, including but not limited to the management of all its personnel and Sub-Contractors, the management of all interfaces with other Contractors of the Project, the Authorities of the Federal Republic of Nigeria, the Owner, Project stakeholders and any other Third Party.
- The Contractor shall manage the execution of the Works in accordance with the requirements of the Contract and in accordance with its Approved Contract Execution Plan.

6.1.1.1. WSG 1.1.1 Overall Project Management

- The Contractor shall establish a dedicated executive Project management entity for the effective overall management of the Project during the performance of the Works.
- The Contractor shall submit the Organization Chart and relevant Job Descriptions for its overall Project management organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.
- The Contractor shall prepare a Contract Execution Plan for the management of the Project, including, but not limited to:
 - a) Project Management Manual;
 - b) Project Management Plan; and
 - c) Project Management Procedures.
- The Contractor's Contract Execution Plan shall be prepared in strict compliance with the overall Project Execution Plan and the Project Milestone Schedule.
- The Contractor shall submit all Project management documents not later than thirty (30) days after the Commencement Date, for Approval by the Owner.
- The Contractor shall establish a reporting system for all executive management affairs. The reporting shall be performed within the Scope of **WSG 1.8.8 "Project Reporting"**.

6.1.1.2. WSG 1.1.2 External Project Management

- The Contractor shall employ a dedicated team for the external representation and management of Project issues and affairs, which includes contacts with the Authorities of the Federal Republic of Nigeria, the general Public, the Owner and other Project stakeholders and Third Parties.

6.1.1.3. WSG 1.1.3 Internal Project Management

- The Contractor shall manage, control, and supervise all aspects of its work performance through the implementation of a management structure and the establishment of relevant inspection, control and management systems as outlined in

this SOW, in accordance with all applicable rules and regulations of the Federal Republic of Nigeria and in accordance with best international practice.

- The Contractor shall provide effective management and supervision to all Sub-Contractors involved in execution of the Works.
- The Contractor shall establish a dedicated organization for the management and control of all Works, as described in the work packages of the Scope of Work.
- The Contractor's internal project management organization shall include dedicated line managers for all work packages, including but not limited to:
 - a) Project Manager;
 - b) Administration Manager;
 - c) Health and Safety) Manager;
 - d) Environmental Manager;
 - e) Security Manager;
 - f) Quality Assurance/Quality Control (QA/QC) Manager;
 - g) Project Control Manager;
 - h) Technical Integrity and Project Reviews Manager;
 - i) Engineering Manager;
 - j) Procurement Manager;
 - k) Construction Manager;
 - l) Mechanical Completion Manager;
 - m) Completions Manager;
 - n) Commissioning Manager;
 - o) Operations Manager; and
 - p) Training Manager.
- On each main working Site of the Contractor the relevant line managers shall be represented by an appointed and dedicated representative to ensure proper management, control, and reporting lines on the relevant work Site.

6.1.2. Work Group 1.2 Health and Safety

- The Contractor is responsible for the provision of all Health and Safety Services for the Project.
- Further detailed Health and Safety requirements for the Project are specified in Exhibit C, Annex B "Project HSE Plan".

6.1.2.1. WSG 1.2.1 Health and Safety Management

- The Contractor shall establish a dedicated H&S entity which is fully competent for the provision of H&S services and the management of all H&S aspects of the Project during the entire Works in the scope of work. Numbers of H&S inspectors/advisors shall be at least one H&S advisor for every fifty (50) workers or at least one for each separate activity for smaller, remote crews.
- The Contractor's Health and Safety Manager shall be reporting directly to the Contractor's Project Manager.
- All the Contractor's health and safety personnel shall be properly licensed or accredited for the provision of such services by the Authorities of the Federal Republic of Nigeria.

- The Contractor shall submit the Organization Chart and relevant Job Descriptions for its health and safety organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.
- The Contractor shall prepare all required documents for the health and safety management of the Project including, but not limited to:
 - a) Health and safety policy;
 - b) Health and safety manual;
 - c) Health and safety plan; and
 - d) Health and safety procedures.
- The health and safety manual, Plan and Procedures shall reflect the Owner's and the Contractor's commitment to provide an accident and incident free environment to all Project personnel and all Project stakeholders and the Environment.
- The Contractor shall submit its health and safety documents not later than thirty (30) days after the Commencement Date, for Approval of the Owner.

6.1.2.1.1. Health and Safety Reporting

- The Contractor shall establish a system for the reporting of all health and safety affairs. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".
- Further detailed requirements are specified in Exhibit C, Annex B "Project HSE Plan".

6.1.2.2. WSG 1.2.2 Crisis and Emergency Management

- Contractor shall submit a crisis and emergency management procedure to the Owner for Approval within thirty (30) days prior to mobilization to the Site. This procedure shall also cover proper crisis management applications such as total evacuation, headquarters involvement, resources, media relations, etc.
- Additionally, Contractor shall submit a specific emergency response plan to the Owner for Approval within fifteen (15) days after the mobilizations with detailed H&S arrangements, including emergency team members, emergency phone numbers, fire detection and fighting applications, etc.

6.1.3. Work Group 1.3 Security

- The Contractor is responsible for the provision of security services at all the Sites covering all its personnel, Sub-Contractors, the Owner's personnel, Consultant's Personnel and any Third Parties visiting the Site including transportation to and from the Sites. The provision of security services shall be coordinated with local Law enforcement agencies and the Security Forces of the Federal Republic of Nigeria.
- The Contractor shall provide its security services in strict compliance with the regulations of the Federal Republic of Nigeria.

6.1.3.1. WSG 1.3.1 Security Management

6.1.3.1.1. Security Organisation

- The Contractor shall establish a dedicated security entity sufficiently staffed to cover

all security aspects of the Project, during the performance of the Works until the Taking over of the Project by the Owner.

- The security entity shall be conversant with marine operations and the locality of the project sites.
- The Contractor shall submit the organization chart and relevant job descriptions for its security organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.
- The Contractor's security personnel shall be properly licensed or accredited for the provision of such services by the Authorities of the Federal Republic of Nigeria.

6.1.3.1.2. Security Management Documents

- The Contractor shall prepare as part of the Contract Execution Plan, all required security management documents for the Project including, but not limited to:
 - a) Security manual;
 - b) Security plan; and
 - c) Security procedures.
- The Contractor shall submit the security management documents not later than thirty
- (30) days after the Commencement Date for Approval by the Owner.

6.1.3.1.3. Security Reporting

- The Contractor shall establish a system for the reporting of all security affairs. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.1.3.2. WSG 1.3.2 Security Analysis

- The Contractor shall conduct and regularly update a security analysis through a reputable national or international organization.
- The Contractor's security analysis shall cover all Project work packages of its Scope of Work, including but not limited to:
 - a) Management and general work items;
 - b) Engineering;
 - c) Procurement (including transportation);
 - d) Construction;
 - e) Completion (including commissioning); and
 - f) Site reinstatement.
- With particular recognition of the work Sites and work procedures involved for the performance of the relevant works.

6.1.3.2.1. Threats

- The Contractor's security analysis shall identify potential threats that are directed towards the Project, including but not limited to threats towards:
 - a) Project personnel including the Contractor's and Sub-Contractor's personnel, the Owner's personnel, Third Party personnel, visitors and any other person involved with any aspect of the Project; and
 - b) Permanent and temporary Project installations and work Sites including but not limited to station location, camps, accommodation facilities, utilities, storage areas, land and water transport and commuter routes for the transport of personnel, Materials, supplies and others.

6.1.3.2.2. Potential Mitigation Measures

- The Contractor's security analysis shall identify and propose measures suitable for the elimination of security threats including, but not limited to:
 - a) Active security measures, such as deployment of guards, patrols and security personnel including the cooperation and coordination with National Law enforcement agencies, security forces of the Federal Republic of Nigeria;
 - b) Passive security measures, such as security fencing barricades, CCTV cameras, protective equipment and other means; and
 - c) Procedural security measures including the establishment of specific work procedures or the alteration or amendment of work procedures through the incorporation and integration of specific security measures and considerations.

6.1.3.3. WSG 1.3.3 Security Measures

- The Contractor shall provide and implement all active, passive and procedural security measures as identified in the security analysis and required to ensure the safety and security of:
 - a) all Project personnel with the Contractor's and Sub-Contractor's personnel, the Owner's personnel, consultants, Third Party personnel, visitors and any other person involved in the performance of the Works; and
 - b) all Project installations including permanent ancillary and temporary installations including all the work Sites, supply and transport routes and any area required or used for the performance of the Works.
- The Contractor's security measures shall adequately address all threats identified in the security analysis and extend to all work packages of the Scope of Work, including:
 - a) Management and general work items;
 - b) Engineering;
 - c) Procurement (including transportation);
 - d) Construction;
 - e) Completion (including commissioning); and
 - f) Site reinstatement.
- The Contractor shall liaise and coordinate its works with Law enforcement agencies,

security forces of the Federal Republic of Nigeria and other EPC Contractors of the Project.

6.1.3.3.1. Active Security Measures

- Within its Scope of Work the Contractor shall employ dedicated security personnel team with sufficient and adequately trained staff to respond to any security threats and to deter any potential assailant.
- This will include trained guard, patrolmen, drivers and others to ensure the security of all Project personnel, Sub-Contractors, the Owner and visitors to the Site.
- All personnel and equipment for the provision of security services shall be properly licensed or accredited in accordance with the requirements of the Federal Republic of Nigeria.

6.1.3.3.2. Passive Security Measures

- Within its Scope of Work the Contractor shall provide all required passive security measures including but not limited to fences (wire mesh and concertina fences), gates road barriers, CCTV, monitoring devices, special equipment, vehicles, materials or installations as defined in the security analysis.

6.1.3.3.3. Procedural Security

- Within its Scope of Work the Contractor shall implement, alter, amend or establish supplementary work procedures considering working time (working during daylight hours), limitation of working spreads or any other aspects of the Contractor's work performance in order to respond to security threats as identified in the security analysis.

6.1.4. Work Group 1.4 Environment

- The Contractor is responsible for the monitoring and supervision of all environmental aspects of the Works.
- Further detailed requirements for the environmental aspects are specified in Exhibit E "EIA Document".

6.1.4.1. WSG 1.4.1 Environmental Management

6.1.4.1.1. Environmental Management Organization

- The Contractor shall establish a dedicated environmental management entity for the management of the environmental aspects of the Project during the performance of the Works.
- The Contractor shall submit the Organization Chart and relevant Job Descriptions for its environmental management organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.4.1.2. Environmental Management Documents

- the Contractor shall prepare all required documents for the environmental management of the Project including, but not limited to:
 - a) Environment manual;
 - b) Environment management plan (EMP); and
 - c) Environment procedures
- The Contractor shall establish its EMP and environmental procedures addressing all recommendations and requirements of the Environmental Impact Assessment (EIA), Governmental or Regulative Authority Requirements of the Federal Republic of Nigeria and any other requirements so as to minimize any negative or adverse impact on the environment during the performance of the Works.
- The Contractor shall submit its environmental management documents not later than 30 days after the Commencement Date, for Approval by the Owner.

6.1.4.1.3. Environmental Management Reporting

- The Contractor shall establish a system for the reporting of all environmental management affairs. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.1.4.2. WSG 1.4.2 EIA Report and associated Proceedings

- The EIA Report has been submitted to the competent Authorities by the Owner.
- Detailed requirements for the management of the environmental aspects of the Project are specified in Exhibit E - "EIA Document".

6.1.4.2.1. Assistance during compilation and completion of the EIA Report

- The Contractor shall provide any required assistance, including but not limited to calculations, documents and drawings, meeting attendance and others for the completion of the environmental impact assessment process and any associated or environmental proceedings of the Federal Republic of Nigeria.

6.1.4.2.2. Implementation of the requirements of the Approved EIA

- The Contractor shall implement all requirements or recommendations of the Approved EIA into its designs for the permanent installations, ancillary installations and temporary
- Installations of the Project as well as any specification procedure or work method statement for the performance of the Works.

6.1.4.3. WSG 1.4.3 Environmental Compliance Monitoring

- The Contractor shall afford for and cooperate with environmental compliance monitoring performed by the Authorities of the Federal Republic of Nigeria, the Owner

or any other nominated Third Party.

6.1.5. Work Group 1.5 Community Liaison

- The Contractor is responsible for the management of the community liaison aspects during the performance of the Works in accordance with the FTO / GMOU put in place by BFPCL

6.1.5.1. WSG 1.5.1 Community Liaison Management

6.1.5.1.1. Community Liaison Organisation

- The Contractor shall establish a dedicated community liaison entity for the community liaison and the interaction with local communities and affected persons, all local entities and Authorities as well as the community liaison team of the Owner and other EPC Contractors during the performance of the Works.
- The Contractor shall submit the organization chart and relevant job descriptions for its community liaison organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.5.1.2. Community Liaison Documents

- The Contractor shall prepare all required documents for the community liaison management of the Project including, but not limited to:
 - a) Community liaison manual;
 - b) Community liaison plan; and
 - c) Community liaison procedures.
- The Contractor shall establish the community liaison plan (CLP) and community liaison procedures addressing all recommendations and requirements of the EIA, Governmental or Regulative Authority Requirements of the Federal Republic of Nigeria and any other requirements so as to ensure the proper consideration of complaints or request of the general Public and Project affected persons during the performance of the Works.
- The Contractor shall submit Community Liaison management documents not later than 30 days after the Commencement Date for Approval by the Owner.

6.1.5.1.3. Community Liaison Reporting

- The Contractor shall establish a system for the reporting of all community liaison affairs. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.1.5.2. WSG 1.5.2 Community Liaison Actions

- The Contractor shall address and implement all actions mandated or requested by the Authorities of the Federal Republic of Nigeria in relation to its performance of the Works.

- All actions shall be Approved by the Owner.
- The Contractor shall address all complaints reasonably made by the general Public, lands and Project affected persons in relation to the performance of the Works.

6.1.5.3. WSG 1.5.3 Community Liaison Register and Data bases

6.1.5.3.1. Commitment Register

- The Contractor shall establish a register of Project commitments towards the general Public, communities, lands or other Project affected persons, identifying the nature, source, time, and initiator of the commitment as well as the actual status of the implementation and close-out.
- The relevant commitments may have been made or received through the Approved EIA study, the Owner, the Authorities of the Federal Republic of Nigeria, or in direct response to concerns voiced from Project affected communities and persons.

6.1.5.3.2. Complaints Register

- The Contractor shall register all complaints or comments that he receives during the performance of the Works, into a central database (the comment and complaint register) tracing the required and initiated actions as well as the status of implementation. The database shall be updated on a daily basis and shall be submitted to the Owner on a regular basis and upon specific request by the Owner.

6.1.5.4. WSG 1.5.4 Community Liaison Compliance Monitoring

- The Contractor shall afford for and cooperate with the community liaison and social compliance monitoring performed by the Authorities of the Federal Republic of Nigeria, the Owner or any other nominated Third Party.

6.1.6. Work Group 1.6 Quality Assurance and Quality Control

- The Contractor is responsible for the quality of the Works and consequently the management of all QA/QC aspects associated with the performance of the Works.
- The Contractor's QA/QC Manager shall be reporting directly to the Contractor's Project Manager.
- Further detailed quality requirements for the Project are specified in Exhibit C, Annex C, Attachment 1 "Quality Management System Requirements".

6.1.6.1. WSG 1.6.1 QA/QC Management

6.1.6.1.1. QA/QC Executive Management Organisation

- The Contractor shall establish a dedicated QA/QC organization being part of the quality management for the Project during the performance of the Works.
- The Contractor shall submit the organization chart and relevant job descriptions for its QA/QC organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.6.1.2. QA/QC Documents

- The Contractor shall prepare all required QA/QC management documents for the Project including, but not limited to:
 - a) Project quality manual;
 - b) Project quality plan;
 - c) Project QA/QC procedures;
 - d) Project QA/QC plans; and
 - e) Project QA/QC records.

6.1.6.1.3. QA/QC Reporting

- The Contractor shall establish a system for the reporting of all QA/QC issues.
- The regular reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".
- Specific QA/QC issue reporting shall be performed by the Contractor as specified in the Scope of Work, Exhibit C, Annex C, Attachment 1 "Quality Management System Requirements".

6.1.6.2. WSG 1.6.2 Contractor's Integrated Quality Management System

- The Contractor shall perform all Works in compliance with its integrated quality system, addressing the aspects of QA/QC during the performance of the Works.
- The Contractor's integrated quality management system shall be organized in 5 levels, recognizing:
 - Level 1 Corporate quality policies laid down in its corporate quality manual;
 - Level 2 Description of the Project specific quality system(s) laid down in the Project quality manual and Project quality plan;
 - Level 3 Project QA/QC procedures and QA/QC plans and Project Work procedures, specifying the requirements for the performance of the Works, including the construction manual, work specifications, work procedures, method statements as well as material Specifications and Vendor inspection plans;
 - Level 4 Inspection and Test Plans, manufacturing record books (pipe log book), punch lists as well as installation operation and maintenance manuals, as part of the as-built
 - Level 5 All Project QA/QC respective forms for QA/QC records including inspection and test related forms.

6.1.7. Work Group 1.7 Technical Integrity, Design and Project Reviews

- The Contractor shall follow-up and implement all actions derived from the performance of all technical integrity, design and Project reviews and provide periodic status reports on the status of such implementation activities, within its Scope of Work under the Contract, regardless of whether such reviews were performed under the Contractor's, the Owner's or any nominated Third Party's coordinative responsibility.
- Further detailed requirements for the management of the technical integrity aspects and the performance of design and project reviews for the Project are specified in

Exhibit C, Annex D, Attachment 2 "Technical Reviews".**6.1.7.1. WSG 1.7.1 Technical Integrity and Project Review Management**

- The performance of technical integrity, design and project reviews will be managed by the Owner or a nominated Third Party at the discretion of the Owner.

6.1.7.1.1. Technical Integrity, Design Management Organisation

- The Contractor shall establish a technical integrity organization that is responsible for the coordination of technical integrity reviews within the Contractor's organization, with the responsibility to acquire required information, arrange for the participation of the required Contractor's personnel and follow-up the implementation and close-out of resulting action items.
- The Contractor shall submit the organization chart and relevant job descriptions for its executive management organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.7.1.2. Technical Integrity Management Documents

- The Contractor shall prepare all required technical integrity management documents for the Project including, but not limited to:
 - a) Technical integrity manual; and
 - b) Technical integrity plan

6.1.7.1.3. Technical Integrity, Design and Project Review Reporting

- The Contractor shall establish a system for the reporting of all technical integrity, design and Project review action items. The reporting shall be performed within the scope of **WSG 1.8.8 "Project Reporting"**.

6.1.7.2. WSG 1.7.2 Technical Safety Reviews

- Technical Safety reviews for the Project include, but are not limited to:
 - a) HAZID (hazard identification);
 - b) HAZOP (hazard and operability);
 - c) SIMOPS (simultaneous operations);
 - d) SIL (safety integrity level / electronic systems);
 - e) PHSER (project health and safety and environmental review);
 - f) Pipeline quantitative risk assessment (QRA);
 - g) Fire safety review; and

6.1.7.3. WSG 1.7.3 Asset Integrity Management

- Contractor shall implement an **assets integrity management system (AIMS)** and perform all the relevant data input for all items received, procured and/or installed within its scope of work.

6.1.7.4. WSG 1.7.4 Project Phase Reviews

- The Owner or its appointed representative will perform selected technical reviews at the beginning, during and/or the end of specific Project work packages, which shall include but are not limited to:
 - a) Design and engineering reviews at 30 %, 60 % and 90 % of progress;
 - b) Procurement and supply reviews at main milestones of procurement plan;
 - c) Construction reviews;
 - d) Completion reviews; and
 - e) Operation Reviews (Training and Operational Readiness).
- Reviews may be performed at the onset, during or at the end of the relevant Project phase or activity in order to assess the “readiness” for entering into the relevant Project phase, the performance during the relevant Project phase or the completion (status of completion) of the relevant phase or work activities.

6.1.8. Work Group 1.8 Project Control

- The Contractor is responsible for the establishment of Project control mechanisms, which include but are not limited to:
 - a) Project structuring;
 - b) Project planning and scheduling;
 - c) Cost control;
 - d) Progress measurement;
 - e) Project reporting;
 - f) Project change control; and
 - g) Project risk assessments.
- Further detailed controlling requirements for the Project are specified in Exhibit C, Annex E, Attachment 1 “Project Control Management”.

6.1.8.1. WSG 1.8.1 Project Control Management

6.1.8.1.1. Project Control Organisation

- The Contractor shall establish a dedicated Project control entity for the effective control of the Project during the performance of the Works.
- The Contractor shall submit the organization chart and relevant job descriptions for its Project control organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.8.1.2. Project Control Management Documents

- The Contractor shall prepare all required management documents for the executive management of the Project including, but not limited to:
 - a) Project control manual, with a dedicated section or an individual manual for each aspect (work-subgroup) of Project control;

- b) Project control plan, with a dedicated section or an individual plan for each aspect (work-subgroup) of Project control; and
 - c) Project control procedures with dedicated procedures for each aspect (work-subgroup) of Project control.
- The Contractor shall submit its Project control documents not later than thirty (30) days after the Commencement Date for Approval by the Owner.

6.1.8.1.3. Project Control Reporting

- The Contractor shall establish a system for the reporting of all Project control matters. The reporting shall be performed within the scope of WSG 1.8.8 “Project Reporting”.

6.1.8.2. WSG 1.8.2 Project Structuring

- The Contractor shall establish project breakdown structures as the basis for the structuring of the Project and the organization of the Works.
- The principle project breakdown structures shall include:
 - a) Project objective structure;
 - b) Project system breakdown structure; and
 - c) Project Work breakdown structure.

6.1.8.3. WSG 1.8.3 Planning and Scheduling

- The Contractor shall employ a dedicated planning and scheduling team to establish the Contract execution schedule and to monitor and to report the actual progress of all Works.
- For planning, scheduling and tracking purposes, all Works shall be broken down to the level of individual work activities and document deliverables.
- The Contractor’s planning shall include, but not limited to:
 - a) Contract execution schedule in accordance with the Program of Works;
 - b) Individual activity and Work phase schedules;
 - c) Resource analysis (as back up for activity durations);
 - d) S-Curve progress charts for main activities; and
 - e) Construction march-chart.
- The individual work phase and activity schedules shall be backed by a resource analysis identifying the number of resources (manpower and equipment) in order to achieve the planned progress rates.
- The Work progress shall be displayed in progress S-curves allowing the comparison of planned and actual progress.
- The Contractor shall employ a specific scheduling software program such as “Primavera”, “MS-Project” (or equivalent) subject to Approval by the Owner for the

preparation of Project schedules and the performance of critical path analysis.

6.1.8.4. WSG 1.8.4 Project Cost Estimation, Budgeting and Cost Control

- The Contractor shall establish its internal system for cost estimation and budgeting and provide a cash flow analysis in regular intervals and upon request of the Owner.

6.1.8.5. WSG 1.8.5 Project Progress Measurement

- The Contractor shall implement an effective Project progress measurement system. The progress weighting factors will be subject to specific Approval by the Owner. Relevant progress figures shall be included as part of the Contractor's regular reporting and shall be submitted as requested by the Owner.
- The progress measurement shall include but not limited to:
 - a) Project baseline
 - b) Progress S-curve tracking (baseline versus actual progress)
 - c) Weighted progress measurement

6.1.8.6. WSG 1.8.6 Project Change Control

- The Contractor shall implement an effective Project change control system, with continuous tracking and monitoring of:
 - a) Project cost/budget;
 - b) Project scope including changes and additions and omissions in the Scope of Work;
 - c) Project schedule;
 - d) Technical changes, with;
 - changes, additions and omissions to the provided design;
 - changes, additions and omissions to the Approved designs; and
 - additions and omissions as part of the construction design.
- Any changes to the Project cost, Project schedule and Project scope are subject to specific Approval by the Owner, for variations to the Scope of Work the provisions in the Contract shall apply.

6.1.8.7. WSG 1.8.7 Project Risk Assessments

- The Contractor shall implement an integrated, pro-active and work specific risk management program administered by qualified risk management professionals with a dedicated risk manager for the preparation of the Project risk management program, including a risk management plan, work specific risk management procedures and the essential risk register.

- All Project risks shall be entered into a risk register administered in a specific risk management data base, allowing the tracking, monitoring and re-evaluation of all risk throughout the life-cycle of the Project.
- Specific requirements for Project risk assessments are defined in Exhibit C, Annex E, Attachment 2 "Risk Management".

6.1.8.8. WSG 1.8.8 Project Reporting

- The Owner shall have the right to specify, change or increase the frequency, level of detail and areas to be addressed in the Contractor's reporting.
 - a) Weekly reports;
 - b) Monthly reports; and
 - c) Special/Specific reports.
- The Contractor shall furnish the Owner with monthly reports for all Works identifying at least, but not limited to:
 - a) The main aspects of the works performed, recognizing all work packages and work groups of the Contractor's Scope of Works;
 - b) The current status of work force and equipment mobilization;
 - c) The actual progress in relation to planned progress;
 - d) Key Performance Indicators for Security, H&S, Environmental, QA/QC works performance; and
 - e) Areas of concern and applied mitigation measures.

6.1.9. Work Group 1.9 Project Administration

- The Contractor is responsible for the management of all administrative affairs in relation to its Scope of Work.
- The Contractor is responsible for the establishment of Project administrative entities, which include but are not limited to:
 - a) Contract administration, including invoicing and payment;
 - b) Document control and data management;
 - c) Human resources;
 - d) Logistics; and
 - e) Mobilization and demobilization assistance.

6.1.9.1. WSG 1.9.1 Project Administration Management

6.1.9.1.1. Project Administration Organisation

- The Contractor shall establish a dedicated Project administration management entity

for the management of all Project administrative affairs during the performance of the Works.

- The Contractor shall submit the organization chart and relevant job descriptions for its project administration organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.1.9.1.2. Administration Management Documents

- The Contractor shall prepare all required documents for the administrative management of the Project including, but not limited to:
 - a) Project administration manual, with a dedicated section or an individual
 - b) manual for each aspect (work-subgroup) of project administration;
 - c) Project administration plan, with a dedicated section or an individual plan
 - d) each aspect (work-subgroup) of Project administration; and
 - e) Project administration procedures, with dedicated procedures for each aspect
 - f) (work-subgroup) of Project administration.
- The Contractor shall submit its administration management documents not later than 30 (thirty) days after the Commencement Date for Approval by the Owner.

6.1.9.1.3. Administration Management Reporting

- The Contractor shall establish a system for the reporting of all administration management affairs. The reporting shall be performed within the **scope of WSG 1.8.8 "Project Reporting"**.

6.1.9.2. WSG 1.9.2 Contract Administration

- The Contractor shall provide effective Contract management for the Contract and all other contracts or sub-contracts for the provision of services and materials for its Scope of Work.
- Contract Administration shall include all activities for invoicing and payment for the Contract and all other contracts and sub-contracts.

6.1.9.3. WSG 1.9.3 Document Control and Data Management

- The Contractor shall maintain a document and data management system for the electronic storage and administration of all data generated or received during the performance of the Works.
- The Contractor shall establish the following data management systems/databases for the management, storage and administration of project or documents and data, namely:
 - a) Document and data management control system;
 - b) Engineering data warehouse; and

c) Specific (Topic) databases and registers.

- The document and data management systems and all Project related data and documentation shall be handed over to the Owner as part of the Project close-out.

6.1.9.3.1. Document Control

- The Contractor shall operate a document control centre (DCC) and implement a document control system for the control of all Project correspondence and documentation. All documentation administered by the DCC shall be stored in hard copy and soft copy (electronic file format) and in English language.
- The Contractor shall maintain separate registers for:
 - a) Correspondence documents;
 - b) Contracts and commercial documents;
 - c) Technical documents;
 - d) Vendor and Third Party documentation; and
 - e) Any other as required.
- Further specific requirements for the review and Approval of Project documents are contained in the Contract (Exhibit C, Annex E, Attachment 1 "Project Control Management" and Exhibit C, Annex F "Documentation Requirements").

6.1.9.3.2. Correspondence Document Register

- The Contractor shall maintain a correspondence register for the registration and administration of all Project correspondence so as to ensure the proper numbering, tracking and distribution of all Project correspondence.

6.1.9.3.3. Contracts and Commercial Document Register

- The Contractor shall maintain a contracts and commercial documents Register for all administered contracts and subcontracts.

6.1.9.3.4. Technical Document Register (TDR)

- The Contractor shall maintain a TDR for the administration of all technical documents. Technical documents shall include but not limited to drawings, specifications, reports, calculations, procedures, datasheets and manuals, as-built drawings and final documentation.
- The Contractor shall submit technical documents particularly drawings and specification sheets in a grouped, orderly fashion for review and Approval for construction. Groups of drawings will normally be sent together, i.e., all civil structural as a package, etc. All technical documents, which are to be submitted to the Owner for review and Approval, shall be recorded in the TDR. The TDR shall be regularly updated and shall be submitted to the Owner on a monthly basis. The TDR shall show among others, the listing of all technical documents, progress, current revision and expected

completion date.

6.1.9.3.5. Vendor Documentation Register

- The Contractor shall maintain a Vendor documentation register for the administration of all Vendor documents. Vendor documents shall include but not limited to drawings specifications, reports, calculations, procedures, datasheets and manuals, and any other documents submitted by Vendors, Manufacturers or Suppliers for the Project. All Vendor documentation shall be acquired in English language.

6.1.9.3.6. Any other registers

- Other documents or data registers shall be established as required for the safe administration and storage of Project data.

6.1.9.3.7. Engineering Data Warehouse and Mechanical Manual (or Project Dossier)

- The Contractor shall implement a database for the storage and administration of all technical and engineering data, such as but not limited to:
 - a) Pipe book;
 - b) Certificates;
 - c) QA/QC documentation;
 - d) Acceptance tests;
 - e) Inspection and test reports;
 - f) Completions data;
 - g) All technical documents including drawings, specifications, procedures;
 - h) Manufacturing record books (MRB);
 - i) GIS data;
 - j) As-built drawings; and
 - k) Others.
- All data in the engineering data warehouse shall be stored and handed over in editable electronic file format. This will form the mechanical manual of the Project.

6.1.9.3.8. Specific Databases and Registers

- Other documents and databases include but are not limited to:
 - a) AIMS database;
 - b) Project risk register;
 - c) Land acquisition database;
 - d) Community liaison databases; and
 - e) Others.

6.1.9.4. WSG 1.9.4 Human Resources

- The Contractor's human resource department shall manage the employment contracts for all Project personnel.
- The Contractor shall establish an employment policy for the hiring (and training) of local

personnel in line with the requirements or stipulations of the EIA.

- The Contractor shall ensure that all employment contracts are in compliance with the relevant regulations of the Federal Republic of Nigeria.

6.1.9.5. WSG 1.9.5 Logistics

- The Contractor shall employ a dedicated logistics team providing logistical assistance for all Project entities during the implementation of the Project.

6.1.9.6. WSG 1.9.6 Mobilisation and De-mobilisation

- The Contractor shall mobilize and demobilize all Project personnel, equipment and other resources in accordance with the Approved Contract execution plan.

6.2. Work Package 2 Location Selection and Acquisition of Rights

6.2.1. Project Work Group 2.1 Route and Station Location Selection

6.2.1.1. WSG 2.1.1 Pipeline Route and Station Location Selection

- The following details regarding pipeline route and station location selection shall be provided:

6.2.1.1.1. Linear Sections

- The information regarding the locations of the pipeline centreline and the ROW, including encroachments, temporary acquisitions and crossings, is provided in the FEED documentation.
- If required, the Contractor shall carry out pipeline re-routings as part of Work Package

6.2.1.1.2. Pipeline Stations

- The locations for all pipeline stations are described in section 2 above. More details are provided in the FEED documentation.

6.2.1.1.3. Ancillary Installations

- The Contractor shall determine the locations for ancillary installations as part of his design for the Pipeline.

6.2.1.1.4. Temporary Installations

- Locations of all other temporary installations such as but not limited to construction camps, temporary work areas, Pipe storage yards (PSY) and temporary access roads and canals as well as any required locations for the installation of temporary utilities and temporary security installations shall be selected by the Contractor and submitted for Approval by the Owner.
- The acquisition of such additional areas is described in work group 2.4 "Acquisition of

Rights to Land”.

6.2.2. Work Group 2.2 Environmental Studies and Programmes

- The Contractor shall provide input for the preparation of the detailed description of the Project as requested by the Owner.
- The Contractor shall provide input as required for the definition of special design and construction methods in environmental sensitive zones.
- The Contractor shall prepare and submit its own EMP following and implementing all requirements of the Approved EIA Study and the Approved EMP.
- The Contractor shall implement all construction and design measures as specified in the EIA and related plans and procedures.
- The Contractor shall accommodate all Authority or Third Party involved in the environmental (including archaeological) monitoring of the design, construction, completions and operation of the Project.

6.2.3. Work Group 2.3 Project Authorizations, Permits and Approvals

6.2.3.1. WSG 2.3.1 Authorization Acquisition Management

6.2.3.1.1. Authorizations Acquisition Organisation

- The Contractor shall establish a dedicated entity to cover the Contractor's responsibilities with regard to the acquisition of authorizations, permits, certificates and Approvals for the Project.
- The Contractor shall submit the organization chart and relevant job descriptions for Approval by the Owner within thirty (30) days after the Commencement Date and as required in respect of the others. Following Approval, the Contractor shall commence with the mobilization of the necessary personnel.

6.2.3.1.2. Authorization Acquisition Management Documents

- The Contractor shall prepare all required management documents for the acquisition authorizations for the Project including, but not limited to:
 - a) Authorizations, permit and approval acquisition manual;
 - b) Authorizations, permit and approval acquisition plan: and
 - c) Authorizations, permit and approval acquisition procedures.
- The Contractor shall submit its authorization acquisition management documents not later than 30 days after the Commencement Date for Approval by the Owner.

6.2.3.1.3. Authorisation Acquisition Reporting

- The Contractor shall establish a system for the reporting of all affairs related to the acquisition of authorization for the Project. The reporting shall be performed within the scope of **WSG 1.8.8 “Project Reporting”**.

6.2.3.2. WSG 2.3.2 Allocation of Responsibilities

- The Contractor shall identify and acquire all permits, Approvals and authorizations required for the performance of its Scope of Work.
- The Contractor shall provide assistance to the Owner as required for the acquisition of any other permits, Approval and authorization to be acquired by the Owner.
- The Contractor shall acquire the requisite certificate issued by an official body, agency or Authority of the Federal Republic of Nigeria certifying compliance with applicable rules and regulations for any part of the Works performed within the Contractor's Scope of Work.
- Further details with regard to the acquisition of permits, approvals, certificates and approvals during the construction phase are provided in Section 10.1.4 WSG 5.1.4 Permits, Authorizations and Third Party Interfaces".

6.2.4. Work Group 2.3 Spatial Planning Approvals

- Spatial planning approvals shall be obtained by the Owner.

6.2.5. Work Group 2.4 Acquisition of Rights to Land

- All areas of the ROW and the Sites for the installation of the Pipeline stations will be acquired by the Owner and handed over to the Contractor as specified in WSG 5.1.7 - "The Construction Site".
- Any additional areas such as additional work spaces such as but not limited to construction camps, temporary work areas, storage yards and temporary access roads shall be acquired by the Contractor subject to the condition that the relevant location(s) are Approved by the Owner and under strict application of the Project requirements for the acquisition of land.

6.3. Work Package 3 Engineering

6.3.1. Work Group 3.1 Feasibility Design

- "Not applicable"

6.3.2. Work Group 3.2 Conceptual Design

- Conceptual design has been performed up to the FEED level and is provided as part of the FEED documents.

6.3.3. Work Group 3.3 FEED

- "Not applicable."

6.3.4. Work Group 3.4 Contractor's Engineering

6.3.4.1. WSG 3.4.1 Contractor's Engineering Management

- The Contractor is responsible for the management of the Contractor's engineering.
- The Contractor shall manage its engineering of the Works in accordance with the requirements of the Contract and in accordance with its Approved Contract execution plan.

6.3.4.1.1. Contractor's Engineering Organisation

- The Contractor shall establish a dedicated engineering entity for the performance of its engineering obligations for the Project.
- The Contractor shall submit the Organization Chart and relevant Job Descriptions for its engineering organization within thirty (30) days after the Commencement Date and commence with the mobilization of the relevant personnel.

6.3.4.1.2. Contractor's Engineering Management Documents

- The Contractor shall prepare all required engineering management documents for the Project including, but not limited to:
 - a) Engineering manual;
 - b) Engineering plan; and
 - c) Engineering procedures.
- The Contractor shall submit its engineering management documents not later than thirty (30) days after the Commencement Date for Approval by the Owner.

6.3.4.1.3. Engineering Management Reporting

- The Contractor shall establish a system for the reporting of all engineering issues. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.3.4.2. WSG 3.4.2 Review and Endorsement of the FEED

- The FEED documents as provided by the Owner have to be reviewed and endorsed by the Contractor as per **Volume 2 – Terms and Conditions of Contract**.
- The Contractor accepts the FEED as the sole and complete Specification for the Project as if it was prepared by him.
- The Contractor is fully responsible and owns the FEED as the basis for its further design development and the performance of the Works.
- The Contractor is responsible to perform detailed engineering, any additional engineering and engineering investigations as required for the completion of the design

of the Project and the performance of the Works.

6.3.4.3. WSG 3.4.3 Authority and Compliance Engineering

6.3.4.3.1. Authority Engineering

- The Contractor's Authority and compliance engineering shall include the active identification, recognition and implementation of all relevant authority and compliance engineering requirements into the specification and design of the Pipeline and the Pipeline Works. Refer to Volume 02 – Annex C – Regulatory Requirements & Permits / Consent Register for the list of some identified regulatory requirements.
- Authority engineering shall include, but not limited to:
 - a) The implementation of requirements of the Approved EIA;
 - b) The implementation of requirements received from the Authorities of the Federal Republic of Nigeria as part of licensing or approval processes, consulatory responses or any other form of official communication;
 - c) The implementation of crossing requirements received from Authorities and the of the facilities to be crossed (such as roads, railways, pipelines, cable, electric power transmission lines, drainage installations and others), features (such as watercourses), special zones and others;
 - d) The implementation of other stakeholder requirements, which includes lands, lending banks and other stakeholders of the Project;
 - e) The identification and preparation of the technical contents of all authorization, permit and approval packages required for the acquisition of authorizations, permits and approvals. The technical contents shall include any engineering
 - f) documents, drawings, calculations, studies and reports required in order to satisfy application requirements;
 - g) The participation in meetings and reviews requested by Government Authority and s/operators of other services and installations crossed and/or affected by the installation of the Pipeline.
- The Contractor shall list all Authority and Third Party engineering requirements in a separate regulatory requirements register.

6.3.4.3.2. Compliance Engineering

- The Contractor shall perform all engineering works in full compliance with:
 - a) All applicable code and standards and regulations of the Federal Republic of Nigeria;
 - b) All applicable international codes and standards;
 - c) Good international engineering practice.

6.3.4.4. WSG 3.4.4 Contractor's Detailed Engineering

6.3.4.4.1. General Engineering Activities

- The Contractor shall perform general activities including:
 - a) Witness performance and acceptance tests at Vendors or Sub-Contractor's premises for all major equipment;
 - b) Review and incorporation of all Vendor data into the design.

6.3.4.4.2. Contractor's Route and Station Location Engineering

- The Contractor shall review the pipeline route as provided in the FEED documents with regard, but not limited to:
 - a) Confirmation of the pipeline alignment and crossing locations as provided in the alignment sheets, including:
 - (i) authority requirements, (in particular requirements determined during the authority approval process and permitting of crossings);
 - (ii) authority requirements stipulated in the Approved EIA;
 - (iii) land acquisition requirements, (in particular requirements for the orientation of the Pipeline along property boundaries and as expressed by lands during initial negotiations for the acquisition of the ROW);
 - (iv) requirements derived from the performance of detail engineering studies, e.g. the orientation of the Pipeline in zones of potential liquefaction and lateral spreading and other;
 - (v) requirements derived from the selection of special crossing methods or techniques;
 - (vi) requirements derived from the definition and incorporation of construction elements, such as field and factory bends;
 - (vii) any other requirements surfaced during the performance of the detailed engineering for the Pipeline.
 - b) The confirmation of all pipeline station locations (Block Valve, Pigging, Metering and Terminal Gas stations);
 - c) Confirmation of the topography, geotechnical and ground water conditions along the route of the Pipeline;
- All changes to the location of the pipeline centreline are subject to prior Approval by the Owner.

6.3.4.4.3. Interface Engineering

- The Contractor shall actively identify and coordinate all internal technical interfaces

with other EPC Contractors and all external technical interfaces with Third Parties. Physical interfaces and preliminary interface plans are defined in Exhibit C, Annex G "Interface Plan and Matrix - EPC Pipeline and Stations ".

- The Contractor shall perform all engineering required for the definition of all interfaces to other EPC Contractors and Third Parties addressing, but not limited to:
 - a) Implementation of a systematic approach for the identification of internal and external interfaces;
 - b) Definition of the battery limits of the systems or installations at the interface;
 - c) Definition of the connection and connection parameters at the battery limit
 - d) Description of the connection of the systems or installations of the interfacing entities at the interface point;
 - e) Allocation of responsibilities of the interfacing entities for the performance of the connection.

6.3.4.4.4. Utilities Engineering

- The Contractor shall perform utility engineering including temporary and permanent utilities (water and electric power supply as well as liquid and solid waste treatment/disposal systems).

6.3.4.5. WSG 3.4.5 Contractor's Detail Engineering –Pipeline and Stations

6.3.4.5.1. Contractor's Engineering Studies and Investigations

Supplementary Geo-hazard Review

- The Contractor shall conduct a geo-hazard investigation for the Project if required for the consolidation of the design, the Contractor shall perform additional geo-mechanical investigations for the quantification of geo-hazard risks and the definition of adequate design and construction measures at station locations and along the route of the Pipeline.

Geotechnical Investigations and Borings

- The Contractor shall perform geotechnical investigations and borings as required for the performance and confirmation of the design of the Pipeline and Stations. This may include, but not limited to:
 - a) Geotechnical soil data through drillings and borings as required for the determination of foundation requirements;
 - b) Geotechnical soil data at selected crossing locations and special zones for the determination of the design parameter and the selection of crossing methods; and
 - c) Geotechnical soil data for the application of special crossing or construction

methods.

Hydraulic Investigations at Watercourse Crossings

- The Contractor shall perform the hydraulic study as performed during the FEED at all major watercourse crossings for the identification of the maximum scour depth, 100 year floodplain and other criteria relevant to the performance of its engineering at the crossing area.

6.3.4.5.2. Supplementary Anti-buoyancy Investigations

- The Contractor shall perform an anti-buoyancy study and calculations in all areas included in the 100 year floodplain of rivers, areas with high groundwater table and other areas susceptible to flooding. Anti-buoyancy studies shall include actual soil density measurements as well as the assessment of soil erosion susceptibility and potential loss of pipeline cover, thereby providing the basis for the anti-buoyancy calculations and the confirmation of the design.

6.3.4.5.3. Soil Resistivity Measurements

- The Contractor shall perform soil resistivity measurements at discrete locations along the Pipeline route, at Pipeline stations and CP-stations to determine the design criteria for the temporary and permanent cathodic protection (CP) system of the Pipeline, including but not limited to the number, location and size of anode ground-beds the design and power requirements for CP-stations, the potential requirement for soil exchange in areas of high resistivity soils, as well as the layout and design of anode groundbeds and the requirements for CP and grounding installations in pipeline stations.

6.3.4.5.4. High Voltage Influence Investigation / Calculation

- Contractor shall carry out a detailed investigation with calculations of the influence of high voltage lines of 11kV, 132 kV and above within of proximity of less than 1000m and present his findings in a report to be issued to the owner. This report shall guide the owner in the final discussion on how to install the armored power cable from Akabel methanol plant to Elepa and BBOU manifold Planforms.
- The high voltage interference calculation shall form the basis for the design of the AC mitigation system.

6.3.4.5.5. Study of Erosive Potential and Properties in Slopes

- The Contractor shall perform a study of the erosive potential and soil properties in slope areas (longitudinal and transversal slopes), for the definition of the relevant design criteria and design measures.
- The Contractor shall design and specify all trench and surface erosion protection and

prevention measures, the relevant installations and their installation location, as well as the appropriate re-vegetation measures.

6.3.4.5.6. Other Studies

- The Contractor shall perform any other study as identified during the progressing of the Contractor's engineering, required to determine additional design parameter and to satisfy the requirements originated or stipulated by:
 - a) Government authorities;
 - b) Good Industry practice;
 - c) The Owner;
 - d) Project lenders and other Project stakeholders; and
 - e) Other Third Parties.

6.3.4.5.7. Contractor's Engineering - Linear Sections

- The Contractor shall verify all linear section engineering supplied by the Owner and supplement the provided engineering as required in order to meet the design intent of the Project.
- The Contractor's linear section engineering shall include, but not limited to:

Calculations and Designs

- a) Pipeline anti-buoyancy calculations and engineering of buoyancy protection measures;
- b) Stress analysis (calculations) for special crossing sections (e.g. river crossing, HDD crossings, zone with potential ground movements and other areas as required for the performance of the detail engineering of the Pipeline) and engineering of special crossing methods;
- c) Engineering at crossings including but not limited to existing and future roads, watercourses, railways, channels crossings, electric transmission lines, pipelines and any other features and installations;
- d) Engineering calculations and engineering for pipeline anchor blocks, expansion loops and appurtenances;
- e) Engineering at special zones including but not limited to rocky areas, steep slopes, swamps, parallel sections or proximity to other pipelines and other geo-hazard areas;
- f) Engineering and definition of trench erosion protection measures;
- g) Engineering and definition of surface erosion protection, reinstatement and re-vegetation measures;

- h) Material selection reports and specifications (steel line pipes, lining and coatings, insulation joints);
- i) Detail design of river Niger crossing, including but not limited to the performing supplementary geotechnical and hydrological investigations if required, structural analysis of pipeline etc.

Drawings

- a) Update of Pipeline alignment sheets;
- b) Update/preparation of detailed crossing drawings for all major crossings;
- c) Update/preparation of all required typical drawings; and
- d) Thematic drawings (as required).

Lists

- a) List of crossings (update);
- b) MTO for line pipe material (update);
- c) MTO for casing pipes;
- d) MTO for factory bends; and
- e) Material requisitions for all procurement items of the linear sections.

Documents

- a) Verification and update of all specifications for Materials, equipment and works as provided in the FEED Project Specification;
- b) Preparation and submission of additional specifications for all Materials, equipment and works required to specify the Contractor's engineering requirements for the Pipeline; and
- c) Preparation of all procurement specifications and datasheets for Materials, equipment and purchase items.

6.3.4.5.8. Contractor's Engineering – Pipeline Stations

- After endorsement of FEED, the Contractor shall verify the FEED engineering supplied by the Owner and supplement the provided engineering as required in order to meet the design intent of the Project.
- The Contractor shall perform all Stations engineering comprising of all engineering activities related to the engineering and specification of the Pipeline stations and as per the requirements defined in **Exhibit C, Annex G, Attachment 1 "Engineering Management Requirements"**.
- The Contractor's Pipeline station engineering shall include, but not limited to:

General

- The preparation of all required calculations, drawings, layout drawings, layouts, plans, schedules, lists, including electrical load lists, diagrams including single line diagrams, cable routing plans; hook-up drawings, datasheets, equipment lists, material take-offs, standard drawings, specifications, procedures and others required to provide the detailed specification of the Pipeline stations of the Project, allowing the procurement, construction, completion and operation of the Pipeline stations of the Project.

Process

- Preparation all piping engineering, including but not limited to:
 - a) Update process flow diagrams (PFD), piping and instrumentation diagrams (P&ID); and
 - b) Endorsement and update of all piping, and equipment sizing and associated process calculations.

Piping

- Preparation of all piping engineering, including but not limited to:
 - a) Confirmation of engineering calculations for all piping;
 - b) Development of piping installation drawings from the initial layouts of all Pipeline stations
 - c) Material selection report and specifications;
 - d) Piping class tables.

Mechanical

- Preparation of all mechanical engineering, including but not limited to:
 - a) Performance verification and endorsement of all piping stress calculations;
 - b) Preparation of installation drawings for all mechanical equipment;
 - c) Submission of all engineering calculations for all mechanical equipment, layout and designs;
 - d) Submission of all engineering for all mechanical equipment including skids and packages;
 - e) Preparation of equipment lists for all mechanical equipment;
 - f) Preparation of specifications and datasheets for all mechanical equipment; and
 - g) Preparation of all work specifications and procedures for mechanical works.

Civil / Structural Steel / Shelters

- Preparation of all civil engineering including but not limited to:
 - a) Preparation of general arrangement drawings;
 - b) Preparation of design for structural steel supports and installations;
 - c) Preparation of design of all foundations for equipment, supports and others with rebar schedules and concrete requirements;
 - d) Preparation of design for all pipe supports, pipe racks, steel structures and miscellaneous fabrications;
 - e) Preparation of design for all large diameter valve supports;
 - f) Preparation of design for all access platforms, ladders and staircases;
 - g) Preparation of design for access and station roads;
 - h) Preparation of design of perimeter fences and gates;
 - i) Preparation of material specifications and datasheets for all civil materials; and
 - j) Preparation of work specifications and procedures for civil works.

Coating, Painting and Insulation

- Preparation of all coating, painting and insulation piping engineering, including but not limited to:
 - a) Preparation of coating and painting plans;
 - b) Preparation of material specifications and data sheets for coating and painting materials; and
 - c) Preparation of work Specifications and procedures for painting and coating works.

Heating Ventilation Air Conditioning (HVAC)

- a) Preparation of the design for all ventilation and air conditioner systems for the stations with regard to the Guard hut, TRU and Office cabins;
- b) Preparation of material specifications and data sheets for equipment and Materials; and
- c) Preparation of work specifications and procedures for the performance of above mentioned works.

Electrical

- Preparation of all electrical engineering, including:
 - a) Preparation of earthing and lightning protection systems design (incorporating containers and other equipment supplied by others);

- b) Preparation of all design and engineering for earthing systems of above ground pipe work including integration of earthing systems into other existing facilities;
- c) Preparation of material specifications and data sheets equipment and Materials; and
- d) Preparation of work specifications and procedures for the performance of above mentioned works.

Cathodic Protection

- “Not applicable.”

Instrumentation and Control Systems

- Preparation of all instrumentation and control systems engineering, including but not limited to:
 - a) Preparation of the design and engineering for all Instrumentation and control systems inside Pipeline stations;
 - b) Preparation of all design, detail drawings and typical drawings for instrumentation and control systems inside Pipeline stations;
 - c) Preparation of material specifications and data sheets for equipment and Materials (in cooperation with EPC Contractor for SCADA); and
 - d) Preparation of work specifications and procedures for the performance of above mentioned works.

Fire Safety

- a) Preparation of the design, drawings and engineering for all fire safety systems;
- b) Preparation of all design, detail and typical drawings for fire safety systems;
- c) Preparation of material specifications and data sheets for equipment and Materials; and
- d) Preparation of work specifications and procedures for the performance of above mentioned works.

HSE

- Verification and implementation of specific H&S and EIA requirements for the design and engineering of pipeline stations.

Electronic Security and Surveillance

- a) Preparation of the design, drawings and engineering for electronic security and surveillance systems;
- b) Preparation of material specifications and data sheets for equipment and Materials; and

- c) Preparation of work specifications and procedures for the performance of above mentioned works.

Utilities

- a) Preparation of the design, drawings and engineering for all utilities installations;
- b) Preparation of material specifications and data sheets for equipment and Materials; and
- c) Preparation of work specifications and procedures for the performance of above mentioned works.

6.3.4.5.9. Contractor's Engineering - Ancillary Installations

- The Contractor shall perform the engineering and design of all ancillary installations. The Contractor's ancillary installations engineering shall include, but not limited to:

Cathodic Protection system

- The Contractor shall prepare the design and engineering of the temporary (sacrificial anode) and permanent (impressed current) cathodic protection system for the Pipeline.
- The design and engineering of the temporary (sacrificial anode) and permanent (impressed current) cathodic protection system for the Pipeline shall include, but not limited to:
 - a) The performance of all calculations for the overall system layout and the sizing of CP-equipment including sacrificial anodes, anode ground-beds, transformer rectifiers and others;
 - b) Preparation of the design and engineering for the cathodic protection systems inside pipeline stations, including the isolation of above and below ground pipe work;
 - c) Preparation of material specifications and data sheets for all electrical and cathodic protection equipment and materials; and
 - d) Preparation of work specifications and procedures for the performance of electrical (and cathodic protection) works.

High Voltage Influence Protection System

- The Contractor shall perform the engineering for the high voltage interference and overvoltage protection system.
- The design and engineering of the high voltage influence protection system for the Pipeline shall include, but not limited to:
 - a) The performance of all calculations for the overall system layout and the sizing of relevant equipment and devices;
 - b) Preparation of material specifications and data sheets for all electrical and

cathodic protection equipment and materials; and

- c) Preparation of work specifications and procedures for the performance of electrical (and cathodic protection) works.

Marker and Measuring Posts

- The Contractor shall perform the design and engineering for marker and measuring posts.

Fibre Optic Cable and Conduit

- The Contractor shall perform the design and engineering of the fibre optic cable conduit including appurtenances and the fibre optic cable itself.

6.3.4.5.10. Contractor's Engineering - Temporary Installations

- The Contractor shall perform the design and engineering for the temporary installations, which include, but not limited to:
 - a) Temporary Site facilities, including but not limited to construction camps, House Boat, manufacturing yard, storage areas and others;
 - b) Temporary Site utilities;
 - c) Temporary access roads;
 - d) Temporary safety installations;
 - e) Temporary security installations; and
 - f) Others.
- The Contractor's temporary installation engineering shall include, but not limited to:
 - a) The preparation of all engineering including calculations drawings, layout plans and others required for the Approval of the installations by the Authorities of the Federal Republic of Nigeria;
 - b) The preparation of all engineering and calculations required for and the hook-up of temporary utility connections as required by the Authorities or Regulatory Agencies of the Federal Republic of Nigeria;
 - c) The preparation of material specifications and data sheets for all required equipment and materials; and
 - d) The preparation of work specifications and procedures for the performance construction, installation and hook-up works.

6.3.4.5.11. Contractor's Engineering - Inspection and Testing

- The Contractor shall perform all engineering associated with or required for the performance of inspection and testing works, which includes but is not limited to:

- a) The specification of supplementary inspection and test requirements in order to ensure the quality of the Works;
- b) Definition of Procedures for Inspection and Testing aligned with the requirements of:
 - Requirements stipulated by the Authorities of the Federal Republic of Nigeria;
 - Applicable codes and standards;
 - The Contractor's quality management system; and
 - Additional requirements stipulated by the Owner.
- c) Determination/calculation of hydrostatic test sections;
- d) Performance of critical engineering assessments for the interpretation of inspection and test results; and
- e) Any other engineering in relation to the inspection and testing of the Works.

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6.3.4.6. WSG 3.4.6 Contractor's Engineering – Access Canal Dredging

- "Not applicable" Access canal is covered by another contract.

6.3.4.7. WSG 3.4.7 Contractor's Engineering – SCADA and Telecom

- The Contractor shall coordinate interfaces with the SCADA Contractors as specified in the document Exhibit C, Annex G, Attachment 2 "Interface Plan and Matrix".

6.3.4.8. WSG 3.4.8 Contractor's Engineering – Completions

- The work to be conducted by the Contractor under this Work Sub-Group comprises of all aspects of engineering preparations required for Work Package 6 "Completions".
- Detailed requirements are provided in Annex I, Attachment 1 "Completions Management Requirements – EPC Pipeline".
- Completions engineering shall include, but not limited to the:
 - a) development of the Project completions methodology;
 - b) preparation of a completions programme and schedule for the Pipeline, considering all completions phase activities (completion phases and steps) durations and milestone completion dates in compliance with the Contract execution plan, including but not limited to:
 - (i) Detailed lists of all activities and durations;
 - (ii) Detailed checklists of items and test records;

- (iii) Detailed list of required personnel and equipment;
 - (iv) Requirements for testing of instrument functionality and calibration; and
 - (v) Material take-offs for all of materials, spare parts and consumables needed for commissioning tests (e.g. nitrogen, lubricants, and others).
- c) Preparation of initial completions management documents, including:
 - (i) Completions manual;
 - (ii) Completions plan; and
 - (iii) Completions procedures.
- d) Preparation of initial completion documents, document formats and checklists, including but not limited to:
 - (i) Completions system / sub-system breakdown;
 - (ii) Item register;
 - (iii) Checklists and test sheets/records;
 - (iv) Punch listing and categorization (The categorization of punch list items is also addressed in Exhibit C, Annex I, Attachment 1 "Completion Management Requirements");
 - (v) Program Phase Close-out Dossiers;
 - (vi) Development of a completions activities program for the Pipeline covering all completion steps (mechanical completion, pre-commissioning, commissioning, start-up, initial operation, trial operation and performance test) of all Pipeline sub-systems under responsibility of the Contractor; and
 - (vii) Definition and establishment of a database for the administration of all completion phase activities, as well as test and performance values for all systems, sub-systems and components. The program shall be compatible with the Owner's software requirements and shall be subject to Approval by the Owner.
- The Contractor shall coordinate the completions engineering with the requirements of Sub-Contractors and Vendors and in particular the completions engineering activities performed by the EPC Contractor SCADA.
- All commissioning program activities shall be coordinated with the EPC Contractor for the SCADA.
- All procedures shall comply with the Owner's procedures and technical specifications.
- The Contractor shall submit all completions engineering including a detailed schedule and programme for all phases and steps of the completions programme, not later than one-hundred-twenty (120) days prior to the mechanical completion of the first system or sub-system of the Project.

6.3.4.9. WSG 3.4.9 Contractor's Engineering – Operation & Maintenance

- The Contractor shall perform operation and maintenance engineering for the Project.
- As part of the operation and maintenance engineering for the Project, the Contractor shall review and endorse the **Exhibit C, Annex J, Attachment 1 "Operation Management" and Exhibit C, Annex J, Attachment 2 "Maintenance Management"**. The Contractor's operations and maintenance engineering shall include, but not limited to the preparation of the following documentation:
 - a) Operation and Maintenance Documentation:
 - (i) Operation and maintenance manual including relevant Vendor information;
 - (ii) Operation and maintenance plans;
 - (iii) Operation and maintenance procedures (for different types of operation);
 - (iv) Job performance manuals;
 - (v) Job descriptions;
 - (vi) Emergency response plans; and
 - (vii) Reference to the final documentation in the mechanical manual.
 - b) AIMS Data Management
 - (i) Contractor shall expressly recognize the requirements of the Owner's asset integrity management system (AIMS) and perform all required data inputs so as to deliver a fully functional asset integrity management system including a comprehensive AIMS database to the Owner.
- All operation and maintenance procedures shall be compliant with equipment Vendor procedures and are subject to review and Approval by the Owner.
- **The Contractor shall manage the input for the SCADA regarding the operation and maintenance of the Pipeline for definition of the overall operation and maintenance requirements of the Project.**

6.3.5. Work Group 3.5 Construction Engineering

- The Works to be conducted under this Work Group shall include all aspects of engineering associated with the construction of the Pipeline and the development of the construction engineering documents.

6.3.5.1. WSG 3.5.1 Construction Engineering Management

- The Contractor is responsible for the performance and management of the construction engineering.

- The Contractor shall manage the construction engineering works in accordance with the requirements of the Scope of Work, the Contract and in accordance with its Approved Contract execution plan.

6.3.5.1.1. Construction Engineering Organisation

- The Contractor shall establish a dedicated construction engineering entity for the performance of the construction engineering of the Project.
- The Contractor shall submit the organization chart and relevant job descriptions for its construction engineering organization thirty (30) days prior to the start of any construction activities on the Site and commence with the mobilization of the relevant personnel.

6.3.5.1.2. Construction Engineering Management Documents

- The Contractor shall prepare all required engineering management documents for the Project including, but not limited to:
 - a) Construction engineering manual;*
 - b) Construction engineering plan;*
 - c) Construction engineering procedures:*
- The Contractor shall submit construction engineering management documents not later than thirty (30) days prior to the start of any construction activities on the Site.
- Note* Documents may be submitted as part of or included in the management documents for the Contractor's engineering as per WSG 3.4.1 "Contractor's Engineering Management".

6.3.5.1.3. Construction Engineering Management Reporting

- The Contractor shall establish a system for the reporting of all construction engineering issues. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.3.5.2. WSG 3.5.2 Construction Engineering Pipeline and stations

6.3.5.2.1. General

- The Contractor shall provide all required technical support for the implementation and execution of the Project designs on the construction Sites which shall include, but not limited to:
 - a) The provision of technical support to the construction teams;
 - b) Provide all engineering for the adaptation of construction design solutions;

- c) Provide all bending engineering;
 - d) Design verification in accordance with actual Site conditions;
 - e) Design adoptions in accordance with actual Site conditions; and
 - f) Continuous mark-up of engineering drawings (Red mark-ups).
- Any changes or modifications to the Approved design are subject to the Owner's Approval and shall be properly documented in the "Red mark-ups" of the "Issued for Construction drawings" for later incorporation into the "As-Built drawings".

6.3.5.2.2. Construction Calculations

- The Contractor shall provide all required construction calculations for the implementation and execution of the Project designs, which shall include, but not limited to:
 - a) stress and strain calculations for specific construction practices;
 - b) bending engineering;
 - c) calculations and sizing utilities at TGS and MS; and
 - d) other engineering calculations.

6.3.5.2.3. Construction Drawings and Plans

- The Contractor shall provide all required construction drawings and plans for the implementation and execution of the Project designs on the construction Site which shall include, but not limited to:
 - a) Erection drawings and plans;
 - b) Scaffolding plans;
 - c) Shop drawings for pre-fabrication, fabrication, erection and installation works;
 - d) Painting, coating and insulation plans; and
 - e) Layout plans for all temporary facilities.

6.3.5.2.4. Construction Documents

- The Contractor shall provide all required construction documents for the implementation and execution of the Project designs on the construction Site which shall include, but not limited to:
 - a) Fabrication and construction manual. The fabrication and construction manual shall include all documents, procedures, plans, drawings and pertinent information related to the performance the fabrication and construction works;

- b) Construction Work plans;
- c) Construction Work procedures;
- d) Construction method statements;
- e) Inputs for construction Work risk assessments;
- f) Input for environmental procedures;
- g) Inputs for safety procedures;
- h) Inputs for quality procedures; and
- i) Others.

6.3.6. Work Group 3.6 Engineering Follow-up

- Engineering follow-up will be performed by the Owner or a designated Third Party. The Contractor shall fully cooperate with requirements established as part or during the engineering follow-up.
- Engineering follow-up will include but not limited to:
 - a) The performance and participation in technical reviews;
 - b) The review and Approval (vetting) of the Contractor's design and engineering works;
 - c) The review and Approval (vetting) of the Contractor's construction engineering;
 - d) The review and Approval (vetting) of the Contractor's completions engineering;
 - e) The review and Approval (vetting) of the Contractor's operations and maintenance engineering; and
 - f) The review and Approval (vetting) of the Contractor's as-built and final documentation.

6.4. Work Package 4 Procurement

- Within this work package the Contractor shall undertake the procurement of all Services, Materials, equipment, tools, consumables for the performance of the Works and as defined in Section 4 “Scope of Procurement”.
- For all procurement of Services and Materials the Owner requires the Contractor to:
 - a) Maximise, wherever possible and practical, the involvement of local and regional resources in the execution of the Works;
 - b) Prepare and agree the Vendor’s List with the Owner. Contractor will comply with BFPCL Approved Vendor List (AVL). For items not in the AVL, Contractor may propose vendors for BFPCL approval
 - c) Register all manufacturers, Vendors, Suppliers and Sub-Contractors with the Contractor’s QA/QC department and procurement department;
 - d) All proposed Vendors and Sub-Contractors shall be pre-qualified with regard to their capability to provide the required materials or services in compliance with overall Project Quality, H&S and Environmental requirements and in accordance with Exhibit C – Annex H, Attachment 2 “Procurement & Subcontracting Plan” and Attachment 3 “Criteria for Selection of Vendors and Sub Contractors”;
 - e) Additional Vendors shall be pre-qualified;
 - f) Standardize the provision of Materials and Services as far as possible; and
 - g) Obtain at least three (3) competitive bids for the requested procurement, where applicable.

6.4.1. Work Group 4.1 Procurement Management

- The Contractor is responsible for the management of all procurement aspects of the Works.

6.4.1.1. Procurement Organisation

- The Contractor shall establish a dedicated procurement entity for the effective management of all procurement activities of the Project.
- The Contractor shall submit the organization chart and relevant job descriptions for its procurement organization and commence with the mobilization of the relevant personnel within sixty (60) days after the Commencement Date.

6.4.1.2. Procurement Management Documents

- The Contractor shall develop the following main documents to manage and control its procurement activities under the Contract:
 - a) Procurement manual;

- b) Procurement plan; and
- c) Procurement procedures.
- The Contractor shall develop its procurement plan under the overall Contract execution plan. The Contractor shall ensure that:
 - a) The requirements arising out of implementation of the Contract execution plan are addressed and all materials are procured in the proper sequence to complement the planned execution of the Works;
 - b) All purchase orders for the required Materials and Services are placed and all Materials and Services are delivered to the Site on, or before their "Required at Site" dates; and
 - c) All legal and governmental requirements for the procurement of Materials equipment and Services in the Federal Republic of Nigeria are recognized and implemented throughout all stages of its procurement activities.
- The Contractor shall develop procurement procedures for all procurement activities as defined in this work package.
- The Contractor shall submit procurement management documents not later than thirty (30) days after the Commencement Date for Approval by the Owner.

6.4.1.3. Procurement Reporting

- The Contractor shall establish a system for the reporting of all procurement activities. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting".

6.4.2. Work Group 4.2 Procurement of Services (Sub-Contracting)

6.4.2.1. WSG 4.2.1 Pre-qualification of Sub-Contractors

- The Contractor shall not sub-contract any portion of the Works, in addition to the portions already defined in the proposal, without the Approval of the Owner.
- The Contractor shall be fully responsible for the due performance of all the Sub-Contractors whether proposed by the Owner or pre-qualified by the Contractor.
- If the Contractor desires to engage other than pre-qualified Sub-Contractors for the performance of any portion of the Works, the Contractor shall perform a pre-qualification of experienced Sub-Contractors, subject to the Approval of the Owner.
- The pre-qualification of Sub-Contractors shall be performed with regard to the following criteria:
 - a) Availability of sufficient capacity and resources to perform the sub-contracted work portion within the Approved Contract execution plan and the overall milestone time schedule for the Project;
 - b) Sound financial capabilities;

- c) Track record / satisfactory references for the previous supply of similar services;
 - d) Reputation for high quality and reliability;
 - e) Meet essential requirements of ISO 9001, ISO 9001:2008, Exhibit C, Annex C, Attachment 1 "Quality Management Systems Requirements";
 - f) Have the capability to perform the Works under full integration into the Contractor's Quality, H&S and Environmental Management Systems;
 - g) Have the capability to provide Services throughout the Period of Maintenance of the Project;
 - h) Agree to a pre-award inspection which the Owner may attend;
 - i) Satisfy the requirements and regulations of any regulatory agency or Governmental Authority of the Federal Republic of Nigeria; and
 - j) For the provision of Services inside the Federal Republic of Nigeria provide proof of appropriate registration for the provision of such Services in the Federal Republic of Nigeria.
- The Contractor shall establish a shortlist of (minimum three) pre-qualified Sub-Contractors for the provision of relevant services for Approval by the Owner. The Owner may reject any shortlisted Sub-Contractor for the provision of any portion of the Works at its own discretion.

6.4.2.2. WSG 4.2.2 Tendering and Contract Award

- The Contractor shall be required to perform a tendering procedure for the award of sub- contracts. Such tender shall satisfy all aspects of best commercial practices for transparent and competitive sourcing. Records of the tendering shall be retained for future reference and provided to the Owner if required.
- If the Contractor selects to perform a tendering process for the sub-contracting of any portion of the Works, all proceedings shall conform to the regulations for procurement and tendering in the Federal Republic of Nigeria.

6.4.2.3. WSG 4.2.3 Direct Contract Award

- Direct contract awards for the procurement of Services shall only be made to pre-qualified Sub-Contractors as per **Section WSG 4.2.1 Exhibit C, Annex H, Attachment 3 "Criteria for Selection of Vendors and Sub-Contractors"**.

6.4.2.4. WSG 4.2.4 Supply of Services

- The Contractor will retain full responsibility for the performance of all Works under the Contract and all Approved Sub-Contractors shall be fully managed and controlled within the Approved management systems, including but not limited to the management systems for QA/QC, H&S and Environmental Management of the Works.

6.4.2.5. WSG 4.2.5 Administration of Service Contracts

- The Contractor shall perform effective contract administration for Service contracts of all Sub-Contractors, performing any portion of the Works. The Contractor's Contract Administration shall include but not limited to:
 - a) Progress measurement;
 - b) Advance payment, retention and performance bond;
 - c) Invoicing and payments;
 - d) Scope changes and Variation Orders; and
 - e) Contract close-out.
- The Owner reserves the right to intervene in the Contractor's management of Service Contracts if required to safeguard overall Project objectives and to comply with all legal requirements of the Federal Republic of Nigeria.

6.4.3. Work Group 4.3 Procurement of Materials

- The Contractor shall undertake all purchasing for all Materials and equipment necessary for the construction of the Pipeline and performance of the Works, including special tools, and spares (commissioning spares, operating spares and capital spares). The detailed scope of the Contractor's procurement is specified in **Section 4 "Scope of Procurement"**.

6.4.3.1. WSG 4.3.1 Contractor's Works for Procurement of Materials

6.4.3.1.1. Pre-qualification of Vendors

- All Vendors for the supply of Goods for the Pipeline shall be subject to the Owner's Approval.
- For the supply of certain Goods for the Project certain Vendors have been identified by the Owner. If the Contractor desires to engage other than the preferred Vendors for
- the supply of Goods for the Pipeline, the Contractor shall perform a pre-qualification of experienced Vendors, subject to Approval by the Owner.
- The pre-qualification of Vendors shall be performed with regard to the following criteria:
 - a) Meet all specification requirements for the commodity;
 - b) Capacity and resources to meet quantity and quality requirements;
 - c) Reputation for high quality and reliability;
 - d) The production of the Goods has to meet essential requirements of ISO 9001; ISO 9001:2008; Exhibit C, Annex C, Attachment 1 "Quality Management Systems Requirements";

- e) Provide satisfactory references for the previous supply of similar Goods;
- f) Availability and to provide after sales support in the Federal Republic of Nigeria;
- g) Ability to provide and maintain a stock of spare parts, repair facilities and technical support, if applicable;
- h) Ability to provide training for plant operators and technicians;
- i) Be financially sound and solvent;
- j) Provide or allow standardization with other similar materials for the Project;
- k) Have adequate experience in the particular service, size and specifications of the subject commodity;
- l) Agree to a pre-award shop survey visit for critical items, at which the Owner may attend;
- m) Satisfy the requirements and regulations of any regulatory agency or governmental authority of the Federal Republic of Nigeria; and
- n) For the manufacturing of Goods inside the Federal Republic of Nigeria, provide proof of appropriate registration for the manufacturing of such good inside the Federal Republic of Nigeria.

6.4.3.1.2. Tendering and Contract Award

- The Contractor shall be required to perform a tendering procedure for the award of Sub- contracts. Such tender shall satisfy all aspects of best commercial practices for transparent and competitive sourcing. Records of the tendering shall be retained for future reference and provided to the Owner if required.
- If the Contractor elects to perform a tendering process for the award of any purchase contract for the supply of Goods for the Pipeline all proceedings shall conform to the regulations for procurement and tendering in the Federal Republic of Nigeria as well as the applicable bank standards as advised by the Owner.

6.4.3.1.3. Direct Contract Award

- Direct contract awards for the procurement of Goods for the Project shall only be made to pre-qualified Sub-Contractors as per Section WSG 4.3.1.1 "Pre-qualification of Vendors".

6.4.3.1.4. Supply of Materials

Expediting

- The Contractor shall perform expediting of all Materials, Vendor data and documentation from inquiry to final delivery to Site in order to secure and enforce the deliveries according to the contractual delivery dates and the schedule of the Contract Execution Plan.
- The Contractor shall similarly expedite all replacement Materials which may be subject of guarantees or insurance claims.

Vendor Inspection and Testing

- The Contractor shall be responsible to ensure that all Materials and Services for the Project are procured in sufficient quantities and strictly in accordance with the relevant purchase orders and Project Specifications.

- The Contractor shall be responsible for the control, inspection and test activities of all Vendors and submit a Vendor inspection and test plan for Approval by the Owner. The inspection and test plan shall contain a schedule of all inspection and test activities providing timely notification (minimum fourteen (14) days prior to the performance of the first schedule activity) and allowing for the participation to the FAT of the Owner or the Owner's Representative.
- The Contractor shall ensure the right of the Owner or any authorized representative of the Owner to have full access to all Vendors' Sites and facilities for the purpose of inspection and to participate in any inspection and test activities performed by the respective Vendor.

Material Certification

- All Materials shall be certified in accordance with the specified Material certification requirements.

Materials Control

- The Contractor shall develop procedures and implement a database system for tracking of all Materials from the time of their identification during the engineering stage, throughout to the mechanical completion of the Project, the materials management system (MMS).

Equipment Lists

- The Contractor shall provide equipment lists providing the main data and tag numbers of all installed equipment and main components.
- The equipment lists shall be arranged so that they can be sorted in accordance with the installation location and shall be provided in editable format (excel format), suitable for incorporation into the Owner's maintenance database as specified in Exhibit C, Annex D, Attachment 3 "Tagging System".

Tagging of Materials and Equipment

- The Contractor shall ensure that all materials and equipment that have been assigned tag numbers shall be tagged as specified in Exhibit C, Annex D, Attachment 3 "Tagging System" at the Vendors premises prior to shipment.

Shipping and Customs Clearance

- The Contractor shall be responsible for the transportation, shipment and importation of all materials from their point of supply or manufacture to its material storage yard(s) or the Site of installation inside the Federal Republic of Nigeria.
- The Contractor shall establish a procedure for materials shipment and customs clearance for Approval by the Owner. The shipping and customs clearance procedure shall be according to Exhibit C, Annex H, Attachment 1 "Importation of Equipment and

Materials”.

Material Transportation (International)

- The Contractor is responsible for any international transportation of Materials and equipment.

Shipping Documents

- The Contractor shall bear full responsibility for the accuracy, timeliness, transmission and regulatory compliance of all shipping documents.

Material Dispatch Dossier

- For main Materials or whenever requested by the Owner the Contractor shall ensure that Vendors compile a material dispatch dossier.

Material Inspection and Acceptance

- Upon receipt of any Materials for the Project, the Contractor shall inspect said Materials and furnish a materials inspection report. Defective Materials shall be repaired under the responsibility of the Vendor or returned to the Vendor for repair or replacement.
- For all inspected Materials a material inspection report shall be forwarded to the Owner within three days after inspection of the relevant material inspection took place.

Material Receipt and Tracking

- For all inspected and accepted Materials a materials receipt notice shall be issued and forwarded to the Owner within 3 days after the Materials was officially received and accepted.
- The Contractor shall employ a material tracking system, tracking all Materials throughout all the relevant phases of the Project from the issuing of a material requisition, through manufacturing and fabrication, shipment, receipt, storage and delivery to the Site until incorporation into the Pipeline.

6.4.3.1.5. Material Storage

- The Contractor shall store all Materials in accordance with Material manufacturer's requirements and in compliance with an Approved procedure for Material storage, providing for climate controlled storage facilities whenever specified for the Material by the Vendor. Further details for the storage of Materials are specified in the document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED.

6.4.3.1.6. Material Transportation (Local)

- The Contractor is responsible for all receiving, clearing, loading and the local transportation, of all Materials and equipment. The Contractor shall comply with all customs regulations, import procedures and all applicable Laws regarding permits, escorts, load limits and any other procedures for handling and transportation of Materials.

6.4.3.1.7. Administration of Purchase Contracts

- The Contractor shall perform effective contract administration for Material purchase and supply for any portion of the Works. The Contractor's Contract administration shall include but not limited to:
 - a) Progress measurement;
 - b) Advance payment, retention and performance guarantee;
 - c) Invoicing and payments;
 - d) Scope changes and Variation Orders; and
 - e) Contract close-out.

6.4.3.2. WSG 4.3.2 Supply of Materials and Equipment

- The supply of Materials and equipment shall be based on the requirements outlined in **Section 4 "Scope of Procurement"**. The supply of equipment, Materials and consumables shall include all items required for the performance of the Contractor's Scope of Work.
- This Work Group shall specifically reflect the supply performed by Vendors and Suppliers. Refer to the MTOs for details. Contractor shall note that the MTOs are not exhaustive and serves as a guide for a reasonable estimate of the cost.
- The following list enumerates some key equipment and Material. However, the items not explicitly mentioned in the list but required for the complete Scope of Work shall be included in the supply;
 - a) Line Pipe as specified on the respective data sheets,
 - b) Bends as specified on the respective data sheets,
 - c) Isolation joints
 - d) Cathodic equipment and bulk materials
 - e) Painting and coating material;
 - f) Mechanical bulk material;
 - g) Electrical Bulk Equipment and Material.
 - h) Field instruments; and SCADA equipment
 - i) FOC
 - j) FOC conduit;
 - k) Intrusion detection system;
 - l) F&G detection system;
 - m) SCS and ESD system

- n) RTU and Control equipment shelter.
- o) Security systems (civil, mechanical equipment).
- p) Valves;
- q) Gas metering and flow control package;
- r) Piping;
- s) Vent system;
- t) Close drain Vessel
- u) Open drain Tank
- v) Diesel system;
- w) Civil Equipment and Material
- x) Furniture;
- y) Any other Ancillary (utility) systems;

6.4.3.2.1. Consumables and Materials for first filling

- It is the Contractor's obligation to supply and to provide all consumables and Materials for the first filling of the pipeline systems.
- This includes all following items, but not limited to:
 - a) Oil for machinery and actuators;
 - b) Hydraulic liquids;
 - c) Instrument fillings;
 - d) Fuels;
 - e) Grease for bearings and mechanical equipment; and
 - f) Nitrogen for filling in front of natural gas filling.
- Only the natural gas for filling the Pipelines will be provided by the Owner. The gas in adequate quality and sufficient quantity will be available at the connecting points of the new Pipeline systems with the existing facilities.

6.4.3.2.2. Commissioning parts and spares

- The Contractor's scope for commissioning activities also include the provision of all parts, equipment, facilities gauges, measuring equipment and tools.
- Sets of gaskets and other sealings will have to be provided by the Contractor and consequently an adequate number of spare items for all Materials which may become necessary to perform the commissioning activities in an efficient way.
- The spare parts for the further two years operation of the Pipeline systems are described at another position in the Scope of Works respectively in the schedule of prices within the Contract documents.

6.4.4. Work Group 4.4 - Owner's Procured Items

- "Not applicable".

6.4.4.1. WSG 4.4.1 Novation Agreement

- “Not applicable”.

6.4.4.2. WSG 4.4.2 Material Supply / Handover

- The Materials provided by the Owner to the Contractor must be fully verified by the Contractor at the Owner indicated storage place and a certificate shall be signed by both parties to document the handover. The Contractor will be fully responsible for these Materials / equipment from handover, including the loading in / unloading off transportation vehicles and the transportation to the Contractor's storage yard.

6.5. Work Package 5 Fabrication and Construction

- The Contractor shall perform all Works in compliance with the relevant Laws of the Federal Republic of Nigeria, the Project Specifications and procedures and in accordance with best industry practice.
- The Contractor shall implement all specific provisions for the management and execution of the construction works are specified in document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED.
- Work Group 5.1 Construction Management and General Items

6.5.1.1. WSG 5.1.1 Construction Management

- The Contractor is responsible for the management of all construction activities as part of the Works.
- The Contractor shall perform and manage the construction of the Works in accordance with the requirements of the Contract and in accordance with the Approved Contract execution plan.

6.5.1.1.1. Construction Organisation

- The Contractor shall establish a dedicated construction organization for the effective performance, management and control of the construction works.
- The Contractor shall submit the organization chart and relevant job descriptions for its construction organization and commence with the mobilization of the relevant personnel within thirty (30) days after the Commencement Date.

6.5.1.1.2. Construction Management Documents

- The Contractor shall prepare all required management documents for the performance and management of the construction activities of the Project including, but not limited to:
 - a) Construction and fabrication manual. The fabrication and construction manual shall include all documents, procedures, plans, drawings and pertinent information related to the performance of the fabrication and construction works;
 - b) Construction and fabrication plan;
 - c) Construction resource analysis;
 - d) Construction march-chart (time-line diagram);
 - e) Construction and fabrication procedures; and
 - f) Construction Work method statements.

- The Contractor shall submit Construction Management documents not later than thirty
- (30) days prior to the start of any construction activities on the Site.
- For the preparation of the relevant construction management documents reference shall be made to **Work Package 3.5** "Construction Engineering".

6.5.1.1.3. Construction Management Reporting

- The Contractor shall establish a system for the reporting of all construction management affairs, and in particular the progress of the construction activities. The reporting shall be performed within the scope of **WSG 1.8.8 "Project Reporting"**.

6.5.1.2. WSG 5.1.2 Work Planning, Supervision and Control

- The Contractor is responsible to implement relevant systems and provisions for the adequate planning, supervision and control of the Works.
- The Contractor shall observe the specified Project requirements for the execution of the Works, including but not limited to:
 - a) Work planning;
 - b) Resource planning;
 - c) Materials planning;
 - d) Work supervision;
 - e) Main spread control;
 - f) Field communication;
 - g) Traffic management;
 - h) Pollution abatement;
 - i) Noise control;
 - j) Permit to Work system; and
 - k) Security, health and safety.
- The Contractor shall provide for adequate supervision of the Works. The Owner reserves the right to stop any part of the Works that is performed without being adequately supervised, without any prior notification.

6.5.1.3. WSG 5.1.3 Work Procedures, Method Statements, Job Site Briefings

- The Contractor shall control all Works including works for the rectification of defective works, through:
 - Work procedures (Approved Work procedures and qualified Work procedures);
 - Work method statements; and
 - Job site briefings.

6.5.1.4. WSG 5.1.4 Permits, Authorizations and Third Party Interfaces

6.5.1.4.1. Permits and Authorizations acquired by Contractor

- The Contractor is responsible to identify and obtain all permits and authorizations for the performance of all Works.
- This shall include, but be not limited to:
 - a) Design permits and Approvals for the Contractor's detail design and construction design works;
 - b) Importation permits for the importation of Materials and equipment for the performance of Project activities;
 - c) Transportation permits for the transportation of Materials and equipment on public roads;
 - d) Labour permits and visa for the hiring of local and expatriate workers;
 - e) Work permits for the performance of general or specific work activities;
 - f) Crossing permits for all crossings with other installations or objects (including crossings newly identified during the Contractor's pre-construction investigation);
 - g) Utility permits for the temporary and permanent connection to existing utilities;
 - h) Disposal permits for the disposal of waste and effluents; and
 - i) Any other permit, authorization or Approval required for the performance of its Scope of Work or related to the completion, certification or inspection of Works performed within the Contractor's Scope of Work.
- Permits and authorizations shall be obtained from local, regional or central Governmental Authorities, agencies or entities either affected by, or regulating any aspect of the Works.
- The Contractor shall obtain all permits and authorizations in time, so as to allow the scheduled progressing of the Works.

6.5.1.4.2. Permits and Authorizations acquired by the Owner

- For permits and authorizations to be acquired by the Owner, the Contractor shall prepare all technical documentation and/or studies as requested by the Owner to satisfy the relevant application requirements and as required by the concerned Authorities, Project sponsors or any other Third Party.
- The Contractor shall participate in meetings or hearings arranged or requested by the Authorities, Project sponsors or Third Parties engaged in Project reviews or Approvals, as requested by the Owner.

6.5.1.4.3. Third Party Interfaces and Community Liaison

- The Contractor shall maintain a dedicated community liaison team interfacing with all Third Parties affected by, or having an interest in the Project. All inquiries or contacts made and managed by the Contractor's community liaison team shall be logged-in and administrated in a dedicated Project database, which shall be submitted to the Owner on regular basis and/or upon specific request.

6.5.1.5. WSG 5.1.5 Implementation of EIA and Project Commitments

- The Contractor shall employ a mechanism to demonstrate that all requirements, obligations and commitments derived from the EIA, other Authority, land or other stakeholder requirements and the Owner's commitments have been fully understood and will be fully implemented during the performance of the construction works.
- The Contractor shall maintain a register of all commitments/obligations and provide periodic (weekly/monthly) reports on the (ongoing) status of the implementation of such commitments.

6.5.1.6. WSG 5.1.6 Land for Construction

6.5.1.6.1. Land provided by the Owner

- The following land shall be provided by the Owner:
 - a) the permanent ROW for the Pipeline as identified on the Pipeline alignment sheets;
 - b) additional ROW for route changes, Approved by the Owner;
 - c) land for the Pipeline stations, including block valve, pigging, metering and terminal gas stations and ancillary installations, e.g. cathodic protection installations (anode ground beds). For works which the Contractor shall perform inside working areas of other Project Contractors or Sub-Contractors, the Contractor will be granted access to such work location, subject to confirmation by the Owner); and
 - d) land for permanent access roads to Pipeline stations;
 - Any other land shall only be provided by the Owner, if specifically advised by the Owner.
 - All land acquired by the Owner will be delivered to the Contractor as specified in the relevant Project procedures.

6.5.1.6.2. Land to be acquired by Contractor

- Any additional areas required by the Contractor for the performance of the Work, such as, but not limited to, the areas identified as "Main pipe storage yards, temporary Site facilities, utilities and access roads" and any additional work areas that are not provided by the Owner shall be acquired by the Contractor.
- The location of any such temporary Site facilities or additional work spaces is subject

to Approval by the Owner.

- All temporary Site facilities shall be selected and acquired in compliance with the relevant EIA requirements, Project procedures and policies.
- The Contractor shall furnish the Owner with written evidence that all areas for the installation of temporary Site facilities have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures, i.e. for the performance of negotiations and the calculation and disbursement of compensation payments.

6.5.1.7. WSG 5.1.7 The Construction Site

- The Contractor shall observe all specific provisions for:
 - a) Access to the Site including:
 - Transfer of land / possession of the Site;
 - Land entry agreement; and
 - Permission to start the Work.
 - b) Pipeline installation tolerance, route deviations and route changes, including:
 - Installation tolerances;
 - Route deviations; and
 - Route changes.
 - c) The Return of the Site including:
 - Completion of construction Works; and
 - Land exit agreement.
 - as defined in document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED.
 - WSG 5.1.8 In-Country Resources
- The Contractor is required to maximize the use of in-country resources which includes personnel, equipment, services and materials.
- Prior to the importation of any resources, the Contractor shall demonstrate that equivalent resources could not be acquired on the local market.

6.5.1.8. WSG 5.1.9 Training of Construction Personnel

- The Contractor shall provide adequate training for all Project personnel including Sub- Contractors. The training shall include:
 - a) General induction training with regard to H&S, environment, security, quality, communication and other Project requirements and procedures; and
 - b) Work trainings and qualifications
- The Contractor shall establish a training plan and keep documented evidence of all trainings as well as the participating staff members.
- All Project personnel is required to carry a Project ID card, identifying all received

Project qualifications and trainings, at all times.

6.5.1.9. WSG 5.1.10 Construction QA/QC

- Reference is made to **Section 2.5.9 “Quality Requirements”**.

6.5.1.10. WSG 5.1.11 Construction HSE

- Reference is made to **Section 2.5.7 Health Safety and Security Requirements** and Section 0 “Environmental Requirements”.

6.5.2. Work Group 5.2 Preparatory and Supporting Activities

6.5.2.1. WSG 5.2.1 Assessment and Upgrading of Infrastructure

- The Contractor shall assess all existing infrastructure required to support the performance of the Works, including but not limited to the mobilization and demobilization of personnel and equipment, the transportation of Materials and the daily support of the Works.
- The Contractor shall submit its logistic study for review and Approval by the Owner, within forty-five (45) days after the Commencement Date.
- In the Logistics Study the following shall be identified:
 - a) Whether the existing installations are sufficient to sustain the anticipated usage during the construction of the Project;
 - b) The required measures for the upgrade of existing infrastructure to sustain the anticipated usage during the construction of the Project; and
 - c) Additional temporary or permanent installations or services that have to be established to support the construction activities.
- The Contractor shall coordinate and seek the Approval for the use and adopt all measures for the use, upgrading and maintenance of existing infrastructure installations with the relevant Authorities of the Federal Republic of Nigeria.
- In its logistics study, the Contractor shall include a record of the condition of all infrastructure installations i.e. accesses, roads and bridges and others, properly documented through drawings, calculations and photographs, prior to their use.
- All works related to the upgrading of infrastructure installations shall be performed as directed by the relevant Authorities and in strict compliance with applicable rules and requirements, Project Specifications and procedures, as well as applicable international good practice standards.
- If the construction of additional infrastructure installations is required, this shall be realized in close cooperation with the local Authorities and the Owner, with the objective to facilitate any required regulatory Approvals and land acquisition.

- Additional infrastructure installations shall be designed so as to provide long term benefits to affected communities and shall be handed over to local communities or
- respective Authorities after completion of the Project, if so desired by the local communities, the Authorities of the Federal Republic of Nigeria and the Owner.

6.5.2.2. WSG 5.2.2 Mobilisation and Demobilisation of Construction Resources

- The Contractor shall perform the mobilization and demobilization of construction resources (personnel and equipment) in accordance with Approved, mobilization and demobilization plans, aligned with the overall Project schedules and the Contractor's resource planning. During the mobilization and demobilization of personnel and equipment the Contractor shall recognize all Laws, regulations, rules and requirements of the Federal Republic of Nigeria.
- The Contractor's mobilization and demobilization plan is subject to the Owner's Approval.
- WSG 5.2.3 Inspection and Maintenance of Equipment and Machinery
- All Equipment and Machinery proposed for the use during construction works shall have a valid certificate, released by an authorized Inspection Entity, which allow operations.
- The Contractor shall perform regular maintenance for all equipment and machinery as per the recommendations of the relevant machinery and equipment manufacturers. The relevant maintenance records shall be kept at the construction Site and shall be readily available for verification by the Owner or its authorized representative.
- The Contractor shall initiate regular inspection of all equipment and machinery as per the legal requirements of the Federal Republic of Nigeria.

6.5.2.3. WSG 5.2.4 Waste Management

- The Contractor shall manage all waste (liquid, solid and gases) in accordance with an Approved waste management plan and in strict compliance with the requirements of the Federal Republic of Nigeria, the EIA, good international practice and as directed by the Owner. Waste Management shall be included in the Environmental Management Plan.

6.5.2.4. WSG 5.2.5 Construction Investigations

- Prior to the commencement of the actual construction activities in the field, the Contractor shall perform the following, but not limited to Construction Investigations:
 - a) Pre-Construction survey (for the accurate determination of construction quantities);
 - b) Additional hydrological and geological investigations (as per 8.4.5.1);

- c) Additional geo-hazard review (as per 8.4.5.1); and
- d) Additional environmental and archaeological screenings and salvage operations (as per 8.4.5.1).

6.5.2.4.1. Pre-Construction Survey

- Prior to the commencement of construction activities the Contractor shall perform a pre- construction survey (investigation) for the determination and assessment of detailed construction quantities, specific construction methods, required resources and equipment as well as overall planning aspects for the performance of the Works.
- During the performance of the pre-construction survey the Contractor shall identify and verify:
 - a) Special construction measures intended for the use in the Project;
 - b) Special construction zones and encroachment; and
 - c) Crossings and allocation of specific crossing methods.
- During the pre-construction survey, the Contractor shall document the existing conditions of the ROW including, but not limited to, all existing services, installations and structures (above and below ground) as well as ground vegetation and other features of the land.
- The pre-construction survey shall be documented in a “Pre-construction Survey Report” that shall be submitted for review by the Owner.
- Based on the results and findings of its pre-construction survey, the Contractor shall notify the Owner for any additional requirements.

6.5.2.4.2. Hydrological and Geological Investigations

- The Contractor shall perform a geological and hydrological review of the Pipeline route for:
 - a) The verification of the FEED documents provided by the Owner;
 - b) The development of specific detailed or construction designs on the basis of the design provided by the Owner; and
 - c) The development of detailed specifications, detailed drawings, work procedures and method statement for the performance of the Work.
- The Contractor shall verify in particular the hydrological conditions with regard to the Specification of: designs, construction methods and work procedures for watercourse crossings, areas with high groundwater table and aquifers. This may include but not limited to the application of pipeline protection measures, anti-buoyancy measures and special crossing measures.

- The Contractor shall verify in particular the geological and geo-technical conditions with regard to the required designs, construction measures and work procedures (i.e. for civil works), including, but not limited to trenching and required soil and ground stabilisation measures.

6.5.2.4.3. Geo-hazard Review

- The Contractor shall perform a Geo-hazard Review of the Pipeline route and Pipeline station Sites. The Geo-hazard Review shall include, but not limited to the following aspects:
 - a) Ground stability, potential landslide zones and potential mudslides;
 - b) Potential ground settlements;
 - c) Potential water logging and erosion areas; and
 - d) Areas of potential soil liquefaction.
- Any findings from the geo-hazard review shall be reported to the Owner and shall be implemented in to the Contractor's designs and construction procedures for the Project.

6.5.2.4.4. Environmental / Archaeological Screening and Salvage

- The Contractor shall implement all Project requirements with regard to environmental, archaeological findings and cultural heritage as identified in applicable Project Specifications, the EIA, the EMPs, and as stipulated by the relevant Authorities or as directed by the Owner.
- In case of findings the Contractor shall cease all Works and inform the Owner and the relevant Authorities.
- The Contractor shall perform salvage operations and/or local re-routings as directed by the relevant Authorities and/or the Owner as part of its Scope of Work.

6.5.2.5. WSG 5.2.6 Surveying

- The Contractor shall perform all terrestrial survey works in accordance with the applicable Project Specifications and the relevant regulations of the Federal Republic of Nigeria.

6.5.2.5.1. Survey Control Network (Benchmarks)

- The survey control network (Benchmarks) will be installed by the Contractor during the performance of the Pipeline design works. The coordinates and calculations will be provided to the Contractor in a survey report which will also identify the location of the benchmarks in the field.
- The Contractor shall verify the physical condition of the benchmarks and replace

missing or demolished structures.

- The Contractor shall increase the density of survey monuments as required for the performance of its survey works, including but not limited to:
 - a) The staking and marking of the Pipeline route (and other objects);
 - b) The marking of property boundaries and structures for later reinstatement;
 - c) Topographic survey activities of Sites and areas as required for the performance of its detailed engineering and construction works;
 - d) Construction survey activities;
 - e) As-built surveys activities; and
 - f) Marker and measuring post surveys.
- The Contractor shall maintain the local survey control network and the established benchmarks in the field throughout the construction period. All relevant documentation as well as all physical benchmarks (monuments) shall be handed over to the Owner as part of the handover process.

6.5.2.5.2. Staking and Marking

- The Contractor shall perform staking and marking for the identification and field delineation of the:
 - a) Pipeline centreline;
 - b) Right of way (permanent ROW and temporary ROW (Working strip));
 - c) Acquisition area boundaries;
 - d) Acquisition areas include areas of stations, access roads, other areas for the installation of Project installations and other areas acquired temporarily or permanently for the performance of the Works; and
 - e) Cadastral plot boundaries of parcels within and adjacent to the ROW, allowing the reinstatement of parcel boundaries after completion of the construction works.

6.5.2.5.3. Topographic Survey Activities

- The Contractor shall perform topographic surveys as required to support the Works.

6.5.2.5.4. Construction Survey Activities

- Construction surveys activities include all survey activities performed for the correct positioning and verification of the position of the Works.

6.5.2.5.5. As-Built Survey Activities

- The Contractor shall carry out and document an as-built survey including, but not

limited to: the Pipeline (pipeline welds), Pipeline appurtenances, pipeline crossings, Pipeline stations foreign installations or services inside the ROW and all other installations associated with the performance of the Works.

- Details for the performance of as-built Surveys and the coding of survey data are provided in the Project Specifications included in the FEED documentation.

6.5.2.5.6. Marker and Measuring Posts Surveys

- The Contractor shall perform a survey of all marker and measuring posts, providing for the exact position of every post for registration in the Project Geographic Information System GIS and the provision of the correct identification of the marker post position on each individual marker plate.

6.5.2.6. WSG 5.2.7 Materials Management

- The Contractor shall be responsible for the handling of all Materials required for the performance of its Scope of Work of the Project. This shall include Materials procured by the Owner and handed over to the Contractor. The Materials management shall include, but not limited to:
 - a) Planning;
 - b) Material inspection and receipt;
 - c) Material tracking;
 - d) Material handling;
 - e) Material transportation;
 - f) Material storage;
 - g) Material preservation and maintenance;
 - h) Marking, colour coding and re-stamping of line Pipe;
 - i) Equipment tagging;
 - j) Surplus Material;
 - k) Cut and loss;
 - l) Material balance and return of line pipe material; and
 - m) Line Pipe emergency stock.

6.5.2.6.1. Surplus material

- Any surplus material shall be removed from the ROW and stored at the Contractor's material storage yards, in accordance with the recommended storage conditions.

6.5.2.6.2. Cut and Loss

- The maximum allowable cut and loss factor for the line pipe shall be two-per-thousand (0.2 %) for each pipe wall thickness and diameter.
- The Owner shall only accept full, uncut and undamaged pipe lengths with proper marking and certificates as part of line pipe emergency stock or as return quantities from the Contractor.

- Any unaccounted-for pipe quantities shall be reimbursed by the Contractor.

6.5.2.6.3. Material balance and return of line pipe material

- All surplus material procured by the Owner shall be returned to the Owner after completion of the Works. Materials shall be delivered to the Owner's material storage yard(s) as directed by the Owner.
- Damaged materials will not be accepted for return. Only full lengths of line pipe accompanied by the original manufacturing and test certificates shall be accepted for return.
- All Materials used by the Contractor in excess of the installed Materials and specified allowances as per the above Section "Cut and Loss" shall be reimbursed by the Contractor to the Owner.

6.5.2.6.4. Line Pipe Emergency Stock

- The Contractor shall arrange for an emergency stock of hydro tested line pipe.
- The Contractor shall hydro-test a quantity of 20 pipes per pipeline construction lot as line pipe emergency stock.
- The Contractor shall transport the pre-tested line pipe emergency stock to a storage location or storage locations as advised by the Owner.
- The wall thickness distribution for line pipes allocated to the line pipe emergency stock shall be advised by the Owner.

6.5.3. Work Group 5.3 Pipeline Construction

6.5.3.1. WSG 5.3.1 Main Sections

6.5.3.1.1. Preparatory Works and Maintenance of the ROW

- The Contractor shall perform all Works related to the preparation and maintenance of the ROW, including, but not limited to:
 - 1.Setting out of the Pipeline centreline and the ROW;
 - 2.Exact position determination of existing pipelines in parallel to the Pipeline;
 - 3.Environmental and archaeological salvage operations;
 - 4.Locating of services and installations;
 - 5.Structures;
 - 6.Unexploded ordinance;
 - 7.Protection of trees;
 - 8.(Saleable Timber) if applicable for the Project;
 - 9.Temporary fencing and gates;
 - 10.Disposal of refuse and debris;
 - 11.Temporary access roads to the ROW; and

l) 12.Maintenance of the ROW.

m) The detailed provisions for the preparatory works and maintenance of the ROW are specified in document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED and other applicable Project Specifications.

6.5.3.1.2. Clearing

- The Contractor shall perform all clearing operations in compliance with the relevant Project Specifications and in particular with all conditions specified in the land entry agreements/notification, the requirements stipulated in the EIA, and all other Authority or Project requirements.
- The Contractor shall be responsible for the removal and adequate disposal of all hedges, fences, walls, brush, vegetation, non-saleable timber, stumps, tree roots, boulders, debris and other items in accordance with the Project requirements for disposal of waste and excess materials.
- All structures not specifically identified as having been compensated during land acquisition and identified on the land entry agreement and/or encountered during the performance of the construction activities shall be reinstated to a condition equal to or
- better than prior to the removal or demolishing of said structure, by the Contractor, as part of its reinstatement activities.

6.5.3.1.3. Grading

- The Contractor is responsible for the proper grading of the ROW and all permanent and temporary Sites of the Project.
- For the grading of temporary and permanent access roads as well as areas for permanent and temporary installations specified grading levels shall be observed.
- The following requirements and additional measures shall be considered:
 - a) Stripping of topsoil;
 - b) Access passages across the ROW;
 - c) Cut and fill slopes;
 - d) Additional measures at special zones and encroachments; and
 - e) Additional measures at crossings.
- The ROW shall be graded on one level. The grading of the ROW in two levels, so as to reduce the cutting in side slopes is subject to the Owner's specific Approval.

6.5.3.1.4. Double Jointing

- The performance of double jointing of line pipe is subject to Approval by the Owner. The Contractor shall perform a procedure qualification and procedure qualification test for the welding, non-destructive testing (NDT) and coating of the welded pipe

joints.

- The Contractor shall perform double jointing of line pipe in a dedicated fabrication yard.
- The Contractor shall prepare and submit a transportation plan for double pipe lengths demonstrating that the transportation of double jointed pipes can be performed in accordance with the legal (traffic) Laws and requirements of the Federal Republic of Nigeria.

6.5.3.1.5. Hauling and Stringing

- The Contractor shall haul and string all equipment including line pipes, factory bends and fittings along the ROW ensuring location wise distribution of correct line pipes wall thickness and distribution of equipment and Materials to the appropriate locations. The Contractor shall exercise due care and safe handling (loading/unloading and placement) of the line pipe so as to avoid any possible damage to PE coating and pipe itself.

6.5.3.1.6. Bends

- The Contractor shall install all bends (factory bends, field bends and elastic bends), in the vertical and horizontal alignment as specified during the Detail Engineering and Construction Engineering (bending survey).

6.5.3.1.7. Flanges, Fittings Insulation Joints and other Appurtenances

- The Contractor shall install all flanges and fittings, insulation joints and other appurtenances as per the requirements of the detail engineering and/or as identified in the Pipeline alignment sheets.

6.5.3.1.8. Cut-off Pipes, Transition Pieces and Pup Pieces

- The Contractor shall perform the handling and installation of cut pipes, transition pieces and pup pieces and observe the specific requirements for the marking and re-stamping of cut pipes, transition pieces and pup pieces as specified in the in EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED.

6.5.3.1.9. Pipe End Preparation and Line-up

- The pipe end preparation and line-up shall include but not limited to:
 - a) Re-bevelling of all pipe ends;
 - b) Buffing of all pipe ends;
 - c) Internal pipe cleaning;
 - d) Pre-heating of pipe ends; and
 - e) Line-up of pipes.

- Within its Scope of Work, the Contractor shall perform re-bevelling of all pipe ends with an automated machine that is self-centring via an internal alignment clamp to ensure proper pipe end preparation for the employed welding method, regardless of the condition and type of the factory applied pipe end bevels.

6.5.3.1.10. Field Welding

- The Contractor shall perform all welding operations in accordance with the Approved and qualified welding procedures, by qualified personnel. The selection of all employed welding processes is subject to the Owner's Approval.

6.5.3.1.11. Handling of Pipe Strings

- The Contractor shall prepare pipe strings for the subsequent lowering-in operations.

6.5.3.1.12. Electric Resistance Welding / Pin Brazing

- The Contractor shall perform all cable connections to the Pipeline in accordance with the Approved electric resistance welding or pin-brazing method.

6.5.3.1.13. Field Coating

- The Contractor shall perform the coating all field joints, sections of bare pipe and factory bends with a 3-layer heat shrinkable sleeve system.
- The Contractor shall perform the coating of all other materials, equipment and appurtenances of the pipeline in accordance with the Approved Project Specifications.

6.5.3.1.14. Excavating and Trenching

- The Contractor shall prepare the Pipeline trench following the contour of the adjacent land, in such a manner that no part of the Pipeline lowered in the trench remains suspended, allowing the installation of the Pipeline without free span and in accordance with the specified minimum and maximum pipe cover. Any irregularities or non-conformances shall be rectified by the Contractor at its own expense.
- The Contractor shall observe the additional requirements for excavation and trenching in special sections and at crossings as per the relevant Project Specifications and the Approved design.

6.5.3.1.15. Bedding

- The Contractor shall bed the Pipeline onto fine grain material as specified in the Project Specifications. If suitable bedding material is not available or cannot be generated at the Site by the use of specialized equipment (padding machines), the Contractor shall import suitable material to the Site. The use of topsoil for bedding operations is not permitted.
- The use of borrow pits for the importation of materials to the Site is subject to

environmental and Authority Approval as well as the specific Approval by the Owner.

6.5.3.1.16. Pipe lowering

- The Contractor shall perform Pipeline lowering-in operations after relevant pipe string has been completed and shall commence lowering-in only after trench bottom has been inspected and Approved by the Owner in writing.

6.5.3.1.17. Padding

- The Contractor shall perform the Pipeline padding with fine grain materials as specified in the applicable Project Specifications. If suitable padding material is not available or cannot be generated at the Site by the use of specialised equipment (padding machines), the Contractor shall import suitable material to the Site. The use of topsoil for padding operations is not permitted.
- The use of borrow pits for the importation of materials to the Site is subject to environmental Authority Approval as well as the specific Approval by the Owner.

6.5.3.1.18. Backfilling

- The Contractor shall perform backfilling for the entire length of the Pipeline trench as specified in the relevant Project Specification.

6.5.3.1.19. Importation and Disposal of Trench Materials

- Importation of trench material
- The Contractor is responsible for the importation of bedding and backfill material if the existing material does not conform to the Project Specifications.
- The Contractor shall only acquire imported material for bedding and backfilling at licensed borrow pits, Approved by the local Authorities, recognized in the EIA and Approved by the Owner.
- Disposal of excess/displaced trench material
- The Contractor shall haul and dispose all excess or displaced material from the Pipeline trench at licensed disposal sites. The distribution of excess/displaced material on the ROW requires specific Approval of the Owner.
- The Contractor has to dispose any contaminated materials at licensed disposal sites for contaminated wastes in strict compliance with the requirements of the EIA, the Authorities of the Federal Republic of Nigeria and applicable Project requirements.

6.5.3.1.20. Marker Warning / Tape for Pipeline and buried Services

- The Contractor shall supply and install a marker tape above the Pipeline, cables and any other buried services or installations as specified by the design. The type of marker/warning tape is subject to Approval by the Owner.

6.5.3.1.21. Tie-Ins

- The Contractor shall perform all tie-ins of pipe strings and individual pipe sections by

employing qualified personnel and in accordance with the legal and Project requirements for the performance of works in confined spaces.

- The Contractor shall observe the specific requirements for:
 - a) Tie-Ins prior to hydro-testing;
 - b) Tie-Ins after hydro-testing (and prior to Mechanical Completion of the Pipeline); and
 - c) Tie-Ins into commissioned / operating pipelines or systems.

6.5.3.1.22. Clean-up

- The Contractor shall keep the ROW in safe, orderly and tidy condition during the performance of the entire construction Works.
- The Contractor shall haul all wastes, debris and surplus material away from the ROW for storage or disposal in accordance with the relevant Project procedures and requirements.

6.5.3.1.23. Reinstatement

- The Contractor shall develop comprehensive plans and implement all requirements for:
 - a) Slope stabilization and erosion protection;
 - b) Reinstatement of the surface; and
 - c) Bio-restoration.
- of the Pipeline ROW and all additional areas that have been temporarily occupied during the performance of the construction activities.
- During the design and construction of slope stabilization, erosion protection, reinstatement and bio restoration measures the Contractor shall specifically acknowledge the design life requirement of the Project. All measures be carefully selected and implemented to allow the safe operation of the Project throughout the specified design life.

6.5.3.1.24. Pipeline Drying

- After mechanical completion of the Pipeline or specific sections of the Pipeline, the Contractor shall perform drying of the Pipeline to a dew point as specified in document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009 "Pipeline Construction Specification" as part of FEED.
- WSG 5.3.2 Special Construction Measures, Special Zones and Encroachments
- The Contractor shall observe the special conditions and construction measures that shall be applied in special zones and at encroachments of the Pipeline and its installations with other objects and/or installations.

- The Contractor shall perform all design and construction works for the installation of the Pipeline at special zones and encroachments as specified in the design for the Pipeline.
- Contractor shall verify all specified construction and design measures at special zones and encroachments with the existing conditions encountered during the construction of the Pipeline, so as to ensure that the implemented design and construction measures conform to the design intent, the Authority requirements of the Federal Republic of Nigeria, the Project Specifications and best international practice. Any deviations from the specified design are subject to Approval by the Owner.
- Installation of the Pipeline and the Pipeline stations in populated area has to be in accordance to ASME B31.8, Guidelines and Procedures for the Design, Construction, Operation and Maintenance of Oil and Gas Pipeline Systems in Nigeria.
- The Contractor shall obtain all permits and approvals required for the performance of the construction works at special zones and encroachments and for the application of special construction measures.
- The Contractor shall identify additional zones or areas where additional construction measures have to be implemented as part of its pre-bid / pre-construction survey and implement relevant measures upon Approval by the Owner and as part of its Contractor's Scope of Work.

6.5.3.1.25. Special Construction Measures

- Special construction measures include, but are not limited to:
 - a) Weighting and mechanical protection;
 - b) Dewatering;
 - c) Trench erosion protection / ditch breakers;
 - d) Blasting and explosives; and
 - e) Pontoon trenching in swampy area, including swim-in of the pipe string.

6.5.3.1.26. Special Construction Zones

- Special construction zones include, but are not limited to:
 - a) Areas with potential ground subsidence;
 - b) Potential liquefaction zones;
 - c) Potential sliding zones;
 - d) Areas with high groundwater table;
 - e) Steep slope / rocky areas; and
 - f) Other special zones.

6.5.3.1.27. Encroachments

- Encroachments include but are not limited to:

- a) Proximity to houses and occupied dwellings;
- b) Parallel alignment to roads and railways;
- c) Parallel alignment to pipelines or underground services; and
- d) Parallel alignment to high tension power-line.

6.5.3.2. WSG 5.3.3 Crossings

- The Contractor is responsible for the installation of all crossings for the Pipeline regardless whether these crossings are identified on the Pipeline alignment sheets, or identified during Contractor's pre-bid and pre-construction surveys of the Pipeline route or during the performance of the Works.
- The Contractor shall identify additional crossings as part of its pre-bid/pre-construction survey and implement relevant measures upon Approval by the Owner and as part of the Contractor's Scope of Work.
- The Contractor shall observe the special conditions and construction measures that shall be applied at crossings.
- The Contractor shall perform all design and construction works for the installation of the Pipeline at crossings, as specified in the design and as required to obtain the relevant crossing permit from relevant s/operators or Authorities(s) of the crossed object or installation.
- Crossings are described by the crossing method (such as open-cut crossings and trenchless crossings) and the crossing classification (such as road crossings, watercourse crossings, power line crossings, pipeline crossings, Third Party line crossings and others).
- The Contractor shall verify all specified construction and design measures at crossings with the existing conditions, so as to ensure that the implemented design and construction measures conform to the design intent, the requirements specified by the /operator, the Authorities, the Project Specifications and good international practice. Any deviations from the specified design are subject to Approval by the Owner.
- The Contractor shall obtain all permits and Approvals required for the performance of the construction works at the crossing and for the application of any special construction measures.

6.5.3.2.1. Pipeline Crossings General

- Refer to Volume 03 – Exhibit C – Annex A – Crossing List” for the recommended method of crossing for each crossing type.

6.5.3.2.2. Major River Crossing Works

- The Works on Major River crossings require consistent coordination with the

Authorities and the Owner. Prior to the performance of the crossing operation the Contractor shall perform a geotechnical and hydrological investigation. The Contractor shall be responsible for the coordination and the interfacing with the Third Party supervision.

- The works include the following main items without being limited to:
 - a) Installation of HDD rig and adjacent facilities;
 - b) Location and dimensions of boring and receiver pits;
 - c) Details of proposed shoring methods for pit walls and static calculations proving pit stability;
 - d) Security measures and installations;
 - e) Fabrication of the pipe string;
 - f) Drilling, reaming and pipe installation; and
 - g) Reinstatement.

6.5.3.3. WSG 5.3.4 Ancillary Installations

6.5.3.3.1. Cathodic Protection

- The Contractor shall install the temporary and permanent cathodic protection system for the Pipeline in accordance with the Approved design.

6.5.3.3.2. Temporary Cathodic Protection System

- The Contractor shall design, supply and install a temporary cathodic protection system using sacrificial magnesium anodes for the protection of the Pipeline. The Contractor shall arrange for the temporary protection of each Pipeline section not later than thirty (30) days after the completion of the backfilling in the relevant section.

6.5.3.3.3. Permanent Cathodic Protection System

- The Contractor shall design, supply and install a fully functional permanent (impressed current) cathodic protection system for the pipeline, consisting of, but not limited to:
 - a) Anode ground bed (horizontal or vertical anode ground beds);
 - b) CP monitoring facilities;
 - c) Cables; and
 - d) AC mitigation protection.

6.5.3.3.4. AC Mitigation / HV Influence Protection

- The Contractor shall design, supply and install an AC-mitigation/HV influence protection system at all sections where stray currents may be inducted into the Pipeline.

6.5.3.3.5. Control and Instrumentation and Telecom Systems

- "Not applicable."

6.5.3.3.6. Fibre Optic Cable Conduit and Appurtenances

- The Contractor shall supply and install two (2) fibre optic cable conduit (FOCC) on one side of the pipe and along the entire length of the Pipeline (including inside of the Pipeline stations).
- Details of the FOC, FOCC and accessories are specified in the Doc no. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002-R03_Basis of Design

6.5.3.3.7. Fibre Optic Cable (FOC)

- The Contractor shall supply, install and test two (2) fibre optic cables of 48 cores each. up to the main control cabinet of each Pipeline station. One FOC is for data in a composite cable and another strapped on the pipeline for leak detection. Only a portion of the cable cores will be used for either data or leak detection and the rest shall serve for redundancy.

6.5.3.3.8. Cables and Wires

- The Contractor shall supply and install all cables and wires for the Pipeline as per the relevant Project Specifications.

6.5.3.3.9. Marker and Measuring Posts

- The Contractor shall supply and install marker posts and measuring posts on adequate supports, fitted with name plates and aerial marker caps as specified in the detailed engineering and as Approved by the Owner.
- The scope of the work for marker and measuring post installation shall include, but not limited to:
 - a) Marker posts for the Pipeline;
 - b) Marker post for cable installations;
 - c) Measuring posts;
 - d) Post foundations;
 - e) Name plates; and
 - f) Aerial marker caps.

6.5.3.4. WSG 5.3.5 Temporary Installations

- This Work Sub-Group comprises all Works including clearance, grading/levelling, filling/compaction, excavation, fencing, building of road accesses, installation of buildings and utilities for the installation, operation and subsequent removal of all temporary Site facilities, Site utilities and temporary access roads thereto.

6.5.3.4.1. Temporary Site Facilities

- The Contractor shall establish all required temporary Site facilities required for the performance of the Works.
- The Contractor shall be responsible for the acquisition and maintenance of all additional areas required for the performance of the Works, such as:
 - a) construction camps;
 - b) manufacturing and pre-fabrication yards;
 - c) main material storage yards;
 - d) intermediate PSY; and
 - e) other areas used for the performance of the Works.
- The locations for the establishment of temporary Site facilities are subject to the Approval of the environmental protection agency and the Owner.
- Prior to accessing any of these additional areas, the Contractor shall furnish the Owner with written evidence that all of these areas have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures.
- The Contractor shall provide for Site 'housekeeping' on a continuous basis to ensure the Site is free from scrap, debris and rubbish at all times; and provide all required temporary facilities to ensure that construction equipment is properly stored and all buildings or structures are maintained in a clean, tidy, painted and serviceable condition.

6.5.3.4.2. Temporary Site Utilities

- The Contractor shall design, construct and install all temporary utilities at all Sites required for the Contractor to carry out the Works. Such utilities shall include, without limitation the installation, commissioning, maintenance, and subsequent removal of:
 - a) Electrical power supply comprising of temporary power supply and distribution of power throughout the Site as required for construction and commissioning;
 - b) Water Supply including potable water comprising of temporary fresh water supply, cooling, collection, storage, transfer and distribution of all fresh water requirements (including potable water production if necessary);
 - c) Natural gas / heating gas supply (if applicable);
 - d) Liquid waste treatment and disposal, in accordance with all applicable Laws, of all sewage and waste water including the disposal of hydro-test water;
 - e) Solid Waste Disposal including disposal of all construction debris and hazardous waste; and
 - f) Communication Installations and temporary construction of communication facilities (i.e. telephone landlines and wireless, internet, satellite based communication reception, Very High Frequency (VHF) Radio, etc.).

6.5.3.4.3. Temporary Access Roads

- The Contractor shall establish all temporary access roads to the right of way required for the performance of the construction works.
- Prior to construction of such temporary accesses, the Contractor shall furnish the Owner with written evidence that all additional areas required for this purpose have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures.
- The Contractor shall maintain temporary access roads in safe and usable conditions at all times.

6.5.3.5. WSG 5.3.6 Inspection and Testing

- All construction inspections and tests shall be performed in accordance with the requirements as laid down in Exhibit C, Annex C “Quality Management System Requirements” and in accordance with the Approved inspection and test plan. The Owner shall have the right to witness all inspection and testing performed by the Contractor and any of its Sub-Contractors or Vendors supplying Materials, equipment or Services for the Project.
- The Contractor shall submit inspection and test plan in accordance with the requirements specified in Exhibit C, Annex C “Quality Management System Requirements” and as specified in the relevant Project Specifications for Approval by the Owner not later than sixty (60) days after the Commencement Date.
- The Contractor’s Inspection and Test Plan shall address all Works, identifying the inspection and test criteria as specified in the relevant Project Specifications, as required by the Authorities of the Federal Republic of Nigeria, in accordance with best international practice and as required to ensure the quality of the Works.
- The Inspection and Test Plan shall identify all “witness points” and “hold points”, where the Works may only proceed after satisfactory examination or test performance witnessed by the Owner or its authorized representative.
- All tests shall be documented and signed by the Contractors supervision and/or QA personnel responsible for the test performance.
- All test records shall be compiled and become part of the as-built documentation of the Works.

6.5.3.5.1. Inspection and testing of Linear Sections

- The inspection and testing of linear sections shall include but not limited to:

6.5.3.5.2. Examination of Field Bends

- The Contractor shall qualify its field bending procedure and shall examine each

individual field bend. The criteria for the performance of procedure qualification and production tests for field bending shall be specified in the relevant construction specification and the Approved inspection and test plan.

6.5.3.5.3. Holiday Detection Testing

- The Contractor shall perform holiday detection testing of all Pipeline components (i.e. pipes, valves, tees, flanges and fittings) and field joint coatings immediately after field coating application.
- The Contractor shall perform holiday testing of completed pipe strings immediately prior to the lowering in of the pipe string.
- The Contractor shall perform additional holiday detection tests whenever mandated to identify a coating defect.

6.5.3.5.4. Field Joint Coating Examination and Testing

- The Contractor shall qualify the field joint coating application procedure and examine and test each individual field joint coating as specified in the relevant Project Specification and the Approved inspection and test plan.

6.5.3.5.5. Testing of Insulation Joints

- The Contractor shall test all insulation joints as specified in the Approved inspection and test plan.

6.5.3.5.6. NDT of Welds

- All welds performed on the Pipeline shall be subject to 100% NDT. The specific requirements for the NDT of pipe welds are laid down in the relevant Project Specifications.

6.5.3.5.7. Hydro-test

- The Contractor shall perform hydrostatic testing for the Pipeline. The testing of the Pipeline by air testing is not permitted.
- Block valves and block valve assemblies shall be separately hydro-tested and inserted into the line after completion of Pipeline hydro-test.
- The Contractor is responsible for the procurement and discharge of hydro-test water in accordance with the requirements of the EIA and all applicable Laws or regulations of the Federal Republic of Nigeria.
- The Contractor shall perform pre-testing of all pre-assembled pipe sections at crossing locations and valves.
- The Contractor shall perform pre-testing of all line pipes designated for the Owner's emergency pipe stock.

6.5.3.5.8. Geometric Inspection / Electronic Calliper Pigging

- After completion of the hydro- test, the Contractor shall verify the internal geometry of the entire Pipeline with an electronic caliper pig. The procedure for electronic caliper pigging shall be submitted for review and Approval by the Owner.
- The Contractor shall perform the electronic caliper pigging in individual sections, whereas the individual length and location (boundaries) of each section shall be subject to Approval by the Owner.

6.5.3.5.9. DCVG Survey

- The Contractor shall perform a direct current voltage gradient (DCVG) survey of the entire pipeline after completion of backfilling and provisional reinstatement of the ROW. All detected coating defects shall be subject to repair.
- The procedure for DCVG survey shall be submitted for review and Approval by the Owner.

6.5.3.5.10. Testing of Ancillary Installations

6.5.3.5.11. Testing of the Fibre Optic Cable Conduit

- Following the installation of the FOC conduit and after completion of the trench backfill, the Contractor shall perform a pressure and calibration test as specified in the relevant construction specification and in the Approved inspection and test plan.

6.5.3.5.12. Testing of the Fibre Optic Cable

- Following the installation and splicing of the FOC Cable, the Contractor shall perform a test as specified in Doc no. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002-R03_Basis of Design.

6.5.3.5.13. Testing of CP System (Temporary/Permanent)

- The Contractor shall perform testing of the temporary and permanent CP system as specified in the relevant specifications and in the Approved Inspection and Test Plan.

6.5.3.5.14. Cable Continuity Tests

- The Contractor shall perform cable continuity test for all cables installed by the Contractor.

6.5.3.5.15. Other Inspection and Tests

- The Contractor shall perform and document all other inspections and tests in accordance with the Approved inspection and test plan and the relevant Project Specifications, the requirements of the Federal Republic of Nigeria and to ensure the quality of the Works in accordance with good international industry practice.

6.5.3.6. WSG 5.3.7 Final Documentation and As-Built

- The following, but not limited to, documents and data, shall be included in the as-built Documentation package that shall be submitted by the Contractor. All documents shall be submitted in electronic and hardcopy (paper) format as described in Exhibit C, Annex F "Documentation Requirements".
- All documents generated by the Contractor and submitted in electronic file format shall also be submitted in editable electronic file format, whereas the term editable shall refer to as editable in commercially available software programs and file formats, such as but not limited to Microsoft office (Ms-word, Ms-windows, Ms-excel, Ms-project, Ms- power point, Ms-Visio and others), Primavera, latest revision of AutoCAD, ESRI ArcGIS software and others.

6.5.3.6.1. Vendor and Manufacturers Documents, Drawings, Certificates

- Vendor manuals;
- Vendor operating manuals;
- Vendor drawings;
- Material certificates;
- Base Material certificates;
- Factory Acceptance test certificates;
- Importation and shipping documents; and
- Damage and repair reports.

6.5.3.6.2. Updated Project Manuals, Specification and Procedures

- Design manual;
- Construction manual;
- Certification manual; and
- Operation and maintenance manual.

6.5.3.6.3. Project Documentation and Reports

- Project close out report (hand-over);
- Pipe log book with certificates;
- QA/QC documentation;
- HSE documentation;
- Inspection and test reports; and
- Radiographs and hardcopies of ultrasonic results.

6.5.3.6.4. As-Built Drawings and Data

- Red mark-up of all the "Issued for Construction" drawings;
- Final "As-Built" alignment sheets drawings;
- Final "As-Built" pipeline design drawings;
- Final "As-Built" survey data; and
- Final "As-Built" special point / crossing design drawings.

6.5.3.6.5. GIS Data and Data Warehouse

- Geographical information system; and
- Data warehouse.

6.5.3.6.6. Interface/Input to Asset Integrity Management Programme

- This work unit entails the compilation and input of all relevant data into the Owner's asset integrity management programme.

6.5.4. Work Group 5.4 Pipeline Stations Construction

- The Contractor shall construct all stations of the Pipeline.
- The Contractor shall observe the requirements defined in **document Annex G, Attachment 2 "Interface Plan and Matrix "** for the identification of the Contractor's construction responsibilities inside Pipeline stations.

6.5.4.1. WSG 5.4.1 BBOU and Elep Manifold Stations

6.5.4.1.1. WSG 5.4.1.1 Preparatory Works and Site Clearing

- The Contractor shall be responsible to execute all preparatory Works required before starting the construction of the Project installations, including, but not limited to:
- The Dredging of access canal to the site is handled in another contract. However Contractor shall obliged to create temporary water way access to facility the installation of the platforms
 - a) Setting out of station area;
 - b) Pre-construction survey;
 - c) Preparation of temporary access road/water way; and
 - d) Clearing of area affected by construction, camps, storage and pre-fabrication area, workshops, etc.
 - e) Pilling works and foundations.

6.5.4.1.2. WSG 5.4.1.2 Site Preparation Earth Works

- Based on the Contractor's engineering, the Contractor is responsible for execution of earth Works required, considering transitioning from actual topography to the station soil contour defined during design optimizing the volumes of cut and fill. The work comprises the following main items without being limited to:
 - a) Cut of ground to level station, including proper compaction of the subsoil where necessary;
 - b) Separation of cut Material into usable and non-usable (spoil) Material;
 - c) Discharge of soil Material and fill in adjacent areas in accordance with environmental requirements;
 - d) Compaction of fill Material in accordance with Project Specifications.

6.5.4.1.3. WSG 5.4.1.3 Temporary Installations

- This Work Sub-Group comprises all Works including clearance, grading/levelling, filling/compaction, excavation, fencing, building of road accesses, installation of buildings and utilities for the installation, operation and subsequent removal of all temporary Site facilities, Site utilities and temporary access roads thereto.

6.5.4.1.4. Temporary Site Facilities

- The Contractor shall establish all temporary Site facilities required for the performance of the Works.
- The Contractor shall be responsible for the acquisition and maintenance of all additional areas required for the performance of the Works, such as:
 - a) Construction camps;
 - b) Manufacturing and pre-fabrication yards;
 - c) Storage yards; and
 - d) Other areas used for the performance of the Works.
- The locations for the establishment of temporary Site facilities are subject to Environmental Authority and the Owner's Approval.
- Prior to accessing any of these additional areas, the Contractor shall furnish to the Owner written evidence that all of these areas have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures.
- The Contractor shall provide for Site housekeeping on a continuous basis to ensure Site is free from scrap, debris and rubbish at all times; and provide all required temporary facilities to ensure that construction equipment is properly stored and all buildings or structures are maintained in a clean, tidy, painted and serviceable condition.

6.5.4.1.5. Temporary Site Utilities

- The Contractor shall design, construct and install all temporary utilities at all work Sites of the Contractor. Such utilities shall include, without limitation, the installation, commissioning, operation, maintenance, and subsequent removal of:
 - a) Electrical Power Supply comprising of temporary power supply and distribution of power throughout the Site as required for construction and commissioning;
 - b) Water Supply including potable water comprising of temporary fresh water supply, cooling, collection, storage, transfer and distribution of all fresh water according to the Project requirements, including potable water production if necessary;
 - c) Natural Gas / Heating Gas supply if applicable;
 - d) Liquid Waste Treatment and Disposal in accordance with all applicable Laws, of all sewage and waste water including the disposal of hydro test water;

- e) Solid Waste Disposal including disposal of all construction debris and hazardous waste; and
- f) Communication Installations comprising of temporary construction of communication facilities (i.e. telephone landlines and wireless, internet, satellite based communication reception, VHF Radio, etc.). Contractor shall ensure the provided installations are suitable for effective communication.

6.5.4.1.6. Temporary Access Roads

- Prior to accessing any additional area for the construction of temporary accesses, the Contractor shall furnish to Owner written evidence that all of these areas have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures.
- The Contractor shall maintain temporary access roads in safe and usable conditions at all times.

6.5.4.1.7. WSG 5.4.1.4 Station Earthworks

- Once site preparation is complete, the Contractor will execute the earthworks for the station construction, which includes the following items, without being limited to:
 - a) Excavation for foundations;
 - b) Compaction of subsoil;
 - c) Incorporation and compaction of imported fill (gravel) to improve soil conditions;
 - d) Improvement of subsoil with special measures (injections, gravel piles, vibro-compaction, etc.) as required;
 - e) Excavation for underground facilities, like pipe trenches, cable trenches, underground utility (ancillary) systems, culverts, channels, etc.;
 - f) Safeguarding of excavations;
 - g) Drainage of the Site during construction activities;
 - h) Excavation and proper compaction of subsoil for roads, walkways, paved areas, and in general all infrastructure measures;
 - i) Bedding and backfilling of trenches, buildings, etc. with suitable Material including proper compaction; and
 - j) Preparation of the final station surface for reinstatement.

6.5.4.1.8. WSG 5.4.1.5 Foundations

- The Contractor shall build all foundations for equipment (process and service) and buildings. In an early stage of the design, the Contractor shall expedite from equipment
- Suppliers the issuing of all construction and design details (e.g. weight, dimensions, static and dynamic loads, anchor bolts location, etc.) to execute foundations work. The foundation work includes, but is not limited to:
 - a) Execution of special foundations (e.g. piling) as required;
 - b) Foundations for pipe supports;

- c) Verification of subsoil and performed compaction;
- d) Placing of lean concrete as required;
- e) Execution of formwork, reinforcement;
- f) Pouring of concrete, proper vibration, curing of concrete;
- g) Removal of formwork; and
- h) Preparation of adequate interface to mounting of piping, equipment, steelworks, etc.

6.5.4.1.9. WSG 5.4.1.6 Buildings

- The Contractor shall erect several buildings in the station. Buildings in the process area have first priority as far as they will be related to several interdisciplinary Works for power and Instrumentation cabling, etc.
- The following Works shall be included, without being limited to:
 - a) Building civil works;
 - b) Building architectural works;
 - c) Building plumbing and sanitary work;
 - d) Building HVAC works;
 - e) Buildings electrical works;
 - f) Building control and telecommunication systems (Note: building control and telecommunication systems shall be covered under WG 5.5 SCADA and Telecom Construction);
 - g) Building passive fire; and
 - h) Furniture;

6.5.4.1.10. WSG 5.4.1.7 Roads, Infrastructure and Reinstatement

- The Contractor shall be responsible for the construction of roads, pavements, general civil infrastructure, and reinstatement of the Site to final Site conditions. The following main activities are included, without being limited to:
 - a) Permanent access road to the station, including connection to the public road;
 - b) Roads inside the station;
 - c) Paved areas, in areas where vehicle access is required for maintenance, walkways for pedestrian maintenance access;
 - d) Placing of crushed granite of appropriate size in pipe corridors of above ground piping, around equipment and in areas accessible for maintenance;
 - e) Greening of the remaining station area. Seeding and irrigation up to first mowing;
 - f) Landscaping of the area outside the station area; and
 - g) Erection of access platforms, staircases to the platforms, ladders, etc to allow maintenance access to equipment, valves, etc.

6.5.4.1.11. WSG 5.4.1.8 Mechanical Equipment Installation (Process and Service)

- The Contractor shall install all mechanical equipment related to main process as well as for service systems following the instructions provided by equipment

manufacturers in order to ensure their proper operation. The installation of the following main equipment shall be included, without being limited to:

- a) Filter separators;
- b) Line heaters;
- c) Metering and flow control package;
- d) Vent system; and
- e) Diesel system;

6.5.4.1.12. WSG 5.4.1.9 Piping Works

- Piping work comprises of all the activities related to the installation of pipe work inside the station, including the pre-fabrication of pipe spools, the installation of all piping on Site, the installation of valves and any other devices directly connected to different systems as indicated in WSG 5.4.1.8. This piping works shall be carried out as specified in the Project documents. This includes but shall not be limited to:
 - a) Structural steel for pipe supports;
 - b) Storage, preservation and handling of piping equipment;
 - c) Pre-fabrication of pipe spools and pipe supports;
 - d) Transport to Site of spools, piping and valves;
 - e) Installation of pipe supports and alignment of the supports;
 - f) Installation of piping for all mechanical equipment and systems;
 - g) Alignment of the piping, including proper alignment for equipment connections in accordance with Vendor requirements to allow stress free connections;
 - h) Welding (Note: NDE is included in WSG 5.4.1.15);
 - i) Installation of valves;
 - j) Connection of piping to equipment and valves. The piping shall be cleaned before connection to any valve or equipment and the proper alignment shall be verified to execute stress free connections to equipment; and
 - k) Note: Hydro test and other inspection and testing are included in WSG 5.4.1.15.

6.5.4.1.13. WSG 5.4.1.10 Painting/Insulation/Fireproofing

- The Contractor shall perform painting, insulation and fireproofing works for all piping works, equipment and structures for which such requirements are defined in Specifications and the design.

6.5.4.1.14. WSG 5.4.1.11 Electrical Installations

- The Contractor shall execute all electrical Works related to power generation and distribution in the station, as well as lighting, earthing and lightning within the station. The Work shall include, but not limited to:
 - a) Installation of gas generators and associated control panel including interconnection wiring;
 - b) Installation of emergency diesel generator and associated control panel including interconnection wiring;
 - c) Installation of 400V switchgears and MCC's including associated power and

- control, interconnection wiring;
- d) Installation of DC and AC UPS systems, consisting of batteries, battery charger(s), inverter etc.);
- e) Power and lighting distribution panels for lighting, small power receptacles, CCTV etc.;
- f) Installation of PLC based system for generator control, power management, and monitoring including interconnection wiring for remote status and control;
- g) Installation of plant lighting system, area lighting, Street lighting, fence lighting, security lighting etc. including lighting poles, supports, cabling and wiring etc.;
- h) Installation of plant earthing system including necessary connections to equipment e.g., LV MCCs, Panel boards, transformers, lightning protection of buildings and equipment etc.;
- i) Installation of complete cabling system for the power generation area, process area, switchgear, MCCs, power transformers, including interconnection and termination to systems installed in the substation(s); and
- j) Installation of heat tracing system for the process area piping / tubing including power distribution board(s), cabling, interconnection wiring and termination within the Station.

6.5.4.1.15. WSG 5.4.1.12 Instrumentation Installations

- The Contractor shall install instrumentation including cable laying within the stations and all other pipeline related instrumentation, intrusion detection system, F&G detection system, SCS based on (PLC/RTU), ESD system and Control containers (i.e. the cabin that contains all these control equipment); Containerized E-House/Field Auxiliary Room (FAR). There will be RTU room in each manifold that shall house SCADA, Telecom equipment, Power supply etc.

6.5.4.1.16. WSG 5.4.1.13 Fire Safety

- The Contractor shall perform all Works for the fire safety systems, in the station which include, but shall not be limited to:
 - a) Fire fighting system;
 - b) Extinguishing system;
 - c) Mobile fire fighting equipment in buildings; and

6.5.4.1.17. WSG 5.4.1.14 Security Systems, Civil, Mechanical & Electrical

- This work package includes the supply and installation of the permanent security systems of the station, namely the civil, mechanical and electrical installations. The control and telecom installations shall be included in WG 5.5. The following systems shall be installed, without being limited to:
 - a) Security installations; and
 - b) Fences, gates, earth and concrete walls, culverts, etc.

6.5.4.1.18. WSG 5.4.1.15 Ancillary Installations

- The Contractor shall install ancillary (utility) systems as required by the station design. The related systems include, but shall not be limited to:
 - a) Service water supply system;
 - b) Potable water supply system;
 - c) Waste water treatment system;

6.5.4.1.19. WSG 5.4.1.16 Inspection and Testing

- All construction inspections and tests shall be performed in accordance with the requirements as laid down in Annex C, Attachment 1 “Quality Management System Requirements” and in accordance with the approved Inspection and Test Plan. The Owner shall have the right to witness all inspection and testing performed by the Contractor and any of its Sub-Contractors or Vendors supplying Materials, equipment or Services for the Project.
- The Contractor shall submit its inspection and test plan in accordance with the requirements specified in Annex C, Attachment 1 “Quality Management System Requirements” and as specified in the relevant Project Specifications for the Approval by the Owner not later than forty-five (45) days after the Commencement Date and fifteen (15) days prior to the commencement of the Works.
- The Contractor's Inspection and test plan shall address all Works, identifying the inspection and test criteria as specified in the relevant Project Specifications, required by the Authorities of the Federal Republic of Nigeria, in accordance with best international practice and as required to ensure the quality of the Works.
- The inspection and test plan shall identify all “witness points” and “hold points”, where the Works may only proceed after satisfactory examination or test performance witnessed by the Owner or its authorised representative.
- All tests shall be documented and signed by the Contractor's supervision and/or QA personnel responsible for the test performance.
- All test records shall be compiled and become part of the as-built documentation and the mechanical completion records of the Works

6.5.4.1.20. Testing at Station Installations

- The Contractor shall perform all inspection and testing at the station in accordance with the Approved Inspection and Test Plan, relevant Project Specifications, requirements of the Federal Republic of Nigeria and as required to ensure the quality of the Works in accordance with good international industry practice.

6.5.4.1.21. Testing of Ancillary Installations

- Water Supply to the station
- This item refers to all the required testing of the mechanical facilities and piping required for the station water supply.

6.5.4.1.22. Other Inspection and Tests

- The Contractor shall perform and document all other inspections and tests in accordance with the Approved inspection and test plan, relevant Project Specifications, requirements of the Federal Republic of Nigeria and as required to ensure the quality of the Works in accordance with good international industry practice.

6.5.4.1.23. WSG 5.4.1.17 Final Documentation and As-Built

- The following, but not limited to, documents and data, shall be included in the as-built Documentation package that shall be submitted by the Contractor. All documents shall be submitted in electronic and hardcopy (paper) format as described in Exhibit C, Annex F "Documentation Requirements".
- All documents generated by the Contractor and submitted in electronic file format shall also be submitted in editable electronic file format, whereas the term editable shall refer to or editable in commercially available software programmes and file formats, such as but not limited to Microsoft office (MS-word, MS-windows, MS-excel, MS-Project, MS- power point, MS-Visio and others), Primavera, latest revision of AutoCAD, ESRI software and others.

6.5.4.1.24. Vendor and Manufacturers Documents, Drawings, Certificates

- a) Vendor manuals;
- b) Vendor operating & maintenance manuals;
- c) Vendor drawings;
- d) Material certificates;
- e) Base material certificates;
- f) Factory acceptance test certificates;
- g) Importation and shipping documents; and
- h) Damage and repair reports.

6.5.4.1.25. MRB, Updated Project Manuals, Specification and Procedures:

- a) Design manual, Specifications and procedures;
- b) Construction manual, Specifications and procedures;
- c) Certification manual; and
- d) Operation and maintenance manual and procedures.

6.5.4.1.26. Project Control and QHSE Documentation and Reports:

- a) QA/QC documentation;

- b) HSE documentation;
- c) Inspection and test reports; and
- d) Radiographs and hardcopies of ultrasonic results.

6.5.4.1.27. Project Licenses, Certificates and Approvals

- All Project licenses, certificates and Approvals obtained during and for the implementation of the Project.

6.5.4.1.28. As-Built Drawings and Data:

- a) Red mark-up of all the "Issued for Construction" drawings;
- b) Final "As-Built" station design drawings;
- c) Final "As-Built" survey data; and.

6.5.4.1.29. Other Documents

- Any other documentation related to the implementation of the Project and requested by the Owner for inclusion into the final documentation package.

6.5.4.1.30. Data Warehouse

- For further details concerning data warehouse reference is made to Annex E, Attachment 1 "Project Control Management Requirements".

6.5.4.1.31. Interface/Input to Assess Integrity Management Program

- This work unit entails the compilation and input of all relevant data into the Owner's asset integrity management program.

6.5.4.2. WSG 5.4.2 Akabel Pigging Stations

6.5.4.2.1. WSG 5.4.2.1 Preparatory Works and Site Clearing

- The Contractor shall be responsible to execute all preparatory Works required before starting the construction of the Project installations, including, but not limited to:
 - a) Setting out of station area;
 - b) Pre-construction survey;
 - c) Preparation of temporary access road; and
 - d) Clearing of area affected by construction, camps, storage and pre-fabrication area, workshops, etc.

6.5.4.2.2. WSG 5.4.2.2 Site Preparation Earth Works

- Based on the Contractor's engineering, the Contractor is responsible for execution of earth Works required, considering transitioning from actual topography to the station soil contour defined during design optimizing the volumes of cut and fill. The work comprises the following main items without being limited to:
 - a) Cut of ground to level station, including proper compaction of the subsoil;
 - b) Separation of cut Material into usable and non-usable (spoil) Material;

- c) Discharge of soil Material and fill in adjacent areas including landscaping in accordance with environmental requirements. Deposits and landscaping shall be subject to Approval by the Owner;
- d) Placing of fill Material in suitable layers, either by re-use of suitable cut Material or with imported Material. Imported Material shall be taken from suitable borrow pits identified by the Contractor, subject to Approval by Owner; and
- e) Compaction of fill Material in accordance with Project Specifications.

6.5.4.2.3. WSG 5.4.2.3 Temporary Access Roads

- Prior to accessing any additional area for the construction of temporary accesses, the Contractor shall furnish to Owner written evidence that all of these areas have been acquired in full compliance with the legal requirements of the Federal Republic of Nigeria and the relevant Project procedures.
- The Contractor shall maintain temporary access roads in safe and usable conditions at all times.

6.5.4.2.4. WSG 5.4.2.4 Station Earthworks

- Once site preparation is complete, the Contractor will execute the earthworks for the station construction, which includes the following items, without being limited to:
 - a) Excavation for foundations;
 - b) Compaction of subsoil;
 - c) Incorporation and compaction of imported fill (gravel) to improve soil conditions;
 - d) Improvement of subsoil with special measures (injections, gravel piles, vibro-compaction, etc.) as required;
 - e) Excavation for underground facilities, like pipe trenches, cable trenches, underground utility (ancillary) systems, culverts, channels, etc.;
 - f) Safeguarding of excavations;
 - g) Drainage of the Site during construction activities;
 - h) Excavation and proper compaction of subsoil for roads, walkways, paved areas, and in general all infrastructure measures;
 - i) Bedding and backfilling of trenches, buildings, etc. with suitable Material including proper compaction; and
 - j) Preparation of the final station surface for reinstatement.

6.5.4.2.5. WSG 5.4.2.5 Foundations

- The Contractor shall build all foundations for equipment (process and service). In an early stage of the design, the Contractor shall expedite from equipment Suppliers the issuing of all construction and design details (e.g. weight, dimensions, static and dynamic loads, anchor bolts location, etc.) to execute foundations work. The foundation work includes, but is not limited to:
 - a) Foundations for pipe supports;
 - b) Verification of subsoil and performed compaction;
 - c) Placing of lean concrete as required;

- d) Execution of formwork, reinforcement;
- e) Pouring of concrete, proper vibration, curing of concrete;
- f) Removal of formwork; and
- g) Preparation of adequate interface to mounting of piping, equipment, steelworks, etc..

6.5.4.2.6. WSG 5.4.2.6 Buildings

- “Not applicable.”

6.5.4.2.7. WSG 5.4.2.7 Roads, Infrastructure and Reinstatement

- The Contractor shall be responsible for the construction of roads, pavements, general civil infrastructure, and reinstatement of the Site to final Site conditions. The following main activities are included, without being limited to:
 - a) Permanent access road to the station, including connection to the public road;
 - b) Roads inside the station;
 - c) Paved areas, in areas where vehicle access is required for maintenance, walkways for pedestrian maintenance access;
 - d) Placing of washed gravel in pipe corridors of above ground piping, around equipment and in areas accessible for maintenance; and
 - e) Greening of the remaining station area. Seeding and irrigation up to first mowing;
 - f) Landscaping of the area outside the station area.

6.5.4.2.8. WSG 5.4.2.8 Mechanical Equipment Installation (Process and Service)

- The Contractor shall install all mechanical equipment related to main process as well as for service systems following the instructions provided by equipment manufacturers in order to ensure their proper operation. The installation of the following main equipment shall be included, without being limited to:
 - a) Vent stack package; and
 - b) Pig launcher/receiver.

6.5.4.2.9. WSG 5.4.2.9 Piping Works

- Piping work comprises of all the activities related to the installation of pipe work inside the station, including the pre-fabrication of pipe spools, the installation of all piping on
- Site, the installation of valves and any other devices directly connected to different systems as indicated in WSG 5.4.2.8. This piping works shall be carried out as specified in the Project documents. This includes but shall not be limited to:
 - a) Structural steel for pipe supports;
 - b) Storage, preservation and handling of piping equipment;
 - c) Pre-fabrication of pipe spools and pipe supports;

- d) Transport to Site of spools, piping and valves;
- e) Installation of pipe supports and alignment of the supports;
- f) Installation of piping for all mechanical equipment and systems;
- g) Alignment of the piping, including proper alignment for equipment connections in accordance with Vendor requirements to allow stress free connections;
- h) Welding (Note: NDE is included in WSG 5.4.2.15);
- i) Installation of valves;
- j) Connection of piping to equipment and valves
- k) The piping shall be cleaned before connection to any valve or equipment and the proper alignment shall be verified to execute stress free connections to equipment; and
- l) Note: Hydro test and other inspection and testing are included in **WSG 5.4.2.15.**

6.5.4.2.10. WSG 5.4.2.10 Painting/Insulation/Fireproofing

- The Contractor shall perform painting, insulation and fireproofing works for all piping works, equipment and structures for which such requirements are defined in Specifications and the design.

6.5.4.2.11. WSG 5.4.2.11 Electrical Installations

- The Contractor shall execute all electrical Works related to as lighting, earthing and lightning within the station. The Work shall include, but not limited to:
 - a) Power and lighting distribution panels for lighting, small power receptacles, CCTV etc.;
 - b) Installation of plant lighting system, area lighting, Street lighting, fence lighting, security lighting etc. including lighting poles, supports, cabling and wiring etc.; and
 - c) Installation of plant earthing system including necessary connections to equipment e.g., Panel boards, transformers, lightning protection of buildings and equipments etc.;

6.5.4.2.12. WSG 5.4.2.12 Instrumentation Installations

- The Contractor install instrumentation including cable laying within the stations and all other pipeline related instrumentation, intrusion detection system, SCS based on (PLC/RTU), ESD system and control containers (i.e. the cabin that houses all these equipment), Containerized E-House/Field Auxilliary Room(FAR). There will be RTU room in each manifold that shall house SCADA, Telecom equipment, Power supply etc.

6.5.4.2.13. WSG 5.4.2.13 Security Systems, Civil, Mechanical & Electrical

- This work package includes the supply and installation of the permanent security systems of the station, namely the civil, mechanical and electrical installations. The control and telecom installations shall be included in WG 5.5. The following systems shall be installed, without being limited to:
 - a) Security installations; and
 - b) Fences, gates, earth and concrete walls, culverts, etc.

c)

6.5.4.2.14.WSG 5.4.2.14 Ancillary Installations

- “Not applicable.”

6.5.4.2.15.WSG 5.4.2.15 Inspection and Testing

- All construction inspections and tests shall be performed in accordance with the requirements as laid down in Annex C, Attachment 1 “Quality Management System Requirements” and in accordance with the approved Inspection and Test Plan. The Owner shall have the right to witness all inspection and testing performed by the Contractor and any of its SubContractors or Vendors supplying Materials, equipment or Services for the Project.
- The Contractor shall submit its inspection and test plan in accordance with the requirements specified in Annex C, Attachment 1 “Quality Management System Requirements” and as specified in the relevant Project Specifications for the Approval by the Owner not later than forty-five (45) days after the Commencement Date and fifteen (15) days prior to the commencement of the Works.
- The Contractor’s inspection and test plan shall address all Works, identifying the inspection and test criteria as specified in the relevant Project Specifications, required by the Authorities of the Federal Republic of Nigeria, in accordance with best international practice and as required to ensure the quality of the Works.
- The inspection and test plan shall identify all “witness points” and “hold points”, where the Works may only proceed after satisfactory examination or test performance witnessed by the Owner or its authorised representative.
- All tests shall be documented and signed by the Contractor’s supervision and/or QA personnel responsible for the test performance.
- All test records shall be compiled and become part of the as-built documentation and the mechanical completion records of the Works.

6.5.4.2.16.WSG 5.4.2.16 Final Documentation and As-Built

- The following, but not limited to, documents and data, shall be included in the as-built Documentation package that shall be submitted by the Contractor. All documents shall be submitted in electronic and hardcopy (paper) format as described in Exhibit C, Annex F “Documentation Requirements”.
- All documents generated by the Contractor and submitted in electronic file format shall also be submitted in editable electronic file format, whereas the term editable shall refer to or editable in commercially available software programmes and file formats, such as but not limited to Microsoft office (MS-word, MS-windows, MS-excel, MS-Project, MS- power point, MS-Visio and others), Primavera, latest revision of AutoCAD, ESRI software and others.

6.5.4.2.17. Vendor and Manufacturers Documents, Drawings, Certificates

- Vendor manuals;
- Vendor operating & maintenance manuals;
- Vendor drawings;
- Material certificates;
- Base material certificates;
- Factory acceptance test certificates;
- Importation and shipping documents; and
- Damage and repair reports.

6.5.4.2.18. MRB, Updated Project Manuals, Specification and Procedures:

- Design manual, Specifications and procedures;
- Construction manual, Specifications and procedures;
- Certification manual; and
- Operation and maintenance manual and procedures.

6.5.4.2.19. Project Control and QHSE Documentation and Reports:

- QA/QC documentation;
- HSE documentation;
- Inspection and test reports; and
- Radiographs and hardcopies of ultrasonic results.

6.5.4.2.20. Project Licenses, Certificates and Approvals

- All Project licenses, certificates and Approvals obtained during and for the implementation of the Project.

6.5.4.2.21. As-Built Drawings and Data:

- Red mark-up of all the "Issued for Construction" drawings;
- Final "As-Built" station design drawings;
- Final "As-Built" survey data; and.

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6.5.4.2.22. Other Documents

- Any other documentation related to the implementation of the Project and requested by the Owner for inclusion into the final documentation package.

6.5.4.2.23. Data Warehouse

- For further details concerning data warehouse reference is made to Annex E, Attachment 1 "Project Control Management Requirements".

6.5.4.2.24. Interface/Input to Assess Integrity Management Program

- This work unit entails the compilation and input of all relevant data into the Owner's

Asset Integrity Management Program.

6.5.5. WG 5.5 SCADA and Telecommunication Construction

- Interfaces Between Pipeline, SCADA and telecom are defined in Exhibit C, Annex G, Attachment 2 "Interface Plan and Matrix".

6.5.6. WG 5.6 Mechanical Completion

- During the mechanical completion Phase of the construction, the Contractor shall demonstrate that all systems, sub-systems and components of its Scope of Work have been installed in accordance with the design documents and Project requirements and:
 - All equipment, components and materials used during the construction have been properly inspected and are accompanied by the specified certificates and inspection notices;
 - Any deficiency classified in Category A shall be cleared prior to the Approval and the signature of the Mechanical Completion Certificate (MCC) by Contractor's and the Owner's Commissioning authorities. A limited number of minor deficiencies, classified in Category B, may be allowed to remain outstanding, captured via a punch list system. They may be completed later, since they do not prevent the passage to the commissioning phase.
 - The mechanical completion of the Project shall be achieved individually for all systems and sub-systems within the main construction work groups, namely "Pipeline".
 - The pre-requisites and detailed requirements for the performance of the completion program of the Project are defined in Exhibit C, Annex I, Attachment 1 "Completions Management Requirements - EPC Pipeline".

6.5.6.1. WSG 5.6.1 Mechanical Completion Management

6.5.6.1.1. Organization

- The Contractor is responsible for the performance and management of its mechanical completion activities.
- The mechanical completion is performed at the end of the construction period under the responsibility of construction manager, and shall be overseen by a dedicated mechanical completion manager who later-on takes over the management of the completion of the Project (Work Package 6 "Completion").
- The Contractor shall submit the Organization Chart and relevant Job Descriptions for its mechanical completion and as-built management organization and commence mobilization of the relevant personnel at least ninety (90) days prior to the scheduled mechanical completion of the first

system of the Project, for Approval by the Owner.

6.5.6.1.2. Mechanical Completion Management Documents

- The Contractor shall prepare all required documents for the management of the mechanical completion, including but not limited to:
 - a) Mechanical completion manual;
 - b) Mechanical completion plan (system based test and inspection plans); and
 - c) Mechanical completion procedures.
- The Contractor shall submit the completions management documents not later than ninety (90) days prior to the scheduled mechanical completion of the first system of the Project, for the Approval by the Owner.

6.5.6.1.3. Other Mechanical Completion Documents, Registers and Checklists

- The Contractor shall prepare and administer all required documents, registers and checklists for the performance of the completion activities, including but not limited to:
 - a) System / sub-system breakdown;
 - b) System / sub-system status list for mechanical completion;
 - c) Mechanical completion checklists;
 - d) Mechanical completion test sheets and test records;
 - e) Mechanical completion punch lists.
 - f) (including categorization of punch list items into “A” and “B” Items, with “A” Items to be completed prior to achieving the Mechanical Completion and moving on to the next phase);
 - a) Additional registers and lists;
 - b) Mechanical completion dossier (MCD);
 - c) Mechanical completion certificate (MCC); and (ready for Approval and signing by the Owner).
- The Contractor shall submit all other completion documents, registers and checklists not later than thirty (30) days prior to the achieving of the scheduled mechanical completion of the first system of the Project for Approval by the Owner.

6.5.6.1.4. Mechanical Completion Certification Database

- The Contractor shall implement a mechanical completion and certification database which shall be integral part of the completions database for the administration of the entire completions program as specified in work package 6 “Completion”. The relevant data administered by the completions database shall include, but not limited to:
 - a) test records, certificates, material certificates, factory acceptance test certificates for all materials and components incorporated into the Project;
 - b) all records for the examination, inspection and testing of the construction and installation works of the Project;
 - c) test (completion) status of each system / sub-system; and
 - d) any other records or data obtained during the mechanical completion of the Project.

6.5.6.1.5. Mechanical Completion Interface Management

- The Contractor is responsible for the management of all interfaces to other Contractors, the Owner, Authorities and official Entities of the Federal Republic of Nigeria or any other stakeholders during the performance of the mechanical completion activities for the Project.

6.5.6.1.6. Mechanical Completion Reporting

- The Contractor shall establish a system for the reporting of all mechanical completion and as-built preparation affairs, including but not limited to schedule and progress of the mechanical completion of all relevant Project systems, the planned forecasted and actual achievement of the mechanical completion for all individual Completion Milestones, the production of the relevant as-built documentation, areas of concern, corrective actions. The reporting shall be performed within the framework of **scope of WSG 1.8.8 "Project Reporting"**.

6.5.6.1.7. Inspection and Testing during Construction

- All records of tests and inspections that have been performed during construction and/or as part of the mechanical completion activities and tests shall be compiled as part of the mechanical completion records.
- As part of the mechanical completion all relevant inspection and test records, certificates and documents shall be organised on a system, sub-system basis, so as to demonstrate that the relevant system or sub-system has been constructed and completed in accordance with the Project requirements.

6.5.6.1.8. Mechanical Completion Programme

- Mechanical Completion inspections and tests shall be witnessed in accordance with the Project quality requirements as required by the Regulatory Authorities of the Federal Republic of Nigeria and the Owner.
- In accordance with the involved disciplines mechanical completion activities, checks and verifications be carried out, but not limited to, for:
 - a) Pipeline;
 - b) Piping;
 - c) Mechanical / pressure vessels / tanks;
 - d) Civil / structural;
 - e) Buildings and shelters;
 - f) HVAC;
 - g) Painting/insulation/fire Proofing;
 - h) Electrical;
 - i) Cathodic protection;
 - j) Instrument and control;
 - k) Fire safety;
 - l) HSE; and

m) (Electronic security and surveillance).

6.6. Work Package 6 Completion

- The following section provides an introduction to the Project completion programme.
- The Project completion starts with achieving the milestone of “Mechanical Completion” at the end of the construction of the Project and comprises of 3 distinct programme phases:
 - a) Pre-commissioning;
 - b) Commissioning and start-up; and
 - c) Trial operation.
- During the execution of the Project Completion Programme the following milestones shall be achieved:
 - a) First gas flow; and
 - b) Provisional acceptance (Taking-Over Certificate).
- The Project completion ends with the Taking-Over of the Installation by the Owner.
- With reference to the Conditions of Contract the following “tests on completion” shall be performed during the implementation of the program:
 - a) Pre-commissioning tests;
 - b) Commissioning tests; and
 - c) Trial operation.
- The pre-requisites and detailed requirements for the performance of the completion program of the Project are defined in **Annex I, Attachment 1 “Completions Management Requirements - EPC Pipeline”**.

6.6.1. Work Group 6.1 Completion Management and General Items

6.6.1.1. WSG 6.1.1 Completion Management

6.6.1.1.1. Organization

- The Contractor is responsible for the performance and management of its Scope of Work within the completions program of the overall Project.
- The entire completions program shall be managed by a dedicated completions manager.
- The Contractor shall submit the organization chart and relevant job descriptions for its completions management organization for Approval by the Owner.
- The Contractor shall commence with the mobilization of the relevant key personnel at an early stage, to achieve the requirements defined below for submission of completion documents and to allow a smooth transition from construction (mechanical completion) to completion (pre-commissioning, commissioning and start-up and trial operation) at least 30 (thirty) days prior to the scheduled Mechanical Completion of the first system of the Project.

6.6.1.1.2. Completion Management Documents

- The Contractor shall prepare all required documents for the management of the Project completions activities, including but not limited to:
 - a) Project completions manual;
 - b) Project completion plan; and
 - c) Project completions procedures (including pre-commissioning, commissioning and start-up and trial operation procedures as well as individual procedures for all "Tests on Completion").
- Relevant Completion management documents shall be prepared as part of WSG 3.4.5 Contractor's Engineering – defined in details in Exhibit C, Annex I, Attachment 1 "Completions Management Requirements".
- The Contractor shall submit the Project completions management documents not later not later than ninety (90) days prior to the achieving of the scheduled mechanical completion of the first system of the Project for Approval by the Owner. In case of a phased mechanical completion this requirement is valid for each system/sub-system.

6.6.1.1.3. Other Completion Documents, Registers and Checklists

- The Contractor shall prepare and administrate all required documents, registers and checklists for the performance of the completion activities, including but not limited to:
 - a) System / sub-system breakdown;
 - b) System / sub-system status list for each completions phase;

- c) Items register for each completions phase;
- d) Checklists, test records and activity records for each completions phase;
- e) Punch lists for each completion phase including categorization of punch list items into “A” and “B” Items, with “A” Items to be completed prior to entering into the next completions phase; and
- f) Dossiers, certificates and notices to proceed for each completions phase.

- The Contractor shall submit all other completion documents, registers and checklists not later than thirty (30) days prior to the scheduled mechanical completion of the first system of the Project for Approval by the Owner.

6.6.1.1.4. Interface Management

- The Contractor is responsible for the management of all interfaces to other EPC Contractors, the Owner, Authorities and official Entities of the Federal Republic of Nigeria or any other stakeholder during the performance of the Project Completion activities.

6.6.1.2. WSG 6.1.2 General Items for Project Completion

6.6.1.2.1. Completion Reporting

- The Contractor shall establish a system for the reporting of all completion affairs, including but not limited to Schedule and Progress of Completion activities including the planned forecasted and accrual achievement of completion milestones, areas of concern, corrective actions. The reporting shall be performed within the framework of scope of WSG 1.8.8 “Project Reporting”.

6.6.1.2.2. Engineering Data Warehouse Database

- The Contractor shall implement a completion database for the administration of all completion Data including but not limited to:
 - a) test records, certificates, material certificates, factory acceptance test certificates for all materials and components incorporated into the Project;
 - b) all records for the examination, inspection and testing of the construction and installation works of the Project;
 - c) test (completion) status of each system / sub-system; and
 - d) all completions activities and specific tests performed as part of the Project completion activities.
- The database shall be integrated into the previously established mechanical completion and certification database (see WSG 5.6.1 Mechanical Completion”).

6.6.2. Work Group 6.2 Completions Programme Phases

- The completion programme consists of three distinct programme phases shown in Figure 11-1: “Project Completion with preceding and subsequent Project Phases” and further defined in the following work sub-groups.

6.6.2.1. WSG 6.2.2 Pre-commissioning

- After receipt of the Mechanical Completion Certificate and achieving the Project Milestone “Mechanical Completion”, the Contractor shall proceed with pre-commissioning of the Project.
- The pre-commissioning is performed by the individual EPC Contractors for all installations, systems and sub-systems under their Scope of Work.
- The pre-conditions and detailed requirements for the performance of the Pre-commissioning are defined in Annex I, Attachment 1 “Completions Management Requirements - EPC Pipeline”.
- The Contractor shall account for the presence of the Owner’s operational staff and inspector’s during the performance of the pre-commissioning works.

6.6.2.1.1. Pre-commissioning Activities and Tests

- The Contractor’s detailed pre-commissioning activities and tests shall be specified in its pre-commissioning procedures.
- All pre-commissioning tests shall be witnessed by the Owner or its designated representative.
- The Contractor’s detailed pre-commissioning activities and tests shall include, but not limited to:
 - a) Provision of input to pre-commissioning procedures of other EPC Contractors,
 - b) Completion of flushing, cleaning and drying of piping and vessels;
 - c) Verification of dew point of sections filled with dry air;
 - d) Conformity checks on each individual item, equipment or component;
 - e) Loop tests for electrical and instrument/SCADA/telecom cables, equipment and systems;
 - f) Static Tests including the verification of alignments, set points and calibration of instruments;
 - g) Functional Test (cold/ without the introduction of hydrocarbons) of all systems and sub-systems; and
 - h) Pre-commissioning, commissioning and start-up and of all ancillary, utility, support, safety and other systems required to support the commissioning, start-up and operation of the Project.

6.6.2.1.2. Pre-commissioning Documents and Dossier

- The Contractor shall update all other documents identified in WSG 6.1.1 “Completion Management”, Section “Other documents Registers and Checklists”, demonstrating the successful performance of the pre-commissioning activities.
- All documents shall be included in the pre-commissioning dossier.
- Following the Approval by the Owner, the pre-commissioning dossier shall be attached to the “Notice to Proceed for Commissioning and Start-up”, which shall be issued by the Owner after he is satisfied that all pre-commissioning activities have been successfully completed by the Contractor.

- With issuance of the “Notice to Proceed for Commissioning and Start-up”, the Owner certifies the achievement of the Project milestone “Ready for First Gas-In”.

6.6.2.2. WSG 6.2.3 Commissioning and Start-up

- After the successful completion of the pre-commissioning activities, the receipt of the “Notice to Proceed for Commissioning and Start-up” and the confirmation of the availability of gas for commissioning purposes (including line pack) the Contractor shall proceed with commissioning and start-up of the Project.
- The commissioning and start-up is performed by the individual EPC Contractor under the management and responsibility of the EPC Contractor SCADA.
- The pre-requisites and detailed requirements for the performance of the Commissioning and Start-up are defined in Annex I, Attachment 1 “Completions Management Requirements - EPC Pipeline”.
- The Contractor shall cater for the presence of the Owner’s operational staff, the Owner’s Representative and Third Party inspectors during the performance of the Commissioning and Start-up works.

6.6.2.2.1. Commissioning and Start-up Activities and Tests

- All commissioning and start-up activities and tests as well as the relevant start-up sequence for all installations shall be defined in the relevant commissioning and start-up
- procedures. All commissioning and start-up tests shall be witnessed by the Owner or its designated representative.
- The Contractor’s commissioning and start-up activities and tests shall include, but not limited to:
 - a) Provision of input to the overall Project commissioning procedure and the commissioning and start-up procedures of other EPC Contractors;
 - b) Re-calibration of instruments (if required);
 - c) Hot tests (functional tests) of the equipment;
 - d) Functional test for systems, sub-systems and components;
 - e) Integrated tests of systems;
 - f) Initial filling of the installations with gas;
 - g) Leak detection tests;
 - h) Start-up of all the installations (as per the Approved start-up sequence);
 - i) Operational tests; and
 - j) Test runs.

6.6.2.2.2. Commissioning and Start-up Documents

- The Contractor shall update all other documents identified in WSG 6.1.1 “Completion Management”, “Other documents Registers and Checklists”, demonstrating the successful performance of the commissioning and start-up activities.
- All documents shall be included in the commissioning and start-up dossier.

6.6.2.2.3. Notice to Proceed for Trial Operation

- Following the Approval by the Owner, the start-up dossier shall be attached to the “Notice to Proceed for Trial Operation”, which shall be issued by the Owner after he is satisfied that the commissioning and start-up of all installations has been successfully completed by the Contractor.
- With issuance of the notice to proceed for Trial Operation, the Owner certifies the achievement of the Project milestone “Ready for First Gas Flow”.

6.6.2.3. WSG 6.2.4 Trial Operation

- After the successful completion of the commissioning and start-up of the Project, the receipt of the “Notice to Proceed for Trial Operation” and the confirmation of the operational readiness of the upstream and downstream installations to deliver and receive the gas quantities required during Trial Operation, the Contractor shall proceed with the performance of the Trial Operation of the Project.
- The Trial Operation shall be performed by the individual EPC Contractors under the management and responsibility of the EPC Contractor SCADA of the Project.
- The pre-conditions and detailed requirements for the performance of the Trial Operation period are defined in Annex I, Attachment 1 “Completions Management Requirements – EPC Pipeline”.
- The Contractor shall satisfy the requirement for the preparation of all operation and maintenance documents as part of the activities for the work sub-group “Trial Operation”, activity “Operation and Maintenance Procedures and Documents”.
- The Contractor shall account for the presence of Owner’s operational staff and the Owner’s inspectors during the performance of the Trial Operation period.

6.6.2.3.1. Trial Operation Activities

- During the trial operation period the Contractor shall perform the following activities but not limited to:
- Operation of the Project in all different operation modes including local, remote operation upset and emergency operations;
- Performance of all standard maintenance procedures for the operation and maintenance of the Project; and
- Performance of emergency and safety operations and drills including firefighting, shut down, isolation and evacuation drills.

6.6.2.3.2. Trial Operation Tests

- The Contractor’s Trial Operation tests shall be specified in its Trial Operation tests procedures. All Trial Operation tests shall be witnessed by the Owner or its designated representative and shall include, but not limited to:
 - a) Reliability test;
 - b) Project performance test; and

- c) Performance tests for all other systems and/or sub-systems as defined in the relevant functional and performance requirements.

6.6.2.3.3. Operation and Maintenance Manuals, Plans, Procedures and Documents

- The Contractor shall submit all operation procedures and documents for review by the Owner, at least thirty (30) days prior to the anticipated start of the Trial Operation.
- Operation and maintenance documents shall cover all aspects of the project operation / maintenance and shall include but not limited to:
 - a) Operation and maintenance manual;
 - b) Operation and maintenance plan;
 - c) Operation and maintenance and repair procedures;
 - d) Job descriptions;
 - e) Emergency response plans for operations; and
 - f) Pipeline patrolling plans and procedures.
- Detailed requirements for the preparation of operation / maintenance manuals are specified in Annex J - Attachment 1 "Operation Management - EPC Pipeline" and Annex J, Attachment 2 "Maintenance Management - EPC Pipeline".

6.6.2.3.4. Trial Operation Documents

- The Contractor shall update all other documents identified in WSG 6.1.1 "Completion Management", "Other Documents Registers and Checklists", so as to demonstrate the successful performance of the trial operation.
- All aforementioned documents and in particular the signed and Approved certificates for all tests, including but not limited to reliability and performance tests that have been performed during the Trial Operation period shall be included in the Trial Operation Dossier.
- Following the Approval by the Owner, the Trial Operation Dossier shall be attached to the "Taking-Over Certificate".

6.6.2.3.5. Taking-Over Certificate / Provisional Acceptance

- Any deficiency classified in category A shall be cleared prior to the Approval and signature of the Taking-Over Certificate by the Owner's Commissioning and Operations Authorities. A limited number of minor deficiencies classified in category B may be allowed to remain outstanding, captured via a punch list system. They may be completed later, since they do not prevent the system handover and the start-up phase.
- The Contractor shall prepare the "Taking-Over Certificate", which shall be issued by the Owner after he is satisfied that the Trial Operation and all associated tests have been successfully completed by the Contractor and the Contractor has satisfied all relevant provisions of the Contract for Taking-Over of the Project by the Owner.
- The Contractor shall prepare the Taking-Over / Provisional Acceptance punchlist

and a specific provisional Acceptance dossier. The requirements for the Taking-Over and the Taking-Over/provisional Acceptance Dossier are included in Annex I, Attachment 1 "Completions Management Requirements - EPC Pipeline".

- With issuance of the Taking-Over Certificate of the Project, the Owner certifies the achievement of the Project milestone for "Taking-Over / Provisional Acceptance".

6.7. Work Package 7 O&M Training And Assistance

6.7.1. Work Group 7.1 O&M Training

- The Contractor shall provide operation and maintenance training to the Owners personnel or its designated operator for the Project.
- The training shall be provided well ahead of the mechanical completion of the Project, so that the Owner's fully trained personnel may observe all Completion activities and in particular the Trial Operation period.
- All training shall consider the Approved operation and maintenance documents prepared as part of WSG 6.2.4 "Trial Operation", Section "Operation and Maintenance Procedures and Documents".
- The training shall be organized by category of employees, and calibrated in relation to their duties and functions, such as:
 - a) Management and control of the operations;
 - b) Management and execution of maintenance works; and
 - c) Safety management and monitoring.

6.7.1.1. WSG 7.1.1 Training Management

6.7.1.1.1. Training Organisation

- The Contractor shall establish a dedicated organization for the performance and management of the training of the Owner's operation and maintenance personnel.
- The Contractor shall submit the organization chart and relevant job descriptions for its training and operation assistance personnel one hundred twenty (120) days prior to the Mechanical Completion of the Project. The Contractor shall commence the training activities ninety (90) days prior to the Mechanical Completion of the Project.
- The essential elements of the training shall be completed at least thirty (30) days prior to the Mechanical Completion of the Project.

6.7.1.1.2. Training Management Documents

- The Contractor shall prepare all required documents for the performance and management of the training activities of the Project, including but not limited to:
 - a) Training manual;
 - b) Training plan; and
 - c) Training procedures.
- The Contractor shall submit its training documents not later than one hundred

twenty

- (120) days prior to the Mechanical Completion of the Project for Approval by the Owner.

6.7.1.1.3. Training Course Materials

- The Contractor shall provide all course material in English language and in sufficient quantities, so that each course attendee receives a complete course material package. Course materials shall include, but not limited to:
- Course booklets;
- Presentation materials (slides); and
- Specific manuals, procedures and drawings covering the relevant course topics.

6.7.1.1.4. Training Activities Reporting

- The Contractor shall establish a system for the reporting of all training activities. The reporting shall be performed within the scope of WSG 1.8.8 “Project Reporting” and shall be submitted and updated on a weekly basis.

6.7.1.2. WSG 7.1.2 Training Courses

- The Contractor shall accept all participants nominated by the Owner for the participation in the training program. Training courses shall be organized in modular structure allowing the nomination of attendees in accordance with their relevant job descriptions.
- The modular course structure shall be outlined in the training manual subject to Approval by the Owner.
- The location for conducting the training courses may vary in relation to the nominated participants and shall be subject to the Owner’s Approval.
- The following, but not limited to, course modules shall be included:
 - a) Project overview/project introduction;
 - b) General operation training;
 - c) Pipeline systems (operation and maintenance);
 - d) Metering system;
 - e) Electrical systems (operation and maintenance);
 - f) Mechanical systems (operation and maintenance);
 - g) Utility and support systems (operation and maintenance);
 - h) Corrosion inspection and prevention and cathodic protection system; and
 - i) Others.

6.7.1.3. WSG 7.1.3 External Vendor Trainings

- External Vendor training shall be held by the relevant equipment Vendor or manufacturer and primarily cover operation and maintenance requirements

for the relevant equipment.

- Vendor trainings may be arranged at Vendors premises or at suitable project locations depending on the training requirements and subject to Approval by the Owner.

6.7.1.3.1. Main and Process Equipment

- Metering installations; and
- Valves and actuators.

6.7.1.3.2. Ancillary, Support and Utility Equipment

- Diesel generators;
- Electrical distribution panels and switchgear; and
- HVAC equipment.

6.7.2. Work Group 7.2 Operation and Maintenance Assistance

- The Contractor shall provide operation and maintenance assistance to the Owner after the provisional Acceptance/Taking-Over of the Project by the Owner.
- The operation and maintenance assistance shall be provided during the initial operation period for a duration of one (1) year.
- The following figure provides an overview of the initial operation period following the provisional Acceptance/Taking-Over of the Project by the Owner, until the issuance of the Maintenance Certificate, which constitutes the final acceptance of the Project.

6.7.2.1. WSG 7.2.1 O&M Assistance Management

6.7.2.1.1. O&M Assistance Organisation

- The Contractor shall establish a dedicated organization for the provision of operation assistance to the Owner, thereby supporting the Owner's operation personnel during the initial operation of the Project.
- The Contractor shall submit its organization chart and relevant job descriptions for its O&M assistance personnel sixty (60) days prior to the provisional Acceptance (Taking- Over) by the Owner.
- The Contractor shall complete the mobilization of its operation assistance personnel at least thirty (30) days prior to the provisional Acceptance (Taking- Over) by the Owner.

6.7.2.1.2. O&M Assistance Management Documents

- The Contractor shall prepare all required documents for the management of the O&M assistance to the Owner, including but not limited to:
 - a) O&M assistance plan;
 - b) O&M assistance procedures; and
 - c) Job descriptions.

6.7.2.1.3. Operation Procedures and Documents

- The Contractor's obligation for the preparation of all operation and maintenance documents is included in WSG 6.2.4 "Trial Operation", Section "Operation and Maintenance Procedures and Documents".

6.7.2.1.4. O&M Assistance Activities Reporting

- The Contractor shall establish a system for the reporting of all O&M assistance activities. The reporting shall be performed within the scope of WSG 1.8.8 "Project Reporting" and shall be submitted and updated on a weekly basis.

6.7.2.2. WSG 7.2.2 O&M Assistance Activities

- The operation and maintenance assistance shall be provided on a manpower basis.
- Whereas the operational responsibility lies with the Owner or its designated operator and the Contractor's personnel acts as an adviser under the responsibility of the Owner's line manager.

6.7.2.2.1. Extend and Duration of Service

- The services shall be provided under the Owner's operational shift concept and Contractor shall account for seven (7) days a week and eight (8) hours per day availability of the defined personnel within the framework of the relevant shift rotation.
- The period for the provision of operational assistance is defined as being one (1) year after commencement of commercial operation.

6.7.2.2.2. Qualification Categories

- Operational Assistance shall be provided in two qualification categories:
 - a) Manager; and
 - b) Operator / specialist.
- The exact Job description shall be determined based on the Approved job descriptions for the Project operation and maintenance.

7. Optional Scope: 48 Core x 38.2 km Fibre Optic Cable.

7.1. General Requirements

- Two fibre optic cables (FOC) of 48 cores each shall be laid in an already installed 4" HDPE cable conduit installed by the Pipeline Contractor for the Gas Gathering pipeline System from the control cabinets (RTU) at BBOU to the control cabinets at Elepa down to the Telecoms Radio Equipment Room in Akabel complex, a total distance of about 38.2km. almost the entire length of the cable shall pass through swamp and marshy terrain with few sections on dry land.
- The FOC shall be laid beside the 24" x 6.7km BBOU to Elepa Pipeline system and 36" x 32Km Elepa to Akabel Pipeline system and traversing Akabel methanol complex to terminate at the control cabinet in the complex.
- One 48-core FOC shall be part of the composite power cable and shall be laid at the appropriate safe distance from the pipeline within the Right of Way. The other cable strapped on the pipeline shall serve for leak detection
- The Contractor shall review all the associated FEED deliverables and documentation and update in Detail Engineering that will lead to a fit for purposed documentations required for procurement, installation and commissioning of the FOC and telecommunication systems.
- The Contractor shall be responsible for the design/engineering, manufacture, procure, installation, end terminations, and testing of the optical fibre cables with associated spare accessories necessary for commissioning as well as the 2 year running spares. The design shall provide for extra cable lengths to be coiled up at appropriate distances to facilitate repairs without the need for procurement of additional cable lengths. CONTRACTOR to earmark an extra 10% of the nominal cable length to accommodate this requirement.
- The CONTRACTOR shall conduct an independent survey to gather all necessary data/information and ascertain the various environmental conditions where the FOC shall traverse to determine cable coating/protection specifications, possible installation method, best splice point locations, splice housing specifications, and so on.
- The FOC shall be of 2 nos, 48 core single mode fibres type ITU – T G 652D
- The CONTRACTOR shall determine the method of the FOC installation in relation to the pipeline
- The optic fibre cable shall be constructed with as minimal joints (splice points) as feasible, and also to the relevant IEC 60794 standard
- Optical fibre cable connectors shall be either LC-type (Lucent connectors) IEC

61754-20 or IEC 61754-4 SC-type (subscriber/standard connectors) depending on the equipment interfaces, but not a mixture of connector types.

- Optical fibre cable patch panels shall be of a type suitable for use with the connectors, where conversion of any one connector can be easily changed to the other type within the same panel.
- The patch panel shall include a cover to protect the fibre from physical damage, exposure to moisture or other corrosive elements.
- The patch panel shall have a high density to conserve space but shall conform to cable management requirements.
- Where data cabinets are used, all such cabinets shall have removable and hinge reversible doors with transparent glass panels.
- Side panels shall be removable to permit access to cables and equipment. All equipment shall be connected to the earth bus bar. Enclosed data cabinets shall have ventilation roofs and fans.
- The CONTRACTOR shall make provision for the testing of the FOC with the relevant Owner personnel at the OEM location prior to shipment to Nigeria.

7.2. Installation and Hook-Up of fiber optic cable

- The Scope of Work includes Project Management, Detailed Design, all other necessary Engineering, Procurement for the;
 - a) Manufacture, Testing and supply of the Fibre Optic Cable System
 - b) Load out, Transport, Installation and Pre-commissioning including on-site Commissioning and Start-up support
- The CONTRACTOR shall ensure that the WORK complies with the requirements in Doc no. EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002-R03_Basis of Design, and shall meet the overall performance requirements.
- CONTRACTOR shall provide all necessary Testing of all major System elements, systems and sub-systems; and of the complete, integrated System. CONTRACTOR shall be responsible for isolating and correcting any problems that arise during any testing phase that either preclude the contractual requirements from being met or cannot be explained to the full satisfaction of the OWNER.
- CONTRACTOR shall provide all necessary installation services and equipment required for the successful delivery, and to meet the Performance Requirements for the System, which shall include but not limited to:
 - a) Secure all the construction licenses and/or permits required.
 - b) Perform site surveys of the Pipeline stations and the Right of Way
 - c) Verify existing route survey data and also surveys the cable route.

7.2.1. Manufacturer's Testing

- The Contractor shall be responsible to perform the following tests and witnessed by the Owner representative during manufacture.
 - a) Factory Acceptance
 - b) System Assembly
-

7.3. Cable Manufacture & Procurement

- Contractor shall manufacture and/or supply all the necessary equipment and materials required to meet the Performance Requirements for the System. CONTRACTOR shall complete analysis, tests, and demonstrations as reasonably requested by the OWNER so that OWNER may be confident of the expected system performance and reliability. This may involve a proof of concept demonstration if reasonably requested.
- Acknowledge and response to all reasonable OWNER issues and questions regarding the Work throughout the Contract implementation period to allow OWNER to obtain all information required for the successful operation of the System.
- Contractor shall also secure all the permits required for equipment importation into Nigeria and perform training of a minimum of 3 OWNER personnel on all elements of the FOC System.
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7.3.1. Additional requirement.

- The following items/elements (not exhaustive though) of the fiber optics systems shall be procured with the FOC
 - a) Line Terminal Equipment (LTE)
 - (i) Cabinetries and housings for all LTE suitable for the project environment
 - (ii) Optical distribution frames, racks and shelves
 - (iii) Line Termination Equipment with ITU-compliant interfaces capable of Direct Wavelength Access (DWA)
 - (iv) Distribution frames, alarm concentration panels, and all required interconnection cabling and cable management (including to station ground systems)
 - (i) Alarm concentration panels, and all required interconnection cabling and cable installation hardware
 - (ii) Termination equipment and hardware to support voice communications between the Pipeline Stations

b) Cable and Accessories

- (i) 2 nos. 48 Core Optical Fiber Cable (suitable in marine environment)
- (ii) Cable splices, transitions and cable-to-cable joints protection
- (iii) Cable fixtures, connectors and terminations
- (iv) Cable Coupling, Entry glands and Cable end seals
- (v) Cable Rigging & Anchoring, Bend Restrictors, and Abrasion/impact protection

c) Software and specialized equipment

- (i) The software necessary for the function of the above items
- (ii) The licenses to operate software necessary for the function of the above item
- (iii) Optical Time Domain Reflectometer (OTDR)
- (iv) Fusion Splicer and Kit

d) Equipment/Material Spares

- (i) Appropriate level of all applicable spares (and storage cabinets) at each site (for all supplied terminal equipment (including the Maintenance Controllers) necessary to maintain the System in accordance with contractual specifications and requirements.
- (ii) Appropriate level of all applicable spares for all cable plant, necessary to maintain the System in accordance with contractual specifications
- (iii) Universal joint kits per industry practice (or equivalent Contractor joint kits)

7.3.2. Precommissioning and Testing of 48 Core Submarine Fiber Optic Cable**7.3.2.1. Testing**

- The Contractor shall be responsible to perform the following tests
 - a) System Acceptance Testing, including support of Owner-testing
 - b) System Design Capacity Testing
 - c) Stability Testing

7.3.2.2. Commission Support

- The Contractor shall provide to the Owner other technical supports and services during the commissioning of the FOC which shall as a minimum include but not limited to:
 - a) Support Owner planning for integration of the FOC system with their local onward network facilities on land.
 - b) Support Owner planning and testing of the System interoperability with the Owner's local network facilities (including backhaul to restoration facilities).
 - c) Support of testing of Owner-provided data communication facilities for

communication between various pieces of terminal equipment that may be required via facilities external to the System itself.

- d) Identifying and obtaining all Permits associated with operation of the FOC including permits in principle on behalf of Owner.

END OF DOCUMENT